

Group Theory for Physicists

Lecture Notes 2025-2026

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Contents

1	Group Theory in Physics: Overview	5
1.1	Noether's theorem	5
1.1.1	Symmetries versus conservation laws	5
1.1.2	Quantum mechanics and commuting observables	5
1.2	Relativity	6
1.2.1	Galilean relativity	6
1.2.2	Einsteinian relativity	8
1.3	Nuclear spectra	9
1.4	Molecular vibrations	9
1.5	Symmetry breaking	9
1.6	Crystallography and band structure	9
1.7	Particle physics	10
I	Finite groups	11
2	Finite Groups: Basic Definitions and Properties	13
2.1	Basic definitions, examples, Cayley table	13
2.1.1	First examples	13
2.1.2	Cayley table	15
2.1.3	Direct products	16
2.2	Homomorphisms, isomorphisms and automorphisms	16
2.3	Subgroups, cosets, normal subgroups, quotient groups	18
2.4	Extending groups	20
2.4.1	(Semi)direct products	21
2.4.2	Sequences & extensions	22
2.A	A list of small groups up to order 8	25
2.B	Group cohomology	25
2.C	Computing cohomology groups and cocycles over $A = U(1)$	27
3	Representation Theory of Finite Groups	29
3.1	Representations	29
3.1.1	First examples	31
3.2	Irreducible representations and direct sum representation	33
3.3	Tensor product representation	35
3.3.1	Tensor product of vector spaces	35
3.3.2	Tensor product representation	37
3.3.3	Clebsch-Gordan coefficients	37
3.4	The Great Orthogonality Theorem	39
3.5	Character theory	39
3.5.1	Characters, inner product of characters	40
3.5.2	A finite group has a finite number of irreps	41

3.5.3	Examples and character tables	42
3.6	Real, complex and quaternionic representations, the Frobenius-Schur indicator . .	44
3.7	Projective Representations	46
3.7.1	Constructing projective representations	47
3.A	The group algebra	50
3.B	Representation theory of the symmetric group: Young tableaux	51
3.C	Induced Representations	54
3.D	Application in physics: selection rules	55
3.E	Application in physics: Kramers' theorem	56
II	Lie groups and Lie algebras	59
4	The rotation groups $SU(2)$ and $SO(3)$	61
4.1	Warming up with $SO(2)$ and $U(1)$	62
4.1.1	Rotations in the plane and group theory	62
4.1.2	Irreducible representations $U(1)$ and $SO(2)$	65
4.2	Definitions and parametrizations of $SO(3)$ and $SU(2)$	66
4.2.1	Definitions	66
4.2.2	Parametrizations	66
4.3	The Lie algebras $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$	69
4.4	Topological aspects $SU(2)$ and $SO(3)$	71
4.5	Irreducible representations $SU(2)$ and $SO(3)$	72
4.5.1	Irreducible representations $\mathfrak{sl}(2)$	72
4.5.2	Computing matrix representations of $\mathfrak{sl}(2)$	76
4.5.3	Examples of irreducible representations of $\mathfrak{sl}(2)$	76
4.5.4	Irreducible representations $SU(2)$, the exponential map and the BCH formula	78
4.5.5	Examples of irreducible representations of $SU(2)$	79
4.5.6	Examples of irreducible and projective representations of $SO(3)$	80
4.6	Tensor product representations $\mathfrak{sl}(2)$ and Clebsch-Gordan coefficients	81
4.6.1	Examples	83
4.6.2	Computing Clebsch-Gordan coefficients	83
4.6.3	Coupling of two spinors	84
4.6.4	Coupling of arbitrary spins	86
4.7	Spherical harmonics	87
4.7.1	Motivation	87
4.7.2	Generators	89
4.A	Table of $\mathfrak{sl}(2)$ Clebsch-Gordan coefficients	90
5	Crystallography: Frieze Groups, Wallpaper Groups, Space Groups	91
5.1	Finite subgroups of $SO(3)$	91
5.2	Crystallography	92
5.2.1	Wallpaper groups	95
	References	101

Chapter 1

Group Theory in Physics: Overview

During the last century, it has become more and more clear that symmetry is THE central concept in physics. Starting from the famous theorem of Emmy Noether in classical mechanics, to Wigner's applications of group theory to the quantum mechanical description of molecules, to Landau's concept of symmetry breaking for characterizing phases of matter, all the way to gauge theories in quantum field theory and topological materials in condensed matter physics, it is impossible to understand any of the modern way of doing physics without having a firm grasp of group theory, the mathematical language for describing symmetries.

In this introductory chapter, I will try to highlight some of the direct applications of group theory in a variety of branches in physics. This list is of course by no means exhaustive, and its primary goal is to motivate you to take group theory seriously: a good understanding of group theory will help enormously in most of the courses that you will get in the coming years.

1.1 Noether's theorem

1.1.1 Symmetries versus conservation laws

The concepts of energy, momentum and angular momentum conservation form the pillars of classical mechanics. Emmy Noether, one of the most influential mathematicians ever, had the visionary insight that any such conservation law is a consequence of symmetries of the equations of motion: conservation of energy follows when the equations of motion are symmetric over translations in time (it is, they do not change as a function of time), conservation of momentum / angular momentum when the equations of motion are symmetric over translations / rotations in space.

More generally, classical mechanics can be developed using the so-called Lagrangian (or variational action) principle, in which the trajectories of the particles are the ones which minimize the action. Noether stipulates that to every differentiable symmetry of this action, there will be a corresponding conserved current.

1.1.2 Quantum mechanics and commuting observables

Noether's principle turns out to be even more central when moving to the equations of quantum mechanics. Noether's theorem becomes almost a tautology, and dictates that observables have conserved expectation values when they commute with a time-independent Hamiltonian:

$$[\hat{H}, \hat{O}] = 0 \Rightarrow \frac{d}{dt} \langle \psi(t) | \hat{O} | \psi(t) \rangle = 0 \quad (1.1)$$

Obviously, the Hamiltonian commutes with itself, so we have conservation of energy. In many situations, we also have rotational invariance, expressed by the fact that the total angular momentum operator $\hat{\mathbf{J}}^2$ commutes with the Hamiltonian. Hence the total angular momentum is conserved.

A central problem in quantum mechanics is the characterization of the eigenstates of a time-independent Hamiltonian. A basic fact in linear algebra is that two commuting operators (matrices) can be diagonalized simultaneously, which implies that eigenstates should be labelled by the eigenvalues of all conserved operators.

Example 1.1. *Let us consider an electron in a central potential $V(r)$, for which the corresponding Hamiltonian is rotational invariant. Eigenstates of this Hamiltonian can then be labelled by the eigenvalues E_n of $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\hat{r})$ and $\hbar^2 l(l+1)$ of $\hat{\mathbf{J}}^2$. Using spherical coordinates (r, θ, ϕ) , group theory dictates that the eigenvectors can then be written as $\phi_n(r)Y_l^m(\theta, \phi)$ with Y_l^m the so-called spherical harmonics and m a label characterizing orthogonal eigenvectors with the same l (as a convention, we choose them to be eigenvectors of $\hat{J}_z$). We will derive this later in this lecture, and see that this follows from the representation theory of the group $SO(3)$, the group of rotations in 3 dimensions.*

The above example shows that the eigenvectors of a non-relativistic Hamiltonian in a central potential can be characterized by the three quantum numbers n , l and m , the last two of which are completely determined by the representation theory of the group $SO(3)$. In reality, there is one extra quantum number called the spin s ; this quantum number is a consequence of the fact that quantum states are so-called *rays* (it is only determined up to a phase) and are hence allowed to transform *projectively* under the rotation group, according to the spin 1/2 representation of $SU(2)$. This is of immense importance in physics, as it is precisely this projective feature which enables the existence of fermions, and hence of the stability of matter (as a world with only bosons would be completely unstable). Not to exaggerate: the representation theory of groups is the main key in understanding quantum mechanics. Indeed, the spin-statistics theorem dictates that particles transforming according to half-integer representations of $SU(2)$ are fermions, for which the wavefunction has to be anti-symmetrized. As a consequence, another group of central importance enters the arena: the permutation group of n items. A nontrivial exercise in combinatorics (or representation theory of the permutation group) is the determination of the dimension of the subspace of a completely antisymmetric tensor with n indices where all indices can take $N \geq n$ values, which is the dimension of the Hilbert space of n electrons in N modes. The most important consequence of this group-theoretic fact is of course the Pauli exclusion principle, by far the most pervasive and strongest force in nature.

Example 1.2. *Let us build on the previous example and consider an atom with M electrons; due to the Pauli exclusion principle, we start filling up the different energy eigenstates, giving rise to the perfect periodicity of the periodic table (see Fig. 1.1).*

1.2 Relativity

1.2.1 Galilean relativity

One of the many visionary insights of Galilei was that the laws of physics are completely insensitive to the reference frame that one is using. In his words:

Shut yourself up with some friend in the main cabin below decks on some large ship, and have with you there some flies, butterflies, and other small flying animals. Have a large bowl of water with some fish in it; hang up a bottle that empties drop by drop into a wide vessel beneath it. With the ship standing still, observe carefully how the little animals fly with equal speed to all sides of the cabin. The fish swim indifferently in all directions; the drops fall into the vessel beneath; and, in throwing something to your friend, you need throw it no more strongly in one direction than another, the

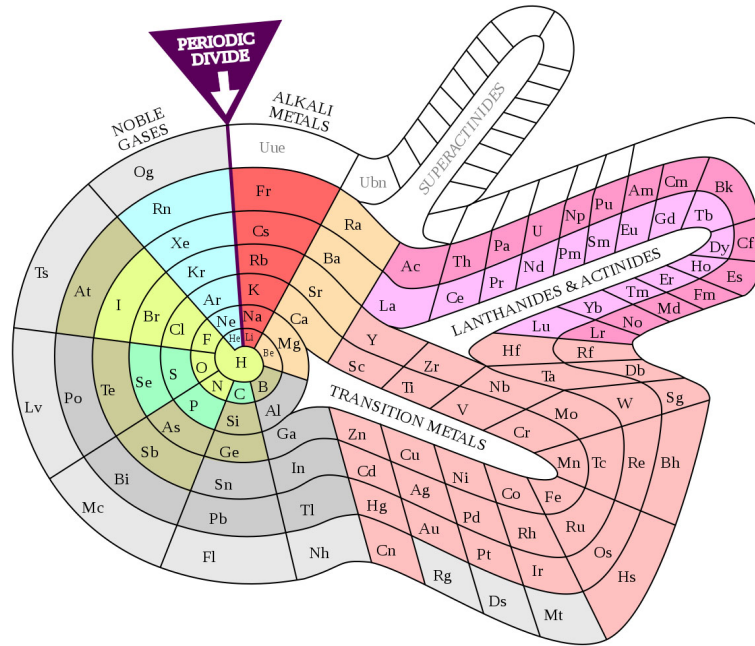


Figure 1.1: The periods 2,8,18,32 in the periodic table follow from the representation theory of the rotation group in 3 dimensions

distances being equal; jumping with your feet together, you pass equal spaces in every direction. When you have observed all these things carefully (though doubtless when the ship is standing still everything must happen in this way), have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still. In jumping, you will pass on the floor the same spaces as before, nor will you make larger jumps toward the stern than toward the prow even though the ship is moving quite rapidly, despite the fact that during the time that you are in the air the floor under you will be going in a direction opposite to your jump. In throwing something to your companion, you will need no more force to get it to him whether he is in the direction of the bow or the stern, with yourself situated opposite. The droplets will fall as before into the vessel beneath without dropping toward the stern, although while the drops are in the air the ship runs many spans. The fish in their water will swim toward the front of their bowl with no more effort than toward the back, and will go with equal ease to bait placed anywhere around the edges of the bowl. Finally the butterflies and flies will continue their flights indifferently toward every side, nor will it ever happen that they are concentrated toward the stern, as if tired out from keeping up with the course of the ship, from which they will have been separated during long intervals by keeping themselves in the air. And if smoke is made by burning some incense, it will be seen going up in the form of a little cloud, remaining still and moving no more toward one side than the other. The cause of all these correspondences of effects is the fact that the ship's motion is common to all the things contained in it, and to the air also. That is why I said you should be below decks; for if this took place above in the open air, which would not follow the course of the ship, more or less noticeable differences would be seen in some of the effects noted.

Galilei is describing Galilean transformations, where one makes a change of variables in the form

$$\vec{x} \rightarrow \vec{x} + \vec{v} \cdot (t - t_0) \quad (1.2)$$

This transformation is parameterized by $\vec{v}$ and t_0 , and gives rise to one of the simplest continuous groups (a so-called Lie group), and is effectively dictating relativity in non-relativistic systems (Einstein's relativity has as an extra ingredient the fact that there is a maximum speed). The fact that the laws of (non-relativistic) physics have to be invariant under the Galilean group transformations has extremely strong consequences: for the case of classical mechanics, it effectively dictates that the Lagrangian must be of the form

$$L(x, \dot{x}) = \frac{m}{2} \dot{x}^2 + V(x) \quad (1.3)$$

as this is the only way of ensuring that a Galileian coordinate transformation leaves the extrema of the action $S_x = \int dt L(x, \dot{x})$ invariant. Indeed, substitution of $y = x + v \cdot (t - t_0)$ relates $L(x, \dot{x})$ and $L(y, \dot{y})$ up to a global derivative, which does not affect the equation of motion.

For the non-relativistic quantum-mechanical case, Galilean invariance completely fixes the kinetic energy term of the Hamiltonian:

$$\mathcal{H} = -\frac{\hbar^2}{2m} \sum_i \nabla_i^2 + \sum_i V_1(x_i) + \sum_{i < j} V_2(x_i - x_j) \quad (1.4)$$

Indeed, the Laplacian is the only term which is compatible with the substitution

$$\psi(x, t) = e^{i\phi(x, v, t)} \tilde{\psi}(x + vt, t) \quad (1.5)$$

and requiring that both ψ and $\tilde{\psi}$ are solutions of the Schrödinger equation (note the necessity of adding a phase factor, again a manifestation of the fact that quantum wavefunctions are described by rays). A simple calculation using the Laplacian gives

$$\nabla \phi = v \quad (1.6)$$

$$\frac{\partial}{\partial t} \phi = -\frac{1}{2m} \left(\nabla^2 \phi + (\nabla \phi)^2 \right) \quad (1.7)$$

and hence

$$\phi(x, v, t) = \frac{m}{\hbar} \left(v \cdot x - \frac{v^2 \cdot t}{2} \right). \quad (1.8)$$

No other term besides the Laplacian would give rise to a possible solution, which is truly remarkable: the kinetic energy term in the Schrödinger equation is hence completely determined by invariance under group transformations!

1.2.2 Einsteinian relativity

Similarly, the equations governing electromagnetism and/or quantum field theory (the theory which marries relativity theory with quantum mechanics) are completely fixed by requiring them to be invariant under Lorentz transformations (or the more general Poincaré group, which is the so-called semidirect product of the translations and the Lorentz group). As you will encounter in your courses of quantum field theory, the representation theory of the Poincaré group forms the foundation of all our understanding of relativistic quantum matter.

1.3 Nuclear spectra

The nuclear shell model explains the structure of the nuclei of atoms, which consist of a combination of protons and neutrons. The model was developed in 1949 following independent work by Eugene Wigner, Maria Goeppert Mayer and collaborators (for which they got the Nobel prize in physics in 1963), and is effectively an application of representation theory of the rotation group.

Although the nuclear interactions are very different from the Coulomb potential, spin-orbit coupling is very important, and the fact that no preferential central point exists, experimental evidence has shown that so-called magic numbers exist for the number of nucleons at which the atomic nucleus is much more stable (namely for 2, 8, 20, 28, 50, 82, and 126 protons and neutrons). This is precisely the number of nuclei filling full shells as in the periodic table, hence group theory explains the stability of those nuclei. As a further example, the core material in the standard textbook “The nuclear shell model” of Kris Heyde of UGent is precisely about the representation theory of the rotation group.

1.4 Molecular vibrations

The understanding of possible or impossible transitions in spectroscopy is also completely based on the representation theory of groups: the coupling of different representations rules out certain transitions.

Molecular vibrations are classified according to the symmetries of the underlying molecule, and the selection rules have to respect that symmetry. For the example of Raman effect, the impinging light produces a dipole moment, whose symmetry properties impose the possible selection rules.

1.5 Symmetry breaking

The concept of symmetry breaking is the central mantra of condensed matter and solid state physics. Different phases of matter are characterized according to their different symmetries; a breaking of a symmetry (or in other words a reduction of the symmetry to a subgroup of that symmetry) cannot happen in a smooth way, and hence such events go hand in hand with phase transitions. Lev Landau was the first physicist who clearly understood that symmetries and symmetry breaking are the key in characterizing phases of matter, and corresponding to this symmetry breaking is the so-called order parameter (like magnetization in a ferromagnet).

One of the simplest examples of such phase transitions is between water and ice: water has the full Euclidean symmetry group (consisting of translations and rotations), but ice has broken this group to a subgroup consisting of the symmetries of its crystal structure (which is only invariant under translations with the unit vector as opposed to any translation).

Equally, a superconducting phase breaks the so-called $U(1)$ symmetry, and acquires an expectation value for $\hat{\psi}$, the order parameter of a ferromagnet is its magnetization (which is zero in the symmetric high-temperature phase), and the Heisenberg spin model acquires long-range order in 3 dimensions.

1.6 Crystallography and band structure

Crystallography is a beautiful topic, and its main aim is to classify all the ways in which the Euclidean space group symmetry can be broken in such a way that a periodic crystal structure emerges. There are exactly 230 ways in which this can happen in 3 dimensions (the 230 space groups), 17 ways in 2 dimensions (the so-called wallpaper groups), and 7 ways in 1 dimension (the so-called frieze groups). In an amazing twist of nature, all those 230 crystal structures are actually realized in some compounds: what is possible, will be exploited!

The crystal structure of a given material is completely determined by the electronic structure of the corresponding atoms; given the s,p,d,... bands, the material organizes itself in such a way

that the different bands are preferably just filled (this also immediately explains why there are so many insulators in nature). The band structure of a given material such as semiconductors is hence completely determined by the symmetries of the underlying crystal.

1.7 Particle physics

The central topic of high energy physics is the characterization of the quantum vacuum and of the elementary excitations (particles) on top of that vacuum. In an amazing series of papers, people such as Pauli, Schwinger, Feynman, Weinberg realized that the action / Hamiltonian of the system exhibits a wide range of symmetries and more precisely gauge symmetries.

Part I

Finite groups

Chapter 2

Finite Groups: Basic Definitions and Properties

2.1 Basic definitions, examples, Cayley table

Definition 2.1. A **finite group** G is a finite set of elements $\{g_i\}$ which transform into each other associatively by a multiplication law $\cdot : G \times G \rightarrow G$. A group must have an identity element 1 (also denoted e) such that $g \cdot 1 = 1 \cdot g = g$, and every g has an (unique) inverse $g^{-1} : g \cdot g^{-1} = 1 = g^{-1} \cdot g$.

Often, the multiplication $g_1 \cdot g_2$ will be written as juxtaposition, ie. $g_1 \cdot g_2 \equiv g_1 g_2$.

The number of group elements is called the **order** or **cardinality** and is denoted $\#G$ or $|G|$. The **order** of a group element is the smallest power $n \geq 1$ one has to take such that it is equal to the identity element $g^n = 1$, where $g^n := \underbrace{g \cdot g \cdots g}_n$.

A group is **Abelian** iff the multiplication is commutative, if $\forall x, y \in G : xy = yx$. For Abelian groups, we will often denote the ‘multiplication’ by addition: $x + y$ instead of $x \cdot y$. The identity will be written as 0 instead of 1 .¹

2.1.1 First examples

A couple of simple first examples are:

$\{1\}, \{0\}, \{e\}$: The group with only one element (which is necessarily also the identity element) and with trivial group multiplication trivially constitutes a group.

Z_n : The cyclic group $Z_n, \mathbb{Z}_n$ or C_n is just the group of integers $\{0, 1, \dots, n-1\}$ with multiplication addition modulo n . It clearly is an abelian group. We can also think of the elements of Z_n as the cyclic permutations of n letters.²

The simplest (and one of the most important) non-trivial cyclic groups is the group Z_2 which has elements $\{0, 1\}$ and which has as group multiplication $a \cdot b := a + b \pmod{2}$.³

S_n : The symmetric or permutation group S_n of n elements. The cardinality is $n!$. The group multiplication is composition of permutations. Often, we will write permutations $\sigma \in S_n$ as

¹The underlying idea is that all representations of Abelian groups are (equivalent to) direct sums of 1-dimensional objects and hence pure phases of the kind $R_g = e^{ix_g}$, such that multiplication amounts to sums of the phases.

²What is the order of the element 1 in Z_n ?

³A simple example of a quantum mechanical system with a Z_2 symmetry is the 1+1D harmonic oscillator defined by the Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$ with Z_2 symmetry $\hat{x} \mapsto -\hat{x}$. As a consequence, the eigenstates come in two classes: those that are odd under the Z_2 symmetry, $\psi(-x) = -\psi(x)$ and those that are even, $\psi(-x) = \psi(x)$.

follows:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix} \quad (2.1)$$

For example:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 & 6 \end{pmatrix} \in S_6 \quad (2.2)$$

is the permutation $1 \mapsto 2 \mapsto 4 \mapsto 1, 3 \mapsto 5 \mapsto 3, 6 \mapsto 6$. It is also possible to write permutations as a collection of **disjoint cycles**, ie. sets of elements that are cyclically permuted. Eg.: the previous example can be written as one 3-cycle and one 2-cycle: $(124)(35)$ (we will not write 1-cycles). The cycle notation is unique up to cyclic permutations within each cycle and the order of the cycles, eg.: $(124)(35) \equiv (53)(241)$.⁴

A_n : The alternating group A_n is the subgroup (see below Def. 2.11) of S_n of even permutations. The cardinality is $n!/2$ and we have $S_n = A_n \rtimes Z_2$ (see below Sec. 2.4.1).

D_n : The dihedral group D_n is the group of symmetries of a regular n -gon, consisting of n cyclic rotations and n reflections through n axes. We have $D_n = Z_n \rtimes Z_2$ (see below Sec. 2.4.1). The order of D_n is $2n$.⁵

Z_n^* : The multiplicative group Z_n^* consists of all elements smaller than n which are co-prime to n , and the group structure is multiplication modulo n . In general, $\#Z_n^* = \phi(n) = \prod_i p_i^{n_i-1}(p_i - 1)$ with $\phi(n)$ the Euler function and $n = \prod_i p_i^{n_i}$ the decomposition of n as a power of primes.

Groups can also concisely be defined by specifying a generating set of group elements, that is a minimal set of group elements whose multiplication with each other, possibly subject to constraints, and those products generate the whole group. The size of the minimal generating set is upper bounded by the logarithm $\log_2(\#G)$ (this bound is only saturated in the case of the direct product of Z_2 's). Examples are:

Example 2.1. $D_n = \langle r, s | r^n = s^2 = (sr)^2 = 1 \rangle$. What is then the order of r , and of s ?

Example 2.2. $\mathbb{Z}_n = \langle a | a^n = 1 \rangle$

This way of defining a group is also called a **presentation** (not to be confused with representation) of a group. In general, a presentation of a group is written as

$$G := \langle S | R \rangle, \quad (2.3)$$

where S is a set of generating group elements and R a number of relations between these generators. It will turn out that given a presentation of a group some concrete computations (eg. computing quotient groups, see below) can be easier than when one has to work with the group defined solely through the Cayley table. However, sometimes the presentation of a group can also be a clumsy way to define a group as for example the order of the group is not readily extracted from the group presentation.

Remark. In accordance with the literature, in presentations we will always use the multiplicative notation, even for abelian groups. So in particular the identity will always be denoted by 1 and not 0.

One of the crowning achievements of 20th century mathematics has been the classification of all finite groups. The main reason that such a fantastic feat is even possible is the central condition that group multiplication is associative; this puts enormous constraints on the possible multiplication rules.

⁴This gives rise to a nice problem: given a random permutation of n elements, what is the chance that it has a cycle of more than $n/2$? What happens in the limit of $n \rightarrow \infty$? [Hint!](#)

⁵In the literature both the notations D_{2n} and D_n are used for the dihedral group of order $2n$.

Small groups

For exercises and concrete computations it is often useful to have an overview of the smallest groups. Such a list is presented in the appendix in table 2.A.

Digression: continuous groups.

Continuous groups are defined in a very similar way to finite groups, but the elements are now continuous variables: a group element is labeled by a tuple $\bar{x} \equiv (x_1, x_2, \dots, x_N)$ with N the rank of the group. The “Cayley table” now consists of N continuous functions $z_i = f_i(\bar{x}, \bar{y})$, and associativity will again impose huge constraints on those functions; as a result, very few of those functions exist, and those correspond to the Lie groups. Lie groups of special interest are $SU(N)$, $U(N)$, the special orthogonal groups $SO(N)$, the symplectic groups, etc. An entire chapter will be devoted to Lie groups.

2.1.2 Cayley table

The **Cayley (multiplication) table** shows how the group multiplication behaves, i.e. the entry (g, h) denotes the element $g \cdot h$; examples are:

Example 2.3. For the cyclic group Z_2 we have

$$C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (2.4)$$

Example 2.4. For the dihedral group D_3 (simplest non-Abelian group, it is a group for which $\exists x, y : x \cdot y \neq y \cdot x$) we can denote the group elements by $g_1, g_2, \dots, g_6$:

$$C = \begin{bmatrix} g_1 & g_2 & g_3 & g_4 & g_5 & g_6 \\ g_2 & g_3 & g_1 & g_6 & g_4 & g_5 \\ g_3 & g_1 & g_2 & g_5 & g_6 & g_4 \\ g_4 & g_5 & g_6 & g_1 & g_2 & g_3 \\ g_5 & g_6 & g_4 & g_3 & g_1 & g_2 \\ g_6 & g_4 & g_5 & g_2 & g_3 & g_1 \end{bmatrix} \quad (2.5)$$

or even more concise:

$$C = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 1 & 6 & 4 & 5 \\ 3 & 1 & 2 & 5 & 6 & 4 \\ 4 & 5 & 6 & 1 & 2 & 3 \\ 5 & 6 & 4 & 3 & 1 & 2 \\ 6 & 4 & 5 & 2 & 3 & 1 \end{bmatrix}. \quad (2.6)$$

Since D_3 is the group of symmetries of a triangle, it is more sensible to denote the group elements by $1, r, r^2, s, r^2s$ where r denotes a (clockwise) rotation by 120° and s a reflection over a reflection axis of the triangle:

$$C = \begin{bmatrix} e & r & r^2 & s & r^2s & rs \\ r & r^2 & e & rs & s & r^2s \\ r^2 & e & r & r^2s & rs & s \\ s & r^2s & rs & e & r & r^2 \\ r^2s & rs & s & r^2 & e & r \\ rs & s & r^2s & r & r^2 & e \end{bmatrix}, \quad (2.7)$$

where the order of the group elements in all the 3 representations of the Cayley table is the same.

Note that every element appears once in every column and row⁶, but there are a lot of hidden constraints associated to the **associativity** of the underlying group multiplication. What is more, associativity is the key reason why groups are so interesting from the physics point of view, the key ingredient which makes it possible to represent groups in terms of matrices, and the crucial element which makes it possible to classify all groups.

2.1.3 Direct products

Definition 2.2. Given two groups G and H , the direct product of those groups, $G \times H$, is a group with elements that can be labeled by the set of tuples (g_i, h_j) for which $(g_i, h_j) \cdot (g_k, h_l) := (g_i \cdot g_k, h_j \cdot h_l)$. Obviously, $|G \times H| = |G||H|$.

Example 2.5. $\mathbb{Z}_n \times \mathbb{Z}_m$ has the following presentation: $\mathbb{Z}_n \times \mathbb{Z}_m = \langle a, b | a^n = b^m = 1, ab = ba \rangle$. The generators a and b then correspond to the group elements $(1, 0)$ and $(0, 1)$, as you can verify. More generally: $a^k b^l \equiv (k, l)$ with $k = 0, 1, \dots, n-1$ and $l = 0, 1, \dots, m-1$ and with the understanding that $a^0 = b^0 = 1 \equiv (0, 0)$. What is then the order of the group element $ab \equiv (1, 1)$?

Remark. For abelian groups A_1, A_2 , also the notation $A_1 \oplus A_2$ is sometimes used for the direct product. So for example: $\mathbb{Z}_n \times \mathbb{Z}_m = \mathbb{Z}_n \oplus \mathbb{Z}_m$.

Remark. We will sometimes write $G^{\times 2}$, G^2 or even $G^{\otimes 2}$ for $G \times G$. Analogously for $\underbrace{G \times G \times \dots \times G}_n$.

In a next section, we will see that there are much more intricate ways to take products of groups (think about the fact that the dihedral group is not just the product of rotations over 120 degrees and reflections ($D_3 \neq Z_3 \times Z_2$), as those do not commute); in fact $D_3 = Z_3 \rtimes Z_2$ with $\rtimes$ the semi-direct product that will be defined below in Sec. 2.4.1.

2.2 Homomorphisms, isomorphisms and automorphisms

Given two groups G_1 and G_2 we can devise maps between these groups by simply mapping the elements of G_1 to those of G_2 . However, there is no guarantee that such a map will preserve the multiplication law. Maps between groups that preserve the multiplication are called homomorphisms:

Definition 2.3. Given two groups G_1 and G_2 a map $\varphi : G_1 \rightarrow G_2$ is called a **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y) \forall x, y \in G_1$.

Remark. Notice that given groups G_1 and G_2 and a homomorphism $\varphi : G_1 \rightarrow G_2$, that $\varphi(g^{-1}) = \varphi(g)^{-1}$ and $\varphi(1_{G_1}) = 1_{G_2}$ where 1_{G_i} is the identity in G_i .

Example 2.6. Given any two groups G_1 and G_2 $\varphi : G_1 \mapsto G_2 : g \mapsto 1$ is a trivial homomorphism.

Example 2.7. Take $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ and $\mathbb{Z}_2 = \{0, 1\}$. A non-trivial homomorphism is $\varphi : \mathbb{Z}_2 \mapsto \mathbb{Z}_4$, $\varphi(0) = 0$, $\varphi(1) = 2$. Check that this is indeed a homomorphism.

The previous examples inspire the following proposition.

Proposition 2.1. The image of a homomorphism $\varphi : G_1 \mapsto G_2$ is a subgroup of its codomain G_2 . $\varphi(G_1) \leq G_2$.

Proof: exercise. □

If a homomorphism is also a bijection, we call it an isomorphism. Isomorphic groups can be thought of as ‘the same’, ie. there is no meaningful way in which they can be distinguished.

⁶See exercises.

Definition 2.4. Two groups G_1 and G_2 are called **isomorphic** if there exists a bijective homomorphism $\varphi : G_1 \rightarrow G_2$. φ is called an *isomorphism of groups*. And we will write $G_1 \simeq G_2$, or $G_1 \cong G_2$.

Example 2.8. Z_{20}^* has 8 elements co-prime to 20: 1, 3, 7, 9, 11, 13, 17, 19. It happens that, modulo 20, anyone of those can be written as $3^a \cdot 19^b$ with $a \in \{0, 1, 2, 3\}$ and $b \in \{0, 1\}$, hence $Z_{20}^* \cong Z_4 \oplus Z_2$.

Example 2.9. $\mathbb{Z}_8^* \simeq \mathbb{Z}_2^2$. Convince yourself!

In case $G_1 = G_2 = G$, an isomorphism of G can be interpreted as a relabeling of the group elements. This leads to the following definition:

Definition 2.5. An **automorphism** of a group G is an isomorphism from G to itself.

We can go yet one step further and consider all the automorphisms of a fixed group G . Automorphisms of G are simply maps between sets that can be composed. Since the composition of maps is associative, there always exists a trivial ‘identity’ automorphism and since every automorphism has an inverse, the automorphisms of a group form themselves a group with group multiplication the composition of maps:

Definition 2.6. The **automorphism group** of G , $\text{Aut}(G)$ is the group of all automorphisms of G with group multiplication the composition of maps.

In other words, an automorphism α of G is a permutation of the group elements such that $\forall g, h : \alpha(g.h) = \alpha(g).\alpha(h)$.

Remark. There is a special class of automorphisms called **inner automorphisms**, which are obtained by doing a relabeling using conjugation with respect to a fixed group element g_i : $g_j \rightarrow g_i g_j g_i^{-1}$. Note that the inner automorphisms form a subgroup of $\text{Aut}(G)$, denoted $\text{Inn}(G)$.

Example 2.10. Let us consider the cyclic group Z_n . Check that its automorphism group is the multiplicative group $\text{Aut}(Z_n) \cong Z_n^*$ (challenging exercise). Note that $\forall n$ Z_2 is a subgroup of Z_n^* (which corresponds to the inversion of the cyclic permutation).

Example 2.11. An interesting example is the automorphism group of $Z_2 \times Z_2$: $\text{Aut}(Z_2 \times Z_2) = S_3$, the permutation group of 3 elements (with cardinality $\#S_3 = 6$). Note that the automorphism group of S_n is itself S_n , except for $n = 2$ for which $\text{Aut}(S_2) = 1$ and $n = 6$ ($\text{Aut}(S_6) = S_6 \rtimes Z_2$).

Let us also peek already into the concept of the automorphism group of the continuous group $U(1)$, where the group elements g_x are labeled by a real number $x \in [0, 2\pi[$ and with multiplication table $g_x \cdot g_y = g_{x+y \bmod 2\pi}$. $U(1)$ can be thought of as the set of complex numbers of norm 1:

Definition 2.7. $U(1) = \{z \in \mathbb{C} : |z|^2 = 1\}$, with group multiplication the multiplication of complex numbers.

As $U(1)$ is an Abelian group, the inner automorphisms are trivial. The automorphisms can be labeled by functions $f(x)$ that necessarily satisfy $\forall x, y : f(x+y) = f(x) + f(y) \bmod 2\pi$ and, in particular, $f(x) + f(2\pi - x) = 2\pi n$ with n an integer. These conditions imply that $f(x) = nx$. Such a function can only be an isomorphism for $n = \pm 1$, hence we identified the automorphism group of $U(1)$ as being Z_2 , the reflection group.

Digression: complexity of finding automorphisms.

How difficult is it to find all automorphisms of a group? Look for all possible generating sets of the group; the total number of minimal generating sets is upper bounded by $\binom{\#G}{\log_2(\#G)}$ which is quasi-polynomial in $\#G$. From the computational complexity point of view, there exists therefore a **quasi-polynomial algorithm** which checks whether two groups are isomorphic, as it is easy to check whether two generating sets give rise to the same Cayley table.

2.3 Subgroups, cosets, normal subgroups, quotient groups

Definition 2.8. A subgroup S of a group G is a subset of the elements of G such that they form a group themselves with the group multiplication that of G restricted to S . We will write $S \leq G$. Or $S < G$ when $S \neq G$.

Example 2.12. $\mathbb{Z}_2 \leq \mathbb{Z}_4$ (see Ex. 2.7).

Example 2.13. $Z_n \leq D_n$. Indeed, staring at the presentation $D_n = \langle r, s | r^n = s^2 = (sr)^2 = 1 \rangle$, we notice that the generator r generates a Z_n subgroup. What is the geometrical interpretation of this subgroup? Can you find other subgroups? [Hint: all subgroups of D_n are either cyclic groups or dihedral groups.]

Example 2.14. Given a direct product group $G \times G$, $G \times G$ always contains a subgroup of the form $\{(g, g) | g \in G\} \leq G \times G$. This subgroup is called the **diagonal subgroup** of $G \times G$.

Note that every group has at least two subgroups, namely the trivial group and itself. These subgroups are called **trivial** subgroups, all other subgroups are called **proper** subgroups.

Just as in the case of identifying the automorphism group, finding all subgroups amounts to identifying all generators of the group, and the complexity is hence quasi-polynomial in the cardinality of the group.

From the point of view of Cayley tables, the task of finding subgroups is to find principal submatrices which are valid Cayley tables themselves. The fundamental theorem concerning the cardinality of subgroups is called **Lagrange's theorem**. We first have to define cosets:

Definition 2.9. Given a subgroup $H \leq G$, the **left cosets** of H in G are the sets $gH := \{gh | h \in H\}$.

Example 2.15. Take $G = Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $H = \{0, 4\} \cong Z_2$. We find the cosets $0 + H = \{0, 4\}$, $1 + H = \{1, 5\}$, $2 + H = \{2, 6\}$, $3 + H = \{3, 7\}$.

Analogously one can define the right cosets of H in G , denoted by Hg . For abelian groups these definitions obviously coincide.

Given gH or Hg , g is called a representative of gH, Hg . The representative of a coset is in general not unique. Note that $g_1 \sim g_2 \iff g_1H = g_2H$, or in other words $g_1g_2^{-1} \in H$, defines an equivalence relation, and the cosets are exactly the equivalence classes.

The collection of all (left) cosets of H in G is written as G/H , and the number of cosets, $|G/H|$, is called the **index** of H in G , sometimes denoted by $[G : H]$ or $|G : H|$. Now comes one of the most important theorems in group theory:

Theorem 2.2 (Lagrange). Let H be a subgroup of G , $H \leq G$; then the elements of G/H are disjoint subsets of G whose union is G , and each coset has the same number of elements $\#H$. Hence $\#H$ divides $\#G$.

Proof: The proof is left as an exercise, but is a simple counting argument. □

Corollary 2.2.1. Z_p with p a prime number has only trivial subgroups.

Corollary 2.2.2. The order of every group element of a finite group divides the order of the group (see exercises).

It is important to notice that G/H doesn't necessarily form a group. This is only true if H is a so called normal subgroup, this will be defined below.

The **centre** or centralizer $Z(G)$ (Z from "Zentrum") of a group G is the (normal, see below) subgroup whose elements commute with all elements of G ; it can trivially be found by searching for zero rows in the matrix $C - C^T$ with C the Cayley table of G .

One easy way of constructing interesting subgroups is by constructing **conjugacy classes** by conjugation: given an element g , we construct a conjugacy class by conjugating it with every other element $h \in G$ hgh^{-1} , and a class is the union of all those elements. Concretely:

Definition 2.10. Given a group G , the **(conjugacy) class** of group elements associated to a given group element $g \in G$ is defined as $C_g := \{hgh^{-1} | h \in G\}$.

We can therefore partition the group in different classes, and every element is contained in exactly one class. The number of different classes plays a crucial role in representation theory. Note that the order of all elements in a class is constant (check this!). Also note that the representative g is not unique, and belonging to the same class is again an equivalence relation on G . If we now construct a subgroup entirely out of complete classes, it will be closed under conjugation with any element of the group, giving rise to the notion of normal (invariant) subgroups.

Definition 2.11. A subgroup $N \leq G$ is **normal** if it is invariant under conjugation with all group elements of the larger group G : given a normal subgroup N , we have $Ng = gN$ for all elements $g \in G$. Said differently, if you conjugate any element of N with any of G you get again an element in N . We will write $N \trianglelefteq G$.

Example 2.16. A very important example is the following (see exercises). Given a homomorphism $\varphi : G_1 \rightarrow G_2$, $\ker(\varphi)$ is a normal subgroup of G_1 .

Example 2.17. A simple example of a normal subgroup is the centralizer of the group $Z(G)$ (check this!).

A group is called **simple** if its only normal subgroup is the one consisting solely of the identity element, the trivial group. Associated to a normal subgroup is the **quotient group** (or factor group) G/N . This is the group of cosets aN : indeed, $(aN)(bN) = a(Nb)N = a(bN)N = (ab)N$ clearly forms a group which is homomorphic to the original one G (note that the left and right coset give rise to same group).

Example 2.18. Z_p with p prime is simple. Indeed, since Z_p only has trivial subgroups and these are both normal, Z_p is simple.

Definition 2.12. Given a group G and a normal subgroup $N \trianglelefteq G$, we define the **quotient (or factor) group** G/N as the group with group elements the cosets gN and group multiplication $g_1N \cdot g_2N := (g_1g_2)N$. Clearly, $|G/N| = |G|/|N|$.

Example 2.19. Take S_4 ; it has two normal subgroups: $Z_2 \times Z_2$ and the alternating group A_4 . This $N_1 := Z_2^2$ subgroup contains the group elements $N_1 = \{(), (12)(34), (13)(24), (14)(23)\}$. Proving that this is a normal subgroup from the definition is tedious. One has to check that for any $g \in S_4$ and any $n_1 \in N_1$ gn_1g^{-1} is again an element of N_1 .

Showing that A_4 is normal is easier. Recall that $N_2 := A_4$ are all even permutations of 4 letters. Consider now gn_2g^{-1} with $g \in S_4$ an arbitrary (can be even or odd) permutation, and n_2 an arbitrary even permutation in A_4 . Notice that g^{-1} then has the same parity as g . gn_2g^{-1} is therefore always even, or thus $gN_2g^{-1} = N_2$, proving that N_2 is normal.

The associated factor groups are (isomorphic to) $S_4/Z_2^2 \simeq S_3$ and $S_4/A_4 \simeq Z_2$.

Since Z_2^2 is normal in S_4 , the quotient S_4/Z_2^2 is well defined. The order of S_4/Z_2^2 is $24/4 = 6$. There are 2 groups of order 6: S_3 and Z_6 (see list of small groups). The possibility Z_6 is easy to rule out by a direct construction of the quotient.

Since A_4 is normal in S_4 , S_4/A_4 is well defined. The order of S_4/A_4 is 2 and the only group of order 2 is (isomorphic to) Z_2 , so we conclude that $S_4/A_4 \simeq Z_2$.

A normal subgroup is called a **maximal normal subgroup** if there exists no other nontrivial subgroups containing it; as a consequence, the factor group of a maximal normal subgroup is simple.

A normal subgroup of special importance is the **commutator group** $[G, G]$, i.e. the group generated from all group elements $xyx^{-1}y^{-1}$ (note that the product of two such elements is not necessarily of that form). The quotient $G/[G, G]$ is called the **Abelianization** of G .

Definition 2.13. $[G, G] := \langle [g_1, g_2] | g_1, g_2 \in G \rangle$ and where $[g_1, g_2] := g_1g_2g_1^{-1}g_2^{-1}$.

Example 2.20. The commutator group of S_n is A_n , and its Abelianization is $S_n/A_n = Z_2$.

Digression: classification of finite simple groups.

Simple groups can be considered the basic building blocks of all finite groups. The classification of all simple groups was a tremendous effort that took almost two centuries to accomplish. This endeavor was considered completed only in 2004. The “proof” is still being simplified and much is yet to be understood about the classification. The so-called second-generation classification proof (after the first-generation proof which really is a collection of hundreds of research papers) is currently being published in a series of about 14 books totaling around 5000 pages. The first of these books appeared in 1994 (when the proof was still very much under construction!) and 9 books have been published so far. The classification can be summarized as follows. The finite simple groups are:

- An infinite family of cyclic groups of prime order: Z_p , p being prime.
- An infinite family of alternating groups of degree 5 and more: A_n , $n \geq 5$.
- 16 infinite families of groups of so-called Lie type.
- 26 exceptions, the so-called sporadic groups:
 - 20 of these come from the so-called Monster group, the largest sporadic group of order 808,017,424,794,512,875,886,459,904,961,710,757,005,754,368,000,000,000. These 20 constitute the so-called “Happy family”.
 - 6 so-called pariahs.

The fact that the smallest non-abelian simple group is A_5 is in itself a very interesting fact. THE founding father of group theory, Évariste Galois (1811-1832), showed that the reason that only A_n for $n \geq 5$ is simple is the reason why roots of polynomials of order 5 cannot be expressed in radicals, a problem which was unsolved for 350 years. This very deep insight (which he proved when he was 20 years old!) lead to the development of Galois theory.

Digression: solvable groups.

Definition 2.14. A solvable group G is a group G for which there exists a **subnormal sequence** of subgroups in which every subgroup is a normal subgroup of the previous one, terminating with the trivial group:

$$1 = A_0 \triangleleft A_1 \triangleleft \cdots \triangleleft A_n = G, \quad (2.8)$$

where additionally all the factor groups A_i/A_{i-1} are abelian.

Example 2.21. Every abelian group is trivially solvable.

Example 2.22. Every dihedral group $D_n = Z_n \rtimes Z_2$ is solvable as follows:

$$1 \triangleleft Z_n \triangleleft D_n, \quad (2.9)$$

Example 2.23. A counterexample is S_5 : A_5 is normal in S_5 , but since A_5 is simple (see above) and non-abelian, there is no way in which the subnormal sequence 2.8 terminates with the trivial group.

2.4 Extending groups

As anticipated, there exist a number of ways in which groups can be constructed from given (smaller) groups. The simplest example that we introduced above is the direct product $G_1 \times G_2$.

This section is devoted to more intricate constructions. ⁷

2.4.1 (Semi)direct products

The direct product of two groups G, H is just the tensor product in matrix language: just associate the group elements (g, h) to them, and respect the pointwise multiplication. Using the concept of normal subgroups, we can also define a more sophisticated version of this: the semidirect product.

We can define the (outer) semidirect product of N with respect to any subgroup of its automorphism group $\text{Aut}(N)$. Technically, we take two groups N, H such that $\beta : H \rightarrow \text{Aut}(N)$ is a homomorphism (i.e. equivalent to a subgroup of $\text{Aut}(N)$) and therefore β_h maps elements of N to N according to an element of the automorphism group labeled by h . Then the semidirect product with respect to β is defined as:

Definition 2.15. *Given groups N and H and a homomorphism $\beta : H \rightarrow \text{Aut}(N)$, we define the (outer) semidirect product of N and H with respect to β , $N \rtimes_{\beta} H = \{(n, h) | n, h \in N, H\}$:*

$$(n_1, h_1) \cdot (n_2, h_2) := (n_1 \cdot \beta_{h_1}(n_2), h_1 \cdot h_2). \quad (2.10)$$

Verify for yourself that the group axioms (Def. 2.1) are indeed fulfilled. You can check that $(n, h)^{-1} = (\beta_{h^{-1}}(n^{-1}), h^{-1})$.

Remark. *Notice that the subgroup $\{(n, 1) | n \in N\} \simeq N$ is normal in $N \rtimes_{\beta} H$ since $(n, h)(n', 1)(n, h)^{-1} = (n, h)(n', 1)(\beta_{h^{-1}}(n^{-1}), h^{-1}) = (n'', 1) \in \{(n, 1) | n \in N\} \simeq N$ for some $n'' \in N$. We leave it as an exercise to see that the corresponding factor group is H .*

Remark. *For the choice $\beta = \text{id}_N$, the semidirect product coincides with the direct product.*

Remark. *Recall the definition of inner automorphisms under Def. 2.6. Consider $(1, h)(n, 1)(1, h)^{-1}$ in $N \rtimes_{\beta} H$: $(1, h)(n, 1)(1, h)^{-1} = (1, h)(n, 1)(1, h^{-1}) = (\beta_h(n), 1)$. This means that the general automorphism β_h of N is in $N \rtimes_{\beta} H$ ‘promoted’ to an inner automorphism in $N \rtimes_{\beta} H$.*

Example 2.24. *As a first example, take the group $Z_n = C_n$, whose automorphism group is $\text{Aut}(Z_n) = Z_n^*$. Z_2 is always a subgroup of Z_n^* and corresponds to the inversion of the cycle C_n , meaning that β acts on C_n as $\beta_0(g) = g, \beta_1(g) = g^{-1}$. Hence we can define $Z_n \rtimes Z_2 \cong D_n$ which is the dihedral group.*

Knowing this, we can write down a presentation of D_n as follows. D_n has two generators r and s where r generates a normal subgroup isomorphic to Z_n , i.e. $r^n = 1$, and s generates the Z_2 subgroup, i.e. $s^2 = 1$. With the homomorphism given above it can be checked that $srs = r^{-1}$. Hence, $D_n = \langle r, s | r^n = s^2 = (sr)^2 = 1 \rangle$.

Example 2.25. *As another example, take $G = Z_2 \times Z_2$ with $\text{Aut}(Z_2 \times Z_2) = S_3$ the permutation group on three letters. The subgroups of S_3 are Z_2, Z_2, Z_2 and Z_3 . It is hence possible to define the groups $(Z_2 \times Z_2) \rtimes Z_2, (Z_2 \times Z_2) \rtimes Z_3$ and $(Z_2 \times Z_2) \rtimes S_3$. The last two play an important role in the identification of symmetries of the platonic solids. We can actually identify $(Z_2 \times Z_2) \rtimes Z_3$ with A_4 and $(Z_2 \times Z_2) \rtimes S_3$ with S_4 ; Z_3, S_3 are indeed subgroups of S_4 , while $Z_2 \times Z_2$ and A_4 are its only normal subgroups. More generally, $S_n = A_n \rtimes Z_2$.*

Notice that the example $D_n \cong Z_n \rtimes Z_2$ is somewhat special: the action β of Z_2 on Z_n is given by conjugation in D_n . Concretely, by looking at the presentation $D_n = \langle r, s | r^n = s^2 = (sr)^2 = 1 \rangle$ we see that $\beta_s(r) = srs^{-1} = srs = r^{-1}$. This appears so often in concrete examples of semidirect products that a special name is reserved for this kind of semidirect product:

Definition 2.16. *Given a group G with a normal subgroup N and subgroup H . G is called the inner semidirect product of N and H if G is the outer semidirect product of N and H where the action of H on N is given by conjugation in G .*

⁷A reference is Chap. IV of [7].

2.4.2 Sequences & extensions

A central role in group theory is played by the notion of **exact sequences** of the form

$$G_1 \xrightarrow{\phi_1} G_2 \xrightarrow{\phi_2} G_3 \longrightarrow \dots \quad (2.11)$$

Here, all the ϕ_i are group homomorphisms, and they are subject to the property that the image of the homomorphism $\phi_{k-1} : G_{k-1} \rightarrow G_k$ must be the complete kernel of the homomorphism $\phi_k : G_k \rightarrow G_{k+1}$, or in other words $\phi_k(\phi_{k-1}(x)) = 1, \forall x$ ⁸. Similarly, we can define **short exact sequences** as exact sequences of the form:

$$1 \longrightarrow N \xrightarrow{\phi_1} G \xrightarrow{\phi_2} G/N \longrightarrow 1 \quad (2.12)$$

Using the defining property of exact sequences, you should be able to check that ϕ_1 is an injective map, and ϕ_2 is surjective.⁹ This means that ϕ_1 defines an embedding of N in G . You should also be able to check that N is a normal subgroup of G , hence G/N is a well defined group (see Ex. 2.16). ϕ_2 thus defines a projection of G on the cosets G/N .

Example 2.26. An example of such a sequence is¹⁰ $0 \rightarrow 2\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}_2 \rightarrow 0$; we can e.g. choose $\phi_1(x) = x$, $\phi_2(x) = x \bmod 2$, for which we indeed have $\phi_2(\phi_1(x)) = 1$.

Example 2.27. Another example is $1 \rightarrow \mathbb{Z}_n \rightarrow D_n \rightarrow \mathbb{Z}_2 \rightarrow 1$.

More interesting cases can easily be obtained for continuous Lie groups¹¹.

Two short sequences

$$1 \longrightarrow N \xrightarrow{i} G_1 \xrightarrow{\pi} H \longrightarrow 1 \quad (2.13)$$

$$1 \longrightarrow N \xrightarrow{i'} G_2 \xrightarrow{\pi'} H \longrightarrow 1 \quad (2.14)$$

are called **equivalent** iff there exists a group isomorphism $T : G_1 \rightarrow G_2$ that makes the diagram

$$\begin{array}{ccccccc} & & & G_1 & & & \\ & & \nearrow i & \downarrow T & \searrow \pi & & \\ 1 & \longrightarrow & N & & H & \longrightarrow & 1 \\ & & \searrow i' & \downarrow & \nearrow \pi' & & \\ & & & G_2 & & & \end{array} \quad (2.15)$$

commute, meaning that every path with the same start - and end point defines the same map.

A natural question is to figure out all possible groups G which are compatible with groups N and H defined in the short sequence

$$1 \rightarrow N \rightarrow G \rightarrow H \rightarrow 1.$$

It turns out that this is a hard problem. There is of course the trivial extension obtained by taking the direct product of $H \times N$, but we want to find all other possible ones. Some special cases are the following.

⁸In this section, following the literature, we use 1 (or 0 in the case of abelian groups) to both denote the identity element and the trivial group.

⁹This also means that exact sequences of the form $1 \longrightarrow A \xrightarrow{\phi} B \longrightarrow 1$ imply that $A \cong B$.

¹⁰See exercises for a definition of $\mathbb{Z}$ and $2\mathbb{Z}$.

¹¹For example: $1 \rightarrow SU(N) \rightarrow U(N) \rightarrow U(1) \rightarrow 1$. The map $\phi : U(N) \rightarrow U(1)$ is obtained by taking the determinant, for which the kernel is indeed given by all matrices in $SU(N)$. Note that $U(N) \neq SU(N) \times U(1)$, as elements in $SU(N)$ involve N 'th roots of the identity matrix; the projective unitary group $PU(N)$ is defined as the quotient group $PU(N) = U(N)/Z(U(N))$, which is not equal to $SU(N)$ but equal to $PSU(N) = SU(N)/Z_N$ (for example, $PU(2) = SO(3) = PSU(2)$). Other exact sequences are $\mathbb{Z}_n \rightarrow SU(N) \rightarrow PSU(N)$ and $U(1) \rightarrow U(N) \rightarrow PSU(N)$.

Splitting Extensions If we require that the sequence “splits”, meaning that there exists a group homomorphism ϕ (recall: $\phi(h_1 h_2) = \phi(h_1) \phi(h_2)$) from H to G and π from G to H whose composition $\pi \circ \phi = \text{id}_H$ is the identity map, then this problem has been solved: all such constructions can be obtained by invoking the semi-direct product of $N \rtimes_{\beta} H$, and hence they are completely classified by homomorphisms of H to the automorphism group $\text{Aut}(N)$; those are called **splitting extensions**.¹² Example 2.27 is of this form (check this!).

Example 2.28.

$$1 \longrightarrow A_n \xrightarrow{i} S_n \xrightarrow{\pi} \mathbb{Z}_2 \longrightarrow 1 \quad (2.16)$$

i is here just the embedding of A_n in S_n . Indeed recall that A_n is a (normal) subgroup of S_n . And π maps every permutation of S_n to its parity which is even (0) or odd (1). This is indeed a short exact sequence because the parity of all the permutations in A_n is even by definition of A_n and thus $\pi \circ i$ maps all elements of A_n to the identity 0 in $\mathbb{Z}_2$.

Central Extensions Central extensions G of a group H by an Abelian group A are defined via the short sequence

$$1 \longrightarrow A \xrightarrow{i} G \xrightarrow{\pi} H \longrightarrow 1 \quad (2.17)$$

where we require that the Abelian group A , or more precisely $i(A)$, is a subgroup of the centralizer $Z(G)$, and $H \simeq G/A$. Central extensions are completely classified by the so called **second cohomology group** $H^2(H, A)$. This means that all the central extensions up to equivalence are in one-to-one correspondence with the group elements of $H^2(H, A)$. A detailed explanation of group cohomology and in particular the second cohomology group is given in the appendix, but we sketch the classification and construction of central extensions and the correspondence with $H^2(H, A)$ below.

Without proof: All central extensions can be obtained by defining elements (a, h) and a group multiplication rule

$$(a_1, h_1) \cdot (a_2, h_2) := (a_1 + a_2 + \omega(h_1, h_2), h_1 h_2) \quad (2.18)$$

where we use the additional notation for the Abelian group elements. From the point of view of exact sequences, this representation is natural due to the existence of homomorphisms $i : A \rightarrow G$ and $\pi : G \rightarrow H$ such that $\forall a : \pi(i(a)) = 1$. Without loss of generality, we can make the choice $i(a) = (a, 1)$ and $\pi(a, h) = h$. As all elements $(a, 1)$ must lie in $Z(G)$ (i.e. commute with everything in G), we define all group elements of G as tuples $(a, h) = (a, 1) \cdot (0, h)$. Group multiplication is then completely defined by defining $(0, h_1) \cdot (0, h_2) = (\omega(h_1, h_2), h_1 h_2)$ or $(a_1, h_1) \cdot (a_2, h_2) = (a_1 + a_2 + \omega(h_1, h_2), h_1 h_2)$, and it remains to specify conditions that ω has to satisfy such that this is consistent. It can readily be checked (important exercise!) that associativity of group multiplication implies the condition

$$\omega(h_1, h_2) + \omega(h_1 h_2, h_3) = \omega(h_1, h_2 h_3) + \omega(h_2, h_3) \quad (2.19)$$

where ω is valued in the group A ; this is called the **2-cocycle condition** of the group H for trivial group action on A . The solutions ω of this equation are called **2-cocycles**. Notice that a trivial solution is always (again: check that this is a solution!):

$$\omega(h_1, h_2) = \varphi(h_1 h_2) - \varphi(h_1) - \varphi(h_2). \quad (2.20)$$

Here, $\varphi : H \rightarrow A$ is an arbitrary function (not even a homomorphism in general). If ω is of this form, it's called a **coboundary**. Two cocycles which can be related by adding a coboundary are called **cohomologous** and ‘being cohomologous’ is simply seen to be an equivalence relation. The second cohomology group $H^2(H, A)$ is then defined as the set of cohomology classes $[\omega]$ where

¹²Note that neither $1 \rightarrow SU(N) \rightarrow U(N) \rightarrow U(1) \rightarrow 1$ nor $1 \rightarrow PU(N) \rightarrow U(N) \rightarrow U(1) \rightarrow 1$ are splitting extensions, showing that the existence of a homomorphism ϕ is way too strong a condition for finding all interesting extensions.

$\omega \sim \omega'$ if and only if $\omega(h_1, h_2) = \omega'(h_1, h_2) + \varphi(h_1 h_2) - \varphi(h_1) - \varphi(h_2)$ for some φ , with group multiplication $[\omega_1] \cdot [\omega_2] := [\omega_{12}]$ where $\omega_{12} \sim \omega_1 + \omega_2$. Indeed, notice that the sum of two 2-cocycles is again a 2-cocycle!¹³

Equivalently, $H^2(H, A)$ can be constructed as $Z^2(H, A)/B^2(H, A)$. Where $Z^2(H, A)$ and $B^2(H, A)$ are respectively the abelian group of (all) 2-cocycles and its (normal) abelian subgroup of coboundaries.

In most physics applications, $A = U(1)$ and those equations are hence defined over the real numbers modulo the integers ($U(1) \cong \mathbb{R}/\mathbb{Z}$). The cocycle conditions form a set of $\#G^3$ linear equations for $\#G^2$ unknowns valued in A ; indeed, we get an equation for every triple (g_1, g_2, g_3) . Such linear equations can be solved using linear algebra; in the case of $A = U(1)$ for example, all solutions can be found using the so-called Smith normal form, which is discussed in the appendix. This Smith normal form characterizes all solutions of this 2-cocycle condition modulo trivial solutions.

Example 2.29. *Let us find all central extensions G of $\mathbb{Z}_2$ by $\mathbb{Z}_2$:*

$$1 \longrightarrow \mathbb{Z}_2 \xrightarrow{i} G \xrightarrow{\pi} \mathbb{Z}_2 \longrightarrow 1. \quad (2.21)$$

Step 1: solving the 2-cocycle equation. As it turns out the 2-cocycle equation 2.19 has two solutions:

$$\omega_1(g, h) = 0 \quad \forall g, h, \quad (2.22)$$

$$\omega_2(g, h) = \begin{cases} 1 & g = h = 1, \\ 0 & \text{else.} \end{cases} \quad (2.23)$$

Step 2: writing down the group multiplication 2.18 and identifying the groups. For the trivial cocycle the group multiplication is given by $(a_1, b_1) \cdot (a_2, b_2) := (a_1 + a_2, b_1 + b_2)$, where $a_i, b_i \in \mathbb{Z}_2$. This corresponds to the trivial central extension

$$1 \longrightarrow \mathbb{Z}_2 \xrightarrow{i_1} \mathbb{Z}_2 \oplus \mathbb{Z}_2 \xrightarrow{\pi_1} \mathbb{Z}_2 \longrightarrow 1. \quad (2.24)$$

Here we can choose for example: $i_1(0) = (0, 0)$, $i_2(1) = (1, 0)$ and $\pi_1((a, b)) = b$, and $\pi_1 \circ i_1 = 0$ as it should. Note that $i(\mathbb{Z}_2)$ is indeed contained in the center of $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ since $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ is abelian and thus $Z(\mathbb{Z}_2 \oplus \mathbb{Z}_2) = \mathbb{Z}_2 \oplus \mathbb{Z}_2$.

Let us now consider the cocycle ω_2 . It is only non-zero for $\omega(1, 1) = 1$. To construct the central extension this cocycle gives rise to, we consider the multiplication rule for $\omega = \omega_2$ in 2.18. We observe now the following: $(1, 1)$ is a group element of G of order 4. Indeed (remember the addition is always modulo 2):

$$(1, 1) \cdot (1, 1) =: (1 + 1 + \omega(1, 1), 1 + 1) = (1, 0), \quad (2.25)$$

$$(1, 0) \cdot (1, 1) =: (1 + 1 + \omega(1, 0), 0 + 1) = (0, 1), \quad (2.26)$$

$$(0, 1) \cdot (1, 1) =: (0 + 1 + \omega(1, 1), 1 + 1) = (0, 0), \quad (2.27)$$

or thus $(1, 1)^4 = (0, 0)$. This means that this central extension is (isomorphic to) $\mathbb{Z}_4$: this is the only group of order 4 with an element of order 4! The non-trivial central extension thus reads:

$$1 \longrightarrow \mathbb{Z}_2 \xrightarrow{i_2} \mathbb{Z}_4 \xrightarrow{\pi_2} \mathbb{Z}_2 \longrightarrow 1. \quad (2.28)$$

Let us choose $i_2(0) = 0$, $i_2(1) = 2$ and $\pi_2(1) = 1$. From $\pi_2(2) = \pi_2(1) + \pi_2(1) = 1 + 1 = 0$, we conclude that $\pi_2 \circ i_2 = 0$.

Let us finally check that the extension 2.28 does not split. So let's assume there exists a homomorphism $\phi : \mathbb{Z}_2 \mapsto \mathbb{Z}_4$ such that $\pi_2 \circ \phi = \text{id}_{\mathbb{Z}_2}$. Then necessarily $\phi(0) = 0$ and $\phi(1) = 2$ as $\phi(1)^2 \stackrel{!}{=} \phi(0) = 0$. But clearly $(\pi_2 \circ \phi)(1) = 0$, or thus $\pi_2 \circ \phi \neq \text{id}_{\mathbb{Z}_2}$. The extension is thus not a splitting extension.

¹³What is $[\omega]^{-1}$?

2.A A list of small groups up to order 8

Order	Group
1	$\{e\} \cong \mathbb{Z}_1 \cong S_1 \cong A_2$
2	$\mathbb{Z}_2 \cong S_2 \cong D_1$
3	$\mathbb{Z}_3 \cong A_3$
4	$\mathbb{Z}_4$ $\mathbb{Z}_2^{\times 2} \cong D_2$
5	$\mathbb{Z}_5$
6	$\mathbb{Z}_6 \cong \mathbb{Z}_3 \times \mathbb{Z}_2$ $D_3 \cong S_3 \cong \mathbb{Z}_3 \rtimes \mathbb{Z}_2$
7	$\mathbb{Z}_7$
8	$\mathbb{Z}_8$ $\mathbb{Z}_4 \times \mathbb{Z}_2$ $\mathbb{Z}_2^{\times 3}$ $D_4 \cong \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ $\mathbb{Q}$

Table 2.1: Non-abelian groups are in blue. $\mathbb{Z}_n$: cyclic group of order n , S_n : permutation group of n elements, A_n : alternating group on n elements, D_n : dihedral group of order $2n$, $\mathbb{Q}$: quaternion group.

2.B Group cohomology

We will encounter cohomology theory several times in these lecture notes; in the classification of central extensions (2.19), when we will discuss 1-cocycles for the determination of all space group symmetries, and in the discussion of projective representations. In all these situations, there is a clear physics problem which yields a linear set of equations for which we were looking for topologically distinct solutions.

Example 2.30. *In the case of space groups (see later), I am given an integer 3×3 matrix representation N_g of a point group in 3 dimensions, and the goal is to find all possible solutions $\omega(g) \in (\mathbb{R}/\mathbb{Z})^3 \simeq U(1)^3$ to the problem*

$$\forall g, h : N_g \cdot \omega(h) - \omega(g.h) + \omega(g) = 0 \pmod{\mathbb{Z}^3} \quad (2.29)$$

There is a continuous set of solutions to this linear set of equations when constraining the solutions to be over $\mathbb{R}^3$ as opposed to $(\mathbb{R}/\mathbb{Z})^3 \simeq U(1)^3$, and those are called co-boundaries and all of the form

$$\omega_{cb}(g) = N_g \omega_0 - \omega_0 \quad (2.30)$$

with $\omega_0 \in \mathbb{R}^3$ a constant vector. There are also distinct topologically nontrivial solutions characterized by vectors $\omega_i(g)$ when working over $(\mathbb{R}/\mathbb{Z})^3 \simeq U(1)^3$, and the different solutions are characterized by integers n_i . Those are the 1-cocycles of the group G for group action N_g on $\mathbb{Z}^3$. The solutions themselves are characterized by integers, formally known as the first cohomology group of G with respect to $U(1)^3 \simeq (\mathbb{R}/\mathbb{Z})^3$: $H_N^1(G, U(1)^3)$.

Example 2.31. *For the case of 2-cocycles as in eq. (2.19), we start from a group G and look for solutions $\omega(g_i, g_j) \in U(1) \simeq \mathbb{R}/\mathbb{Z}$ satisfying*

$$\omega(g_2, g_3) - \omega(g_1.g_2, g_3) + \omega(g_1, g_2.g_3) - \omega(g_1, g_2) = 0 \pmod{\mathbb{Z}} \quad (2.31)$$

There is again a set of “trivial” co-boundary solutions, this time parameterized by

$$\omega_{cb}(g, h) = \phi(h) - \phi(gh) + \phi(g) \quad (2.32)$$

for an arbitrary function $\phi : G \rightarrow U(1) \simeq \mathbb{R}/\mathbb{Z}$, and a set of nontrivial solutions, characterized by integer indices. Those solutions modulo trivial solutions form a group called the 2-th cohomology group of G with respect to $U(1)$: $H^2(G, U(1))$.

Both of those examples are described by group cohomology, n-cocycles and the n-th cohomology group $H^n(G, A)$. To define group cohomology, we need a group G , an Abelian group A and a group automorphism $\beta_g(a) : G \rightarrow \text{Aut}(A)$ from G to a subgroup of $\text{Aut}(A)$. Let us consider an arbitrary function $f : G^{\otimes n} \rightarrow A$. At the heart of the theory lies the co-boundary operator ∂ which maps any function $f^{(n)} : G^{\otimes n} \rightarrow A$ to another one of one higher degree $(\partial f)^{(n+1)} : G^{\otimes(n+1)} \rightarrow A$ in the following way:

$$\begin{aligned} (\partial f)^{(n+1)}(g_1, \dots, g_{n+1}) &= \beta_{g_1} \left(f^{(n)}(g_2, g_3, \dots, g_{n+1}) \right) \\ &+ \sum_{k=1}^n (-1)^k f^{(n)}(g_1, g_2, \dots, g_{k-1}, g_k \cdot g_{k+1}, \dots, g_{n+1}) + (-1)^{n+1} f^{(n)}(g_1, \dots, g_n) \end{aligned}$$

∂ maps a function defined in an n dimensional space to a function defined in a $n+1$ dimensional space. The crucial property of ∂ is $(\partial \circ \partial)f = 0$ for every function f . This property can readily be checked to follow from the telescopic sum property present in the definition of ∂ and from the fact that β is a homomorphism: $\beta_g(\beta_h(x)) = \beta_{gh}(x)$.

n-cocycles $\omega^{(n)}(g_1, g_2, \dots, g_n)$ of the group G for a group action $\beta_g(a)$ on A are now defined as the solutions $\omega^{(n)}$ of the equation $(\partial \omega^{(n)})^{(n+1)}(g_1, g_2, \dots, g_n, g_{n+1}) = 0$ with 0 the identity element in A . Those solutions form themselves a group under pointwise addition called the **cocycle group of G over A** , $Z^n(G, A)$. Two solutions are deemed equivalent if they can be obtained from each other by adding a **co-boundary** ω_{cb} to it. Those co-boundaries are defined by acting with the boundary operator on any function $f^{(n-1)}(g_1, \dots, g_{n-1})$:

$$\omega_{cb}^{(n)}(g_1, g_2, \dots, g_n) = (\partial f)^{(n)}(g_1, g_2, \dots, g_n) \quad (2.33)$$

As $(\partial \circ \partial)f = 0$, this is of course automatically a solution independent of the functional form of $f^{(n-1)}$, and this is the reason that those co-boundaries gave rise to a continuum of solutions in the examples already discussed. Note that in the case of $n = 1$, we have to start with the constant function f^0 which is independent of the group elements to construct the co-boundary $\omega_{cb}(g) = \beta_g(f^0) - f^0$. Also the coboundaries form an abelian group, the **coboundary group of G over A** , $B^n(G, A)$.

The **n-th cohomology group of G over A** is defined as $Z^n(G, A)/B^n(G, A)$ and denoted $H^n(G, A) \simeq Z^n(G, A)/B^n(G, A)$. To stress to group action we can also write $H_\beta^n(G, A)$.

Equivalently we can define $H^n(G, A)$ as the group with group elements the **cohomology classes** $[\omega]$:

$$[\omega] = [\omega'] \iff \omega \sim \omega' \iff \omega = \omega' + \partial\varphi, \quad (2.34)$$

and with group multiplication defined via

$$[\omega_1] \cdot [\omega_2] := [\omega_{12}], \quad \omega_{12} \sim \omega_1 + \omega_2. \quad (2.35)$$

If $A = U(1)$, $H^n(G, U(1))$ is always a direct product of cyclic groups. This follows trivially from the existence of the Smith normal form (see appendix 2.C). It also follows that $H^n(G, \mathbb{R}) = 1$ with $\mathbb{R}$ the real numbers: this is another way of saying that the null space of the matrix Ω in the linear equations $\Omega \vec{\omega} = 0$ is completely spanned by the co-boundaries $\partial f^{(n-1)}$ where $f^{(n-1)} : G^{\otimes(n-1)} \rightarrow \mathbb{R}$ is any function over the real numbers. In this and the following $\vec{\omega}$ denotes the vectorized form of the function $\omega(g_1, g_2, \dots, g_n)$ and Ω is the matrix representation of the coboundary map ∂ .

As a corollary of this, we get the following interesting connection between different cohomology groups:

$$H^n(G, U(1)) = H^{n+1}(G, \mathbb{Z}). \quad (2.36)$$

Sketch of Proof: The problem of characterizing $H^{n+1}(G, \mathbb{Z})$ is to identify all inequivalent solutions of the cocycle condition, modulo the co-boundaries. As $A = \mathbb{Z}$, this is equivalent to identifying the null space of the corresponding matrix $\Omega \vec{\omega} = 0$ (not modulo $\mathbb{Z}$ as in the $U(1)$ case!). As Ω only contains integers, this null space is certainly spanned by vectors with only integer coefficients, and the problem becomes to identify which ones of those null vectors are co-boundaries and which are not. The co-boundaries are obtained by lifting functions $f^{(n)}$ over $\mathbb{Z}$ as $\partial f^{(n)}$. But there is clearly another class of functions which live in this null space, namely the lifting of the nontrivial cocycles of $H^n(G, U(1))$ defined over $U(1)$: the whole point of those cocycles ω^n is precisely the fact that $\partial \omega^n$ as interpreted as an equation in $\mathbb{R}$ as opposed to $U(1)$ has only integer coefficients in $\mathbb{Z}$, and hence is a valid function from $G^{\otimes n+1} \rightarrow \mathbb{Z}$. As $\partial \partial \omega^n$ is certainly zero, those null vectors are not co-boundaries and precisely span the rest of the null space of the matrix Ω . This proves the formula $H^n(G, U(1)) = H^{n+1}(G, \mathbb{Z})$. $\square$

In summary, group cohomology is providing the mathematical language for classifying essentially/topologically distinct solutions to interesting physics problems. An important remark is that the whole discussion above is both valid for discrete and continuous groups. However, in the case of continuous groups we should allow functions that are not continuous; this is then called Borel group cohomology.

2.C Computing cohomology groups and cocycles over $A = U(1)$

The Smith normal form

The Smith normal form exists for every matrix with integer entries, or more generally for entries in a [principal ideal domain](#). Given a matrix $\Omega \in \mathbb{Z}^{n \times m}$, the Smith normal form of Ω reads

$$\Omega = P \Lambda R. \quad (2.37)$$

Here, $P \in \mathbb{Z}^{n \times n}$, $R \in \mathbb{Z}^{m \times m}$, and both have determinant 1. They can thus both be inverted, and their inverses are also integer valued. Λ is a diagonal $n \times m$ matrix of the form

$$\Lambda = \left(\begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0 \\ \hline 0 & 0 \end{array} \right), \quad (2.38)$$

in which the non-zero elements $d_1, \dots, d_r$ along the diagonal, some of which might be one, are in increasing order, $d_1 \leq d_2 \leq \dots$, and every element is a divisor of the next, $d_i | d_{i+1}$.

In this section we explain how cohomology groups $H^n(G, U(1))$ can be computed using simple linear algebra methods developed in [\[2202.12880\]](#). As a side product, we will also obtain representative cocycles for every cohomology class.

The key insight is that the cocycle condition over $U(1)$ can be written as a linear system of the form

$$\Omega \vec{\omega} = \vec{0}, \quad \text{mod } 2\pi. \quad (2.39)$$

Here $\vec{\omega}$ is the vectorized form of the function $\omega(g_1, g_2, \dots, g_n)$ and Ω is the matrix representation of the coboundary map ∂ . $\vec{\omega}$ is $|G|^n$ -dimensional and Ω is $|G|^{n+1} \times |G|^n$ -dimensional, a consequence of the fact that ∂ maps n -cycles to $(n+1)$ -cycles.

Since Ω only contains integers, we can write Ω in its Smith normal form

$$\Omega = P \Lambda R, \quad (2.40)$$

In this decomposition P and R are respectively $|G|^{n+1} \times |G|^{n+1}$ and $|G|^n \times |G|^n$ matrices that only contain integers and have determinant one (and thus have integer-valued inverses). Λ also

only contains integers, is $|G|^{n+1} \times |G|^n$ -dimensional and is of the form

$$\Lambda = \left(\begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0 \\ \hline 0 & 0 \end{array} \right), \quad (2.41)$$

in which the non-zero elements $d_1, \dots, d_r$ along the diagonal, some of which might be one, are in increasing order, $d_1 \leq d_2 \leq \dots$, and every element is a divisor of the next, $d_i | d_{i+1}$. The Smith normal form Λ is unique. Inserting this decomposition in the system of equations (2.39) gives rise to the solution

$$\vec{\omega} = 2\pi R^{-1} \Lambda^+ \vec{\nu}. \quad (2.42)$$

Λ^+ denotes the (unique) Moore-Penrose pseudoinverse of Λ that satisfies $\Lambda \Lambda^+ \Lambda = \Lambda$ and which is found to be

$$\Lambda^+ = \left(\begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0 \\ \hline 0 & 0 \end{array} \right). \quad (2.43)$$

$\vec{\nu}$ is an arbitrary vector that only contains integers.

Writing the solution (2.42) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{\nu_j}{d_j}. \quad (2.44)$$

Because $\vec{\nu}$ can be chosen freely, one can choose subsequently $\nu_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$ to obtain a basis of the solution space that can be written as

$$(\vec{\omega}_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (2.45)$$

Hence, the n-cocycles $\vec{\omega}$ are found to be the columns of R^{-1} . Since R is full rank, all the solutions $\vec{\omega}$ are linearly independent. In particular, the non-trivial cocycles (below) can not be related by a coboundary, $\vec{\omega}' = \vec{\omega} + \Omega' \vec{\varphi}$, where Ω' denotes the (n-1)-coboundary map and $\vec{\varphi}$ is a $|G|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of Λ .

From (2.44) and the fact that R^{-1} contains only integers, it follows that for every diagonal entry $d_i = 1$, a trivial solution $\omega_i = 0 \pmod{2\pi}$ is obtained. The non-trivial solutions are those that correspond to entries $d_i > 1$. From (2.44) and the fact that the solution space is $\mathbb{Z}$ -linear, it follows that the cocycle $\vec{\omega}_j$ corresponding to some d_j generates a cyclic group. Now note that not all elements of $\vec{\omega}_j$ can be divisible by d_j (or any of its prime factors) as this would be in contradiction with the fact that R^{-1} has determinant one. Hence, the cyclic group generated by $\vec{\omega}_j$ is $\mathbb{Z}_{d_j}$. Finally, the zero entries of Λ can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group G one can write down the matrix representation of the coboundary map Ω that can be brought in Smith normal form $P\Lambda R$. The diagonal entries of Λ , $d_1, d_2, \dots, d_r$, determine the second cohomology group which is then of the form $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$, with the understanding that $\mathbb{Z}_1 = \{e\}$ denotes the trivial group and that all zero diagonal entries can be discarded. The non-trivial n-cocycles in some arbitrary gauge correspond then to the columns of R^{-1} , weighted by the appropriate factor $2\pi/d_j$.

Chapter 3

Representation Theory of Finite Groups

Representation theory is the theory of matrix representations of groups: we want to study groups via their actions on vector spaces. Central concepts are invariant subspaces, irreducible representations, characters,... From the point of physics, group theory and representation theory are basically equivalent: we study group theory as we want to know how wave functions transform under symmetry operations. Wave functions are vectors, and hence those transformations are given by matrices which form representations of the group under consideration. A central point will be that group theory provides a very natural basis for writing wave functions in, namely a direct sum of wave functions all transforming in different ways. All those different subspaces will be labelled by quantum numbers.

Let us for example consider the ammonia molecule, which has the symmetry of the triangle D_3 (also sometimes called C_{3v} in atomic and molecular physics literature), see figure 3.1. As the corresponding Hamiltonian commutes with the symmetry operations of D_3 ($[H, T_i] = 0$ with T_i operators implementing the symmetry operations on the Hilbert space $L_2(\mathbb{R}^3)$), we can simultaneously block-diagonalize H and all T_i ; in this particular case, the set of operators T_i form a representation of D_3 , and as we will see, any such representation can be written in a basis consisting of a direct sum structure

$$T_i = (1 \otimes \mathbb{1}_{n_1}) \oplus \left((-1)^{p(i)} \otimes \mathbb{1}_{n_2} \right) \oplus (D(i) \otimes \mathbb{1}_{n_3}) \quad (3.1)$$

with $p(i)$ the parity of the group element i and $D(i)$ a so-called irreducible 2×2 matrix representation of D_3 ($D(i)D(j) = D(ij)$). The integers n_i are the multiplicities of the corresponding irreducible representations. As a consequence, the eigenstates of the ammonia Hamiltonian can be organized into 3 classes, one of them exhibiting a two-fold degeneracy in the eigenspectrum. The corresponding eigenstates $\sum_{\alpha=1}^2 \xi_{\alpha} \psi_{\alpha}(x)$ transform as $\xi_{\alpha} \rightarrow \sum_{\beta=1}^2 D_{\alpha\beta}(i) \xi_{\beta}$ under the symmetry operations g_i .

From the more abstract point of view, the goal of representation theory is to construct a homomorphism from the group G to a matrix algebra $GL(n; \mathbb{C})$ acting on V , with $V \simeq \mathbb{C}^n$ an n -dimensional complex vector space. The identity element corresponds to the identity matrix, and hence all associated matrices A_g must be invertible (as there must be an inverse), and satisfy

$$A_g \cdot A_h = A_{gh}. \quad (3.2)$$

3.1 Representations

Let us now formally introduce representations of (finite) groups. Recall that for a vector space V $\text{End}(V) \equiv \text{Hom}(V, V)$ denotes the linear maps of V to itself.

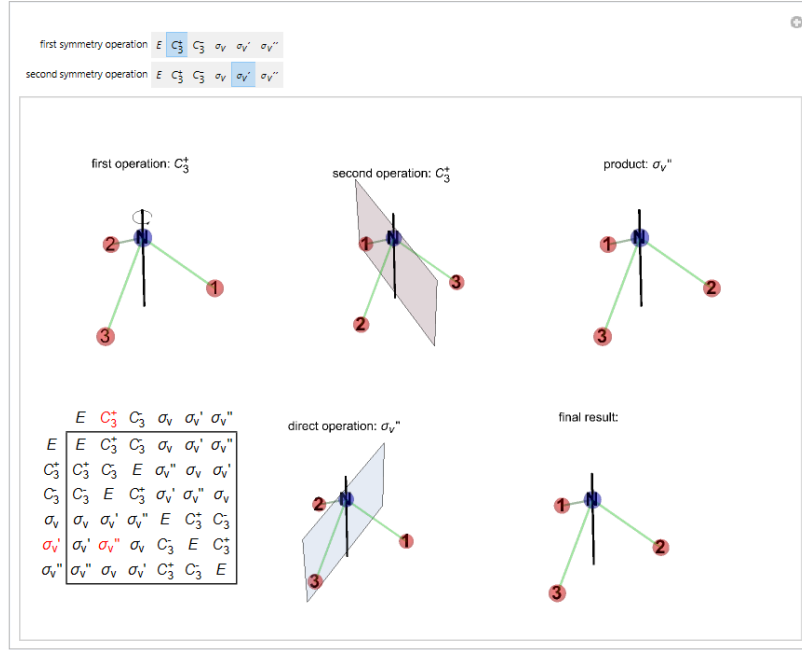


Figure 3.1: Symmetries of the ammonia molecule: $D_3 \simeq S_3$, also denoted C_{3v} in chemistry literature. Taken from [this](#) very useful application! (Best to download the application and execute it with the Wolfram player.)

Definition 3.1 (Representation: basis independent). A **(linear) representation of a finite group** G , (V, ρ) , is a vector space V and a homomorphism $\rho : G \mapsto \text{End}(V)$. In particular, $\rho(1) = \text{id}_V$. The dimension of V is called the **dimensions of the representation**.

Remark. We will later introduce **projective representations**. To distinguish between these projective representations and the ‘ordinary’ representations we just introduced, we shall sometimes call the latter **linear representations**.

We will restrict to finite-dimensional vector spaces over $\mathbb{C}$, and below we shall always assume that this vector space is a complex Hilbert space and thus comes equipped with an inner product. It is very important to notice that even though representation theory can be defined over real vector spaces (or even over vector spaces over finite fields $\mathbb{F}_p$, in which we are dealing with so called modular representation theory), many of the following theorems and examples would in that case fail to hold or need significant modification, examples include Schur’s lemma and Maschke’s theorem.

There are a couple of different notation used to denote representations. We will mostly use Greek letters, $\alpha, \beta, \dots, \Gamma, \dots$, and sometimes $\underline{\alpha}, \underline{\beta}, \dots$ when $\alpha, \beta, \dots$ take a numerical value, for example when it refers to the dimension of the representation, but also $l, l_1, l_2, \dots$ will be used. Usually the vector space on which the group acts is left implicit. Numerous examples will be encountered below and in the exercises.

After choosing a basis for V , one can identify $\text{End}(V)$ with the matrix group $GL(n; \mathbb{C})$ of invertible $n \times n$ complex matrices, where $n = \dim V$.

Definition 3.2 (Representation: basis dependent). A **(linear) (matrix) representation of a group** G is a collection of matrices labeled by the group elements of G , $\{X_g | g \in G\}$ that multiply according to $X_g X_h = X_{gh}$. In particular, $X_1 = \mathbb{1}$.

We will also denote matrix representations as $\{X(g)|g \in G\}$. Different matrix representations will be distinguished by introducing a superscript such as $\{X_g^{(\gamma_1)}|g \in G\}$, $\{X^{(\gamma_2)}(g)|g \in G\}$,...

The reason that matrix representations of groups exist is due to the defining property of a group, namely associativity. In the case of **finite groups**, it turns out that we can always restrict ourselves to unitary matrices (if they are not, we can make them unitary by working in a different basis), and one can also easily prove that any two equivalent representations (i.e. related by a conjugation $A_g = XB_gX^{-1}$) are related to each other by a unitary conjugation X . This will be proven in Theorem 3.1 below. But first:

Definition 3.3. Two representations $\{A_g\}$ and $\{B_g\}$ are called **(unitarily) equivalent representations** if there exists a (unitary) invertible matrix V independent of g which satisfies

$$A_g = VB_gV^{-1}, \quad \forall g \in G. \quad (3.3)$$

V is called a **similarity transformation**.

Definition 3.4. A representation (ρ, V) is called called a **unitary representation** if $\rho(g)$ is a unitary linear transformation for all $g \in G$:

$$\langle \rho(g)\varphi | \rho(g)\psi \rangle = \langle \varphi | \psi \rangle, \quad \forall g \in G, \quad \forall |\varphi\rangle, |\psi\rangle \in V. \quad (3.4)$$

A very important theorem:

Theorem 3.1. Any matrix representation $\{U(g)|g \in G\}$ is always equivalent to a unitary representation.

Proof: Consider

$$A = \sum_g U(g)U(g)^\dagger. \quad (3.5)$$

As A is the sum of positive semidefinite matrices including the identity matrix, A is itself strictly positive definite, and there exists a hermitean and positive definite square root $\sqrt{A} = \sqrt{A}^\dagger$ which is full rank. We can now construct an equivalent representation

$$V(g) = \sqrt{A}^{-1}U(g)\sqrt{A} \quad (3.6)$$

which can be shown to be unitary. Indeed, we have

$$V(g)V(g)^\dagger = \sqrt{A}^{-1}U(g)\sqrt{A}\sqrt{A}U(g)^\dagger\sqrt{A}^{-1} \quad (3.7)$$

$$= \sqrt{A}^{-1}U(g)\sum_x U(x)U(x)^\dagger U(g)^\dagger\sqrt{A}^{-1} \quad (3.8)$$

$$= \sqrt{A}^{-1}\sum_x U(gx)U(gx)^\dagger\sqrt{A}^{-1} \quad (3.9)$$

$$= \sqrt{A}^{-1}\sum_x U(x)U(x)^\dagger\sqrt{A}^{-1} \quad (3.10)$$

$$= \sqrt{A}^{-1}A\sqrt{A}^{-1} \quad (3.11)$$

$$= \mathbb{1} \quad (3.12)$$

□

3.1.1 First examples

Example 3.1. Every group has at least the **trivial representation**. In the trivial representation (V, φ) all group elements act trivially on V , i.e. $\varphi(g)|\psi\rangle = |\psi\rangle$, $\forall g \in G$, $\forall |\psi\rangle \in V$.

Example 3.2. A one-dimensional representation of $\mathbb{Z}_N = \{0, 1, 2, \dots, N-1\}$ is given by $\underline{1}(n) = \omega_N^n$, where $\omega_N := \exp\left(\frac{2\pi i}{N}\right)$. See also example 3.8 below.

Example 3.3. The dihedral group $D_n = \langle r, s | r^n = s^2 = (rs)^2 = 1 \rangle$ has a 2-dimensional representation

$$\rho(r) = \begin{pmatrix} \cos\left(\frac{2\pi}{n}\right) & \sin\left(\frac{2\pi}{n}\right) \\ -\sin\left(\frac{2\pi}{n}\right) & \cos\left(\frac{2\pi}{n}\right) \end{pmatrix}, \quad (3.13)$$

$$\rho(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.14)$$

Indeed: $\rho(r)^n = \rho(s)^2 = \rho(r)\rho(s)\rho(r)\rho(s) = \mathbb{1}_2$.

Example 3.4. One obvious way of making a representation of a finite group of dimension $\#G = n$ is by constructing the **regular representation**, which is obtained by constructing $n \times n$ permutation matrices exactly encoding the information of the rows/columns of the Cayley table. We first introduce a basis for $V \simeq \mathbb{C}^{|G|}$ that we will write as $\{|g\rangle | g \in G\}$ with the inner product $\langle g|h\rangle = \delta_{g,h}$, where $\delta_{g,h} = 1$ if $g = h$ and 0 else. The **left regular representation** is then given by

$$L_g = \sum_h |gh\rangle\langle h| \quad (3.15)$$

for which it holds that $L_x L_y = L_{xy}$, while the **right regular representation** is given by

$$R_g = \sum_h |hg^{-1}\rangle\langle h| = \sum_{h'} |h'\rangle\langle h'g| \quad (3.16)$$

for which $R_x R_y = R_{xy}$. Note that all matrices L_x commute with all matrices R_y , hence they can be simultaneously be block-diagonalized. Those blocks will exactly correspond to the irreducible representations that we will introduce below.

Example 3.5. The regular representation of $\mathbb{Z}_2$ is given by:

$$L_0 = \mathbb{1}_2 \quad (3.17)$$

$$L_1 = \sigma_X, \quad (3.18)$$

where σ_X is one of the famous Pauli matrices:

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3.19)$$

Example 3.6. Given a representation of a group G , (ρ, V) , and a subgroup $H \leq G$, you can always restrict ρ to H to obtain a representation of H called the **restricted** - or **subduced** representation $(\rho|_H, V)$. Note that the restricted representation acts on the same vector space V as the original one.

Given a representation (π, W) of $H \leq G$ one can also lift this representation to construct the **induced** representation of G from π , $(\text{Ind}_H^G(\pi), W)$. This construction is explained in the appendix.

Example 3.7. Given a representation $\{X_g\}$, the **complex conjugate representation** is given by $\{\overline{X_g}\}$ or $\{X_g^*\}$, where the overline or $\star$ denotes complex conjugation. This defines still a representation since $\overline{X_g X_h} = \overline{X_{gh}}$. Sometimes the conjugate representation is equivalent to the original one, and sometimes not. We will come back to this issue below.

Of particular interest are representations for which distinct group elements have distinct actions on the underlying vector space:

Definition 3.5. A **faithful** representation (V, ρ) is one for which ρ is injective.

3.2 Irreducible representations and direct sum representation

As it will turn out every finite group has a finite number of so-called irreducible representations (irreps), which are representations with no invariant subspaces. We will show in this section that every representation can be written as the direct sum of such irreps.

From your course on linear algebra you should recall the definition of the direct sum of vector spaces. We then define:

Definition 3.6. *Given two representations (ρ, V) and (ρ', V') of a group G , the **direct sum representation** $(\rho \oplus \rho', V \oplus V')$ is defined via*

$$(\rho \oplus \rho')(g) := \rho(g) \oplus \rho'(g). \quad (3.20)$$

You should check that $(\rho \oplus \rho')(g)(\rho \oplus \rho')(h) = (\rho \oplus \rho')(hg)$.

In terms of matrices the direct sum representation thus maps every group element to a block diagonal matrix. The blocks corresponding to the given representations of which the direct sum is constructed.

We will show that:

Theorem 3.2. *(Maschke's theorem) Every representation of a finite group is equivalent to the direct sum representation of a number of irreducible representations with no non-trivial invariant subspaces.*

A subspace Y of the vector space is called a left invariant subspace if it is left invariant by left multiplication by all matrices in the algebra, i.e. $\forall g : A(g)Y \subset Y$. Similarly, a right invariant subspace is one for which the right multiplication is left invariant: $\forall g : YA(g) \subset Y$. For matrix algebras with left invariant subspaces, there always exist a basis in which we can write all those matrices in block diagonal form

$$A(r) = \begin{bmatrix} B(r) & D(r) \\ 0 & C(r) \end{bmatrix} \quad (3.21)$$

and the left invariant subspace is then spanned by vectors of the form $\begin{bmatrix} X \\ 0 \end{bmatrix}$. Similarly for the case of right invariant subspaces, there is a basis for which

$$A(r) = \begin{bmatrix} B(r) & 0 \\ D(r) & C(r) \end{bmatrix}, \quad \forall r \in G. \quad (3.22)$$

Let us now assume that we have a representation of a group $A(g)$ with a right invariant subspace. **Maschke's theorem** then states that there always exists a similarity transformation which makes all matrices in the representation strictly block diagonal, it is we can get rid of the off-diagonal blocks.

This can be proven as follows: given a representation with a right invariant subspace of the form

$$A(r) = \begin{bmatrix} B(r) & 0 \\ D(r) & C(r) \end{bmatrix} \quad (3.23)$$

and hence with the property that $D(ab) = D(a)B(b) + C(a)D(b)$, we want to find a similarity transformation of the form

$$\begin{bmatrix} \mathbb{1} & 0 \\ -F & \mathbb{1} \end{bmatrix} \begin{bmatrix} B(r) & 0 \\ D(r) & C(r) \end{bmatrix} \begin{bmatrix} \mathbb{1} & 0 \\ F & \mathbb{1} \end{bmatrix} = \begin{bmatrix} B(r) & 0 \\ 0 & C(r) \end{bmatrix} \quad (3.24)$$

By making use of the fact that B and C form representations of the group, check that

$$F = \frac{1}{|G|} \sum_a D(a)B(a^{-1})$$

does exactly that, given that $B(1) = \mathbb{1}$ (which clearly has to hold for any representation without invariant subspaces). Any representation is hence equivalent to a direct sum of representations, without off-diagonal blocks.

Any representation $U(g)$ can hence be written in a basis for which

$$U(g) = \begin{pmatrix} \boxed{U_1(g)} & 0 & \cdots \\ 0 & \boxed{U_2(g)} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad (3.25)$$

in which each of the $U_i(g)$ do not have invariant subspaces anymore.

Definition 3.7. An **irreducible representation (irrep)** is a matrix representation of a group without non-trivial invariant subspaces.

Definition 3.8. A **reducible representation** is a matrix representation of a group with non-trivial invariant subspaces.

Remark. In the exercises we will prove that all the irreducible representations of abelian groups are one-dimensional. We will prove this using Schur's (first) lemma (see below).

Example 3.8. The group $\mathbb{Z}_N$ has exactly N distinct irreps. Writing $\mathbb{Z}_N = \langle g | g^N = 0 \rangle$, we have that these N irreps, $\underline{\alpha}$, correspond to N roots of unity: $\underline{\alpha}(g) = \exp\left(\frac{2\pi i \alpha}{N}\right)$ where $\alpha = 0, 1, \dots, N-1$. This will also be the topic of one of the exercises.

Numerically, these invariant subspace and irreps can be calculated by finding the center of the matrix algebra¹. An irrep is the matrix algebra on such an irreducible block. Note also that in general those irreps will be complex. The central task in representation theory is the identification of all possible irreps of a given group.

Of central importance are **Schur's lemma's**, as it are the working horses in representation theory:

Lemma 3.3 (Schur's first lemma). If $U(g)$ is an irrep and $\forall g : XU(g) = U(g)X$, then $X = c\mathbb{1}$ for a constant $c \in \mathbb{C}$.

Lemma 3.4 (Schur's second lemma). Given two irreps $U(g), V(g)$ and a (rectangular) matrix T such that $\forall g : TU(g) = V(g)T$, then either U, V are equivalent and T is unique up to a constant, or $T = 0$.

Proof: The proof is an application of elementary linear algebra. Let us first prove Schur's first lemma. X has a Jordan normal form

$$X = \begin{bmatrix} \lambda_1 J_1 & 0 & \cdots \\ \cdots & \lambda_2 J_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (3.26)$$

with J_i Jordan blocks of dimension n_i . It can easily be shown that, if any of the $n_i > 1$, then $U(g)$ would have an invariant subspace, so all $n_i = 1$. Assume now that $\lambda_i \neq \lambda_j$. Then the $U(g)_{ij}$ element has to be equal to zero, and hence an invariant subspace would again emerge. We conclude that all $\lambda_i = \lambda_j = c$. The first lemma hence basically says that an irrep spans the whole matrix algebra. The second lemma can be proven as follows: given that for any irrep, $U(g)$ is full rank, we can take the pseudo-inverse of $TU(g) = V(g)T$, $U(g)^{-1}T^{\text{PI}} = T^{\text{PI}}V(g)^{-1}$. We then have $U(g)^{-1}T^{\text{PI}}TU(g) = T^{\text{PI}}V(g)^{-1}V(g)T = T^{\text{PI}}T$ and following Schur's first lemma: $T^{\text{PI}}T = c\mathbb{1}$.

¹In practice, this can be achieved by diagonalizing a random linear combination of all U_g , and the irreps then correspond to the degenerate eigenspaces.

Similarly, we directly show that $TT^{\text{PI}} = c'\mathbb{1}$. This is only possible when T is full rank and can be inverted, or when $T = 0$.

Now assume that also S satisfies $SU(g) = V(g)S$ for all g , then we can also take the pseudo-inverse of this equation and combine it with $TU(g) = V(g)T$ to obtain $TS^{\text{PI}} = \tilde{c}\mathbb{1}$. And since also $S^{\text{PI}}S = c_S\mathbb{1}$, we find $T = TS^{\text{PI}}Sc_S^{-1} = \tilde{c}c_S^{-1}S$. $\square$

Schur's (first) lemma also provides a simple (although not always efficient) way to test if a given matrix representation of a finite group is irreducible or not. If one can find a matrix which is not proportional to the identity matrix that commutes with all the representation matrices, that representation is certainly reducible. Can you find an example? Hint: $\mathbb{Z}_2$.

The importance of Schur's lemmas cannot be overstated. As applications of Schur's lemma, we will prove the great orthogonality theorem, introduce characters, and finally prove that there are three types of representations (real, complex and quaternionic).

3.3 Tensor product representation

In physics we are often confronted with systems that transform according to the tensor product of two or more representations. In this section we introduce the notion of tensor product representations and we will argue that these decompose as a direct sum of irreps.

Before introducing the tensor product of representations, we need the notion of the tensor product of vector spaces, which will be the kind of vector spaces on which tensor product representations will act.

3.3.1 Tensor product of vector spaces

Given a d -dimensional vector space V with orthogonal basis $\{|\psi_i\rangle | i = 1, 2, \dots, d\}$, the tensor product $V \otimes V$ is a d^2 -dimensional vector space with orthogonal basis $\{|\psi_i\rangle \otimes |\psi_j\rangle | i, j = 1, 2, \dots, d\}$. We will simply write $|\psi_i\rangle \otimes |\psi_j\rangle \equiv |\psi_i\rangle |\psi_j\rangle \equiv |\psi_{ij}\rangle$.

Given linear operators $\hat{A}$ and $\hat{B}$ both acting on V , $\widehat{A \otimes B}$ acts on $V \otimes V$ according to $(\widehat{A \otimes B})|\psi_{ij}\rangle = \hat{A}|\psi_i\rangle \otimes \hat{B}|\psi_j\rangle$.

$V \otimes V$ contains two particularly interesting linear subspaces:

1. The symmetric or symmetrized subspace $(V \otimes V)^S$ with orthogonal basis vectors $|\psi_{ij}^S\rangle := |\psi_{ij}\rangle + |\psi_{ji}\rangle$, where $i, j = 1, 2, \dots, d$, which is symmetric under $i \leftrightarrow j$: $|\psi_{ij}^S\rangle = |\psi_{ji}^S\rangle$.
2. The antisymmetric or antisymmetrized subspace $(V \otimes V)^A$ with orthogonal basis vectors $|\psi_{ij}^A\rangle := |\psi_{ij}\rangle - |\psi_{ji}\rangle$, where $i, j = 1, 2, \dots, d$, which is skewsymmetric under $i \leftrightarrow j$: $|\psi_{ij}^A\rangle = -|\psi_{ji}^A\rangle$.

$$V \otimes V \simeq (V \otimes V)^S \oplus (V \otimes V)^A.$$

If we are given some matrix representation of $\hat{A}$ and $\hat{B}$, A and B , with respect to the basis vectors $|\psi_i\rangle$, then the matrix representation of $\widehat{A \otimes B}$ with respect to the basis $\{|\psi_i\rangle \otimes |\psi_j\rangle | i, j = 1, 2, \dots, d\}$ is given by the **Kronecker product** of A and B , also denoted $A \otimes B$:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}. \quad (3.27)$$

You can check that $(A \otimes B)(C \otimes D) = AC \otimes BD$. This property is crucial in the definition of the tensor product representation below.

More generally, for two vector spaces V and W with basis vectors $\{|\psi_i\rangle | i = 1, 2, \dots, d\}$ and $\{|\phi_i\rangle | i = 1, 2, \dots, d'\}$. A basis of $V \otimes W$ is given by $\{|\psi_i\rangle \otimes |\phi_j\rangle | i = 1, 2, \dots, d; j = 1, 2, \dots, d'\}$. The

matrix representation of the tensor product of two operators acting on V and W is the again given by the Kronecker product defined above.

Example 3.9. Consider a **qubit**, which is a two-dimensional vector space $V \simeq \mathbb{C}^2$ spanned by the orthonormal vectors denoted $\{|\uparrow\rangle, |\downarrow\rangle\}$ ('spin up and spin down', also called the **sigma Z basis** or **standard computational basis**) we define the **Pauli operators**:

$$\hat{\sigma}_X |\uparrow\rangle = |\downarrow\rangle \quad (3.28)$$

$$\hat{\sigma}_X |\downarrow\rangle = |\uparrow\rangle \quad (3.29)$$

$$\hat{\sigma}_Y |\uparrow\rangle = i |\downarrow\rangle \quad (3.30)$$

$$\hat{\sigma}_Y |\downarrow\rangle = -i |\uparrow\rangle \quad (3.31)$$

$$\hat{\sigma}_Z |\uparrow\rangle = |\uparrow\rangle \quad (3.32)$$

$$\hat{\sigma}_Z |\downarrow\rangle = -|\downarrow\rangle \quad (3.33)$$

We will identify the basis vectors $\{|\uparrow\rangle, |\downarrow\rangle\}$ with column vectors as follows:

$$|\uparrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3.34)$$

The matrix representations of these Pauli operators in this basis then read:²

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (3.35)$$

$$\sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (3.36)$$

$$\sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.37)$$

In the tensor product basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ which spans $\mathbb{C}^2 \otimes \mathbb{C}^2 \simeq \mathbb{C}^4$, we identify the basis vectors with columns vectors as follows:

$$|\uparrow\uparrow\rangle \equiv |\uparrow\rangle \otimes |\uparrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad (3.38)$$

$$|\uparrow\downarrow\rangle \equiv |\uparrow\rangle \otimes |\downarrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad (3.39)$$

$$|\downarrow\uparrow\rangle \equiv |\downarrow\rangle \otimes |\uparrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad (3.40)$$

$$|\downarrow\downarrow\rangle \equiv |\downarrow\rangle \otimes |\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (3.41)$$

We then have for example that

$$\sigma_X \otimes \sigma_Z = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \quad (3.42)$$

²In the context of Lie algebras, we will see that these matrices form a basis of the Lie algebra $\mathfrak{su}(2)$. For that reason you should memorize these matrices as they appear all over the place in physics!

Example 3.10. *Fundamental particles that appear in nature are either fermions (particles with half integer spin $\frac{1}{2}, \frac{3}{2}, \dots$, the matter particles) or bosons (particles with integer spin $0, 1, 2, \dots$, the force carriers such as the photon). See further courses on particle physics, statistical physics, QFT,...*

Suppose we were given two identical non-interacting bosons described by wavefunctions $|\psi^{(1)}\rangle$ and $|\psi^{(2)}\rangle$ respectively, the wavefunction of the combined system boson $1 + 2$ is given by $|\psi^{(12)}\rangle = \frac{1}{\sqrt{2}} (|\psi^{(1)}\rangle |\psi^{(2)}\rangle + |\psi^{(2)}\rangle |\psi^{(1)}\rangle)$ and thus lives in the symmetric subspace.

If, on the other hand, we were given two identical non-interacting fermions described by wavefunctions $|\psi^{(1)}\rangle$ and $|\psi^{(2)}\rangle$ respectively, the wavefunction of the combined system fermion $1 + 2$ is given by $|\psi^{(12)}\rangle = \frac{1}{\sqrt{2}} (|\psi^{(1)}\rangle |\psi^{(2)}\rangle - |\psi^{(2)}\rangle |\psi^{(1)}\rangle)$ and thus lives in the antisymmetric subspace.

3.3.2 Tensor product representation

Definition 3.9. *Given two representations (ρ, V) and (ρ', V') of a group G , the **tensor product representation** $(\rho \otimes \rho', V \otimes V')$ is defined via*

$$(\rho \otimes \rho')(g) := \rho(g) \otimes \rho'(g). \quad (3.43)$$

That this defines a representation follows from the property $(A \otimes B)(C \otimes D) = AC \otimes BD$ of the Kronecker product. Indeed,

$$(\rho \otimes \rho')(g)(\rho \otimes \rho')(h) := (\rho(g) \otimes \rho'(g))(\rho(h) \otimes \rho'(h)) = \rho(gh) \otimes \rho'(gh) =: (\rho \otimes \rho')(gh), \quad (3.44)$$

where in the second equation we used that (ρ, V) and (ρ', V') are representations.

3.3.3 Clebsch-Gordan coefficients

As mentioned above, every representation can be written as a direct sum of irreps with no non-trivial subspaces. Thus also the tensor product of two or more representations is equivalent to a direct sum of irreps appearing with some multiplicity which might be greater than one. In particular, and this may come across as somewhat counterintuitive, the tensor product of irreps is not necessarily an irrep itself. We write this decomposition for representations α, β as:

$$\alpha \otimes \beta \simeq \bigoplus_{\gamma} N_{\alpha\beta}^{\gamma} \gamma. \quad (3.45)$$

The sum is over all irreps of the group. We will later see that every finite group has a finite number of irreps. In this notation $N_{\alpha\beta}^{\gamma}$ are sometimes called the **fusion coefficients** which are integers greater than or equal to 0 and are the multiplicity of the irrep γ in the decomposition of $\alpha \otimes \beta$. Later we will be able to compute these coefficients using character theory.

In this expression $\simeq$ means ‘up to basis transformation’. In matrix notation we can write more explicitly that

$$V (U^{\alpha}(g) \otimes U^{\beta}(g)) V^{\dagger} = \begin{pmatrix} \boxed{U^{\lambda_1}(g)} & & & \\ & \boxed{U^{\lambda_1}(g)} & & \\ & & \ddots & \\ & & & \boxed{U^{\lambda_n}(g)} \end{pmatrix}, \quad (3.46)$$

where V is a unitary basis transformation (which is independent of g !), and every block appears $N_{\alpha\beta}^{\lambda_i}$ times. In general, finding the basis transformation V is not so easy, but is often feasible for small groups.

Proposition 3.5. *The fusion coefficients satisfy:*

1. $N_{\alpha\beta}^\gamma = N_{\beta\alpha}^\gamma$
2. $\sum_\epsilon N_{\alpha\beta}^\epsilon N_{\epsilon\gamma}^\delta = \sum_\epsilon N_{\alpha\epsilon}^\delta N_{\beta\gamma}^\epsilon$

Proof: The first equation follows from $\alpha \otimes \beta \simeq \beta \otimes \alpha$. The second property from $(\alpha \otimes \beta) \otimes \gamma \simeq \alpha \otimes (\beta \otimes \gamma)$. $\square$

For Lie groups and Lie algebras the entries of V appearing in the block decomposition above are called the **Clebsch-Gordan coefficients** (CG) but also in the case of finite groups we will use this terminology. Clebsch-Gordan coefficients appear all over the place in physics in concrete computations.

The Clebsch-Gordan coefficients will be written as $C_{i_1 i_2 i_3 s}^{\alpha \beta \gamma}$ and the notation has the following meaning: as in the above example α, β and γ denote the involved representations where γ is in particular an irreducible representation. i_1, i_2, i_3 denote the indices in the involved representations and can take values $i_1/i_2/i_3 = 1, 2, \dots, \dim(\alpha/\beta/\gamma)$. s finally is a degeneracy label that takes values $1, 2, \dots, N_{\alpha\beta}^\gamma$ and keeps track of which block in the direct sum decomposition you end up in. Concretely, the analogue of 3.46 written out using CGs:

$$U_{i_1 j_1}^\alpha(g) U_{i_2 j_2}^\beta(g) = \sum_{\gamma, i_3, j_3, s} C_{i_1 i_2 i_3 s}^{\alpha \beta \gamma} \bar{C}_{j_1 j_2 j_3 s}^{\alpha \beta \gamma} U_{i_3 j_3}^\gamma(g). \quad (3.47)$$

By grouping the indices of $C_{i_1 i_2 i_3 s}^{\alpha \beta \gamma}$ you obtain the basis transformation denoted as V above. Again, notice the independence of the Clebsch-Gordan coefficients on the group elements!

In terms of basis vectors for the underlying vector spaces we can write $C_{i_1 i_2 i_3 s}^{\alpha \beta \gamma} = \langle \gamma, i_3, s | (\alpha, i_1) \otimes (\beta, i_2) \rangle \equiv \langle \gamma, i_3, s | \alpha, i_1; \beta, i_2 \rangle$. The notation should be self-explanatory.

It can be shown that the Clebsch-Gordan coefficients satisfy:

$$\sum_{i_1, i_2} \bar{C}_{i_1 i_2 i_3 s}^{\alpha \beta \gamma} C_{i_1 i_2 i_3' s'}^{\alpha \beta \gamma'} = \delta_{\gamma, \gamma'} \delta_{s, s'} \delta_{i_3, i_3'}, \quad (\text{orthonormality})$$

and

$$\sum_{\gamma, i_3, s} \bar{C}_{i_1 i_2 i_3 s}^{\alpha \beta \gamma} C_{i_1' i_2' i_3 s}^{\alpha \beta \gamma} = \delta_{i_1, i_1'} \delta_{i_2, i_2'}. \quad (\text{completeness})$$

Example 3.11. Let us again consider the regular representation of $\mathbb{Z}_2$ from Ex. 3.5. For the tensor product of the regular representation with itself we obtain:

$$(L \otimes L)(0) = \mathbb{1}_4, \quad (3.48)$$

$$(L \otimes L)(1) = \sigma_X \otimes \sigma_X = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (3.49)$$

Consider then the basis transformation which brings $\sigma_X \otimes \sigma_X$ in diagonal form:

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}. \quad (3.50)$$

We then compute that:

$$V((L \otimes L)(0))V^\dagger = V \mathbb{1}_4 V^\dagger = \mathbb{1}_4 \quad (3.51)$$

$$V((L \otimes L)(1))V^\dagger = V(\sigma_X \otimes \sigma_X)V^\dagger = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} \underline{0}(1) & 0 & 0 & 0 \\ 0 & \underline{0}(1) & 0 & 0 \\ 0 & 0 & \underline{1}(1) & 0 \\ 0 & 0 & 0 & \underline{1}(1) \end{pmatrix}. \quad (3.52)$$

We thus find that the tensor product of the regular representation of $\mathbb{Z}_2$ with itself decomposes in (see Ex. 3.8) the trivial representation $\underline{0}$ (with multiplicity 2) and the non-trivial ‘sign representation’ $\underline{1}$ (also with multiplicity 2): $L \otimes L \simeq \underline{0} \oplus \underline{0} \oplus \underline{1} \oplus \underline{1}$. In the notation of above: $N_{LL}^{\underline{0}} = 2$ and $N_{LL}^{\underline{1}} = 2$.

3.4 The Great Orthogonality Theorem

A great result in group theory is the fact that the number of inequivalent irreps of a finite group is finite; let's call $\hat{G}$ the set of equivalence classes of irreps and $\hat{N} = \#(\hat{G})$ the number of inequivalent irreps. Let's also define d_α as the dimension of the irrep α . Of crucial importance is the great orthogonality theorem:

Theorem 3.6 (Great orthogonality theorem). *Consider all $\hat{N}$ inequivalent unitary irreps of a group G $D_{ij}^\alpha(x)$, then*

$$\frac{1}{\#G} \sum_x D_{ij}^\alpha(x) \bar{D}_{kl}^\beta(x) = \frac{1}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl} \quad (3.53)$$

This means that the irreps span a $\sum_{\alpha=1}^{\hat{N}} d_\alpha^2$ dimensional orthonormal basis in a $\#(G)$ dimensional vector space (think of a matrix with $\#(G)$ rows and $\sum_{\alpha=1}^{\hat{N}} d_\alpha^2$ columns).

Corollary 3.6.1. *This theorem immediately implies that $\sum_{\alpha=1}^{\hat{N}} d_\alpha^2 \leq \#(G)$; it will actually turn out (which we will prove using character theory in Sec. 3.5) that this is an equality, such that this matrix is unitary.*

Proof: It is a simple application of Schur's lemma: Let's take any $d_\alpha \times d_\beta$ matrix B and define $Q = \sum_x D^\alpha(x) B D^{\beta\dagger}(x)$. Then we have

$$D^\alpha(y) Q = \sum_x D^\alpha(yx) B D^{\beta\dagger}(x) = \sum_{x'} D^\alpha(x') B D^{\beta\dagger}(y^{-1}x') = Q D^\beta(y) \quad (3.54)$$

and Schur's lemma implies that $Q = \delta_{\alpha\beta} \cdot c_\alpha \cdot \mathbb{1}_{d_\alpha}$. Let's now concentrate on the case of equal irreps $\alpha = \beta$. By taking the trace of Q , we see that $c_\alpha = \text{Tr}(B) \frac{\#(G)}{d_\alpha}$. If we now take $B = |j\rangle\langle l|$, we obtain the theorem. $\square$

The importance in quantum physics of this result cannot be underestimated. It basically says that any set of wavefunctions $\{|\psi_i\rangle\}$ transforming into each other like an irrep D_{ij}^α under a unitary

$$\hat{U}_g |\psi_i\rangle = \sum_j |\psi_j\rangle D_{ji}^\alpha(g) \quad (3.55)$$

will be orthogonal to any other set of wavefunctions transforming according to another irrep β :

$$\langle \chi_i | \psi_j \rangle = \frac{1}{\#G} \sum_g \langle \chi_i | \hat{U}_g^\dagger \hat{U}_g | \psi_j \rangle = \frac{1}{\#G} \sum_g \underbrace{\bar{D}_{ik}^\beta(g) D_{jl}^\alpha(g)}_{=0} \langle \chi_k | \psi_l \rangle \quad (3.56)$$

This result is central in the determination of the degeneracy of energy levels in a Hamiltonian with a certain symmetry.

3.5 Character theory

Now that we have defined representations, tensor products and the notion of irreducible representations, the following natural question arises. Given a representation (for example the tensor product of two irreps) what irreps does it contain? A very elegant formalism to answer this question is by computing the character of the representation, see Def. 3.10 below, and projecting it in a basis of irreducible characters. This procedure will be the topic of this section.

3.5.1 Characters, inner product of characters

Irreps are pretty arbitrary as they are only defined up to equivalence. But their traces are of course independent of the basis in which they are written, and are therefore universal. We will therefore define

Definition 3.10. *The irreducible character of the irrep α , $\chi^{(\alpha)}$, is defined as*

$$\chi^{(\alpha)}(x) = \text{Tr} D^{(\alpha)}(x). \quad (3.57)$$

Or more generally:

Definition 3.11. *The character of any representation Γ , $\chi^{(\Gamma)}$, is defined in the same way as*

$$\chi^{(\Gamma)}(x) = \text{Tr} D^{(\Gamma)}(x). \quad (3.58)$$

It is important to notice that the character is constant in a given conjugacy class (recall Def. 2.10). A function $f : G \mapsto \mathbb{C}$ which only depends on the conjugacy class is sometimes called a **class function**. The character is thus an example of a class function. Let's therefore label the classes as C_k and denote N_k as the number of elements in each class.

Given characters we can define following canonical inner product:

Definition 3.12. *The inner product of characters is defined as*

$$\langle \chi^{(a)} | \chi^{(b)} \rangle := \frac{1}{|G|} \sum_g \bar{\chi}^{(a)}(g) \chi^{(b)}(g). \quad (3.59)$$

You can immediately check that this indeed defines an inner product:

1. $\langle \chi^{(a)} | \chi^{(b)} \rangle = \overline{\langle \chi^{(b)} | \chi^{(a)} \rangle}$
2. Linear in second argument.
3. Positive-definite: $\langle \chi^{(a)} | \chi^{(a)} \rangle > 0$. This follows from the fact that for every representation Γ $\chi^{(\Gamma)}(1) = d_\Gamma > 0$.

Now comes a first important theorem:

Theorem 3.7. *The irreducible characters form an orthonormal basis in the space of group elements:*

$$\frac{1}{\#(G)} \sum_x \chi^{(\alpha)}(x) \bar{\chi}^{(\beta)}(x) = \delta_{\alpha\beta} \quad (3.60)$$

Proof: Set $i = j$ and $k = l$ in the great orthogonality theorem 3.6. □

By virtue of this, we can construct the matrix with orthonormal columns $O_{x\alpha}$ defined via

$$O_{x\alpha} = \sqrt{\frac{1}{\#(G)}} \chi^{(\alpha)}(x) \quad (3.61)$$

Notice the matrix $O_{x\alpha}$ has full rank³ since O is an isometry (but not unitary!): $O^\dagger \cdot O = \mathbb{1}$ because of the result above.

We can also introduce the matrix:

$$\tilde{O}_{k\alpha} = \sqrt{\frac{N_k}{\#(G)}} \chi^{(\alpha)}(C_k). \quad (3.62)$$

³Recall that the row/column rank of a matrix is the dimension of the vector space linearly spanned by the rows/columns. A fundamental result in linear algebra is the fact that the row rank of a matrix equals its column rank. Moreover, $\text{rank } AA^\dagger = \text{rank } A$ and obviously $\text{rank } \mathbb{1}_m = m$.

Because of the same reason, also this matrix is an isometry: $\tilde{O}^\dagger \cdot \tilde{O} = \mathbb{1}$. In components:

$$\frac{1}{\#(G)} \sum_k N_k \bar{\chi}^{(\alpha)}(C_k) \chi^{(\beta)}(C_k) = \delta_{\alpha,\beta}. \quad (3.63)$$

This immediately proves that the number of classes is larger or equal to the number of irreps. Below, in (Example 3.12) we will actually prove that this matrix is unitary and hence, that the number of conjugacy classes is equal to the number of inequivalent irreps.

As a trivial corollary of Theorem 3.7, we get:

Corollary 3.7.1. *Given two irreps with the same characters, then they are equivalent!*

Indeed, if they would not be equivalent, then the characters would have to be orthogonal.

From the definition of the character it is also easy to see that:

Proposition 3.8. *Given representations Γ_1 and Γ_2 :*

- $\chi^{(\Gamma_1 \otimes \Gamma_2)}(g) = \chi^{(\Gamma_1)}(g) \cdot \chi^{(\Gamma_2)}(g),$
- $\chi^{(\Gamma_1 \oplus \Gamma_2)}(g) = \chi^{(\Gamma_1)}(g) + \chi^{(\Gamma_2)}(g).$

Proof: exercise. □

We hence obtain a very simple algorithm for counting the number of times a certain reducible representation Γ contains a given irreducible representation γ_i . As we have seen above, every representation is equivalent to a direct sum of irreps. Therefore we find from Prop. 3.8 that $\chi^{(\Gamma)}(g) = \sum_i n_i \chi^{(\gamma_i)}(g)$ for integers n_i and where the sum is over all the irreps. The multiplicity n_i with which each irrep appears in this decomposition is then because of Th. 3.7 given by the inner product $\langle \chi^{(\Gamma)} | \chi^{(\gamma_i)} \rangle$! This is especially interesting in the case of looking for irreps in the decomposition of tensor products of irreps. We will come back to this particular case below.

From this argument it also immediately follows that:

Theorem 3.9. *A representation Γ is irreducible if and only if*

$$\langle \chi^\Gamma | \chi^\Gamma \rangle = 1. \quad (3.64)$$

3.5.2 A finite group has a finite number of irreps

Example 3.12. *In particular, let's take the (left) regular representation (Ex. 3.4), obtained by associating a permutation matrix to every group element:*

$$L(g) = \sum_h |gh\rangle\langle h|. \quad (3.65)$$

Obviously, its characters are $\chi^L(1) = \#(G)$ for the unit element and zero otherwise. Let's count the number of times that it contains a certain irrep λ :

$$n_\lambda = \langle \chi^\lambda | \chi^L \rangle = \frac{1}{\#(G)} \sum_x \bar{\chi}^{(\lambda)}(x) \chi^L(x) = d_\lambda \quad (3.66)$$

Here d_α is the dimension of the irrep, and obviously $\chi^{(\lambda)}(1) = d_\lambda$. This is very cool: the regular representation contains each d_α dimensional irrep $n_\alpha = d_\alpha$ times! As the dimension of the regular representation is $\#(G)$, this implies the celebrated result

$$\sum_{\alpha=1}^{\hat{N}} d_\alpha^2 = \#(G). \quad (3.67)$$

This example immediately proves that the orthonormal matrix obtained in the great orthogonality theorem is square and hence unitary. We can therefore turn the orthogonality relations around and obtain:

$$\frac{1}{\#G} \sum_{\alpha, i, j} d_{\alpha} D_{ij}^{\alpha}(x) \bar{D}_{ij}^{\alpha}(y) = \delta_{x, y}. \quad (3.68)$$

The same applies to the character matrix of classes vs. irreps (3.62), and one can prove using the concept of the group algebra that **the number of irreps is equal to the number of classes**, and

$$\frac{1}{\#G} \sum_{\alpha=1}^{\hat{N}} \sqrt{N_k N'_k} \chi^{(\alpha)}(C_k) \bar{\chi}^{(\alpha)}(C_{k'}) = \delta_{k, k'}. \quad (3.69)$$

This is probably the most important conclusion of this chapter and this result is quite profound, both in the simplicity of the statement as in the simplicity of the proof. It's so beautiful and noteworthy that we repeat it:

Theorem 3.10. *For a finite group, the number of inequivalent irreps is finite and exactly equal to the number of conjugacy classes.*

3.5.3 Examples and character tables

There are plenty of constraints on the characters such that it is very easy to construct them for small groups. Let's for example consider finding all irreps of S_3 .

Example 3.13. S_3 is non-Abelian, has order 6 and has three conjugacy classes, therefore it must at least have one irrep with dimension larger than 1. There is only 1 possibility: S_3 has 2 1-dimensional irreps (one of which is the trivial irrep) and one two-dimensional one, since $1^2 + 1^2 + 2^2 = 6$. The 3 classes correspond to the identity element e , elements with even parity ($N_e = 2$), and one class with all elements with odd parity ($N_o = 3$). The characters of the trivial irrep have to be all equal to 1. The characters of the second 1-D irrep have to be orthogonal to the trivial one ($1+2a+3b=0$), they have to be normalized ($1+2|a|^2+3|b|^2=6$), and clearly $|a|=|b|=1$ (as the representations are unitary). The only solution of those equations is $a=1, b=-1$. The characters of the two-dimensional irrep are now also completely determined, as $\chi(e)=2$, and all other components are uniquely specified by requiring orthogonality to the characters of the 1-D irreps; we get $a_2 = -1, b_2 = 0$. We therefore get the following table, where the group elements label the rows and the irreps (that we will denote by $\underline{1}, \underline{1}', \underline{2}$, see exercises) the columns, and the prefactor is $\frac{1}{\sqrt{\#(G)}}$, in accordance with (3.61):

$$O = \frac{1}{\sqrt{6}} \begin{matrix} & \underline{1} & \underline{1}' & \underline{2} \\ \begin{matrix} 1 \\ s \\ rs \\ r \\ r^2s \\ r^2 \end{matrix} & \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \end{pmatrix} \end{matrix}. \quad (3.70)$$

As you can readily check, this matrix is an isometry:

$$O^{\dagger} \cdot O = \frac{1}{\sqrt{6}} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 \\ 2 & 0 & 0 & -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.71)$$

To conclude, let us also construct $\tilde{O}$ (3.62):

$$\tilde{O} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 1 & 2 \\ \sqrt{3} & -\sqrt{3} & 0 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} \end{bmatrix}, \quad (3.72)$$

which is unitary:

$$\tilde{O}^\dagger \cdot \tilde{O} = \tilde{O} \cdot \tilde{O}^\dagger = \mathbb{1}_3! \quad (3.73)$$

The power of the theory of characters becomes very clear when considering decomposing the tensor product of a representation $V^{\otimes N}(g)$ into its irreps: the corresponding characters are just $\chi(V(C))^N$, and hence we just have to express those vectors in the basis of characters of the irreps to know the multiplicity of any irrep!

Example 3.14. Recall 3.45. We can now express the multiplicity's in terms of the characters of the involved representations as follows:

$$N_{\alpha\beta}^\gamma = \frac{1}{\#G} \sum_g \bar{\chi}^\gamma(g) \chi^\alpha(g) \chi^\beta(g). \quad (3.74)$$

This is a very strong and useful result!

Example 3.15. Let's again consider S_3 with its 3 irreps of dimension 1, 1, 2. Let's now consider the 2-dimensional irrep $\underline{2}$, and take the tensor power $\underline{2}^{\otimes N}$; the corresponding characters are $\chi_N = \chi_{\underline{2}}^N = [2^N, 0, 0, (-1)^N, 0, (-1)^N]$, and hence we can calculate the multiplicity of all irreps as

$$\frac{1}{\#(G)} X^\dagger \cdot \chi_N = \frac{1}{6} \begin{bmatrix} 2^N + 2(-1)^N \\ 2^N + 2(-1)^N \\ 2^{N+1} + 2(-1)^{N+1} \end{bmatrix}. \quad (3.75)$$

This means that

$$\underline{2}^{\otimes N} \simeq \underline{1}^{\oplus \frac{1}{6}(2^N + 2(-1)^N)} \oplus \underline{1}'^{\oplus \frac{1}{6}(2^N + 2(-1)^N)} \oplus \underline{2}^{\oplus \frac{1}{6}(2^{N+1} + 2(-1)^{N+1})}, \quad (3.76)$$

where $\underline{1}$ stands for the trivial 1-D irrep and $\underline{1}'$ for the non-trivial 1-D irrep. Notice that the dimensions on both sides of the decomposition indeed are equal:

$$2^N = \frac{1}{6} (2^N + 2(-1)^N + 2^N + 2(-1)^N + 2 \times (2^{N+1} + 2(-1)^{N+1})). \quad (3.77)$$

So by now you should be convinced of the fact that the characters of representations play a pivotal role in the representation theory of groups. With applications and exercises in mind we introduce:

Definition 3.13. The character table of a group G is a table that lists all the irreps of a group and the character of every irrep on the conjugacy classes.

Remark. It might be tempting to think that a group is uniquely (up to isomorphism) specified by its character table. This is not true. A counterexample is D_4 vs the quaternion group. These group are not isomorphic but have the same character table.

This rather abstract definition is easily clarified with some small examples:

Example 3.16. $\mathbb{Z}_3$ has as character table

$$\begin{array}{c|ccc} & \{0\} & \{1\} & \{2\} \\ \hline \underline{1} & 1 & 1 & 1 \\ \underline{\omega} & 1 & \omega & \omega^2 \\ \underline{\omega^2} & 1 & \omega^2 & \omega \end{array}, \quad (3.78)$$

where $\omega = \exp(\frac{2\pi i}{3})$, $\omega^3 = 1$.

Example 3.17. S_3 has as character table:

$$\begin{array}{c|ccc}
 & \{()\} & \{(123), (132)\} & \{(12), (13), (23)\} \\
 \hline
 \underline{1} & 1 & 1 & 1 \\
 \underline{1'} & 1 & 1 & -1 \\
 \underline{2} & 2 & -1 & 0
 \end{array} \quad . \quad (3.79)$$

Notice that indeed $2 \times 1^2 + 2^2 = 6$, the order of the group.

Example 3.18. The quaternion group (see exercises for a definition) has as character table:

$$\begin{array}{c|ccccc}
 & \{1\} & \{-1\} & \{\pm i\} & \{\pm j\} & \{\pm k\} \\
 \hline
 \underline{1} & 1 & 1 & 1 & 1 & 1 \\
 \underline{1'} & 1 & 1 & 1 & -1 & -1 \\
 \underline{1''} & 1 & 1 & -1 & 1 & -1 \\
 \underline{1'''} & 1 & 1 & -1 & -1 & 1 \\
 \underline{2} & 2 & -2 & 0 & 0 & 0
 \end{array} \quad . \quad (3.80)$$

Can you infer from the character table the order of this group?

Remark. As a last remark, consider the tensor product of two copies of the same irrep; then we can divide the irreps of the combined system into the symmetric ones and antisymmetric ones (as the corresponding subspaces clearly have to transform into themselves under the symmetry operations). The wavefunctions in the symmetric subspace

$$|\psi_{ij}\rangle^+ = |\psi_i\rangle|\psi_j\rangle + |\psi_j\rangle|\psi_i\rangle \quad (3.81)$$

transform as

$$\sum_{ij} U_{i'i} U_{j'j} (|\psi_i\rangle|\psi_j\rangle + |\psi_j\rangle|\psi_i\rangle) = \frac{1}{2} \sum_{ij} (U_{i'i} U_{j'j} + U_{i'j} U_{j'i}) |\psi_{ij}\rangle^+ \quad (3.82)$$

and hence the characters on the symmetric subspace are

$$\chi^+(g) = \frac{1}{2} (\chi(g)^2 + \chi(g^2)) \quad (3.83)$$

Similarly, for the anti-symmetric subspace we get

$$\chi^-(g) = \frac{1}{2} (\chi(g)^2 - \chi(g^2)) \quad (3.84)$$

3.6 Real, complex and quaternionic representations, the Frobenius-Schur indicator

We will now revisit the **(complex) conjugate representation** first encountered in Ex. 3.7. Given an irreducible matrix representation of G , $\{U(g)|g \in G\}$, we define the conjugate representation as $\{\bar{U}(g)|g \in G\}$. It can readily be checked that this again forms a representation of G . Although the precise form of $\bar{U}$ depends on the basis chosen, it can be shown that its equivalence class does not. Also notice that if a representation is irreducible its conjugate is also irreducible.

A representation U is called **self-conjugate** iff $\bar{U}$ is equivalent to U , i.e. if there exists a unitary X s.t. $\bar{U}(g) = XU(g)X^{-1} = XU(g)X^\dagger$. If such an X does not exist, we call the representation **complex**. Even if a representation is self-conjugate, there might be an obstruction in making the representation matrices real. The two ensuing cases are called **real** and **quaternionic**. Let us investigate this in a bit more detail.

3.6. REAL, COMPLEX AND QUATERNIONIC REPRESENTATIONS, THE FROBENIUS-SCHUR INDICATOR

If a representation is self-conjugate, there exists by definition a unitary X such that $\bar{U}(g) = XU(g)X^{-1} = XU(g)X^\dagger$. Taking the conjugate of this equation, we get

$$U(g) = \bar{X}XU(g)(\bar{X}X)^{-1} \quad (3.85)$$

and hence $\bar{X}X = c \cdot \mathbb{1}$ because of Schur's lemma. c has to be real (as the trace of $\bar{X}X$ is real), and thus $X = cX^T$. Taking the transpose of $X = cX^T$ a second time, we get $c^2 = 1$ or $c = \pm 1$. This is the so-called **Frobenius-Schur indicator** and distinguishes the two cases at hand! Let us indeed prove that $c = 1$ implies that we can always make the irrep real, while for $c = -1$, we can bring the representation in the form

$$\bar{U}(g) = (\epsilon \otimes \mathbb{1})U(g)(\epsilon^\dagger \otimes \mathbb{1}) \quad (3.86)$$

with

$$\epsilon = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = i\sigma_Y. \quad (3.87)$$

Of course we could as well use $\epsilon = \sigma_Y$ or any tensor product of σ_Y 's as they are all equivalent to each other. Let's prove this:

$c = +1$ In this case, $X = X^T$, and therefore the Takagi decomposition (singular value decomposition for complex symmetric matrices) assures that we can write X as $X = QQ^T$ for some unitary matrix Q . We want to find a similarity transformation Y such that $\bar{Y}U(g)Y^{-1} = YU(g)Y^{-1}$ or $\bar{Y}XU(g)X^{-1}\bar{Y}^{-1} = YU(g)Y^{-1}$. By choosing $Y = Q^T$, we obtain exactly what we wanted.

$c = -1$ For any unitary $X = -X^T$, the diagonal elements have to vanish and therefore Householder transformations of the kind WXW^T bring this $n \times n$ matrix in $n-1$ steps to an antisymmetric matrix whose only non-zero elements are one up and one down the diagonal ⁴ But such a matrix can only be unitary if every second matrix element vanishes. We can therefore write $X = W(\epsilon \otimes D)W^T$ for ϵ defined above and D a diagonal matrix ⁵. We can now absorb the square roots of D into W to obtain $X = W(\epsilon \otimes \mathbb{1})W^T$. We can now proceed as in the first case and choose the similarity transformation $Y = W^T$, and we obtain

$$\overline{W^T U(g) W} = (\epsilon \otimes \mathbb{1}) W^T U(g) \bar{W} (-\epsilon \otimes \mathbb{1})$$

which finishes the proof. $\square$

There is a simple test based on characters to check whether an irrep is real, complex or quaternionic:

Theorem 3.11. *Let $\chi(x)$ be the characters of an irrep U ; then*

$$\frac{1}{\#(G)} \sum_x \chi(x^2) = \begin{cases} 1 & \text{real} \\ 0 & \text{complex} \\ -1 & \text{quaternionic} \end{cases} \quad (3.88)$$

Proof: It is a simple application of the orthogonality theorem.

$$\frac{1}{\#(G)} \sum_x \chi(x^2) = \frac{1}{\#(G)} \sum_{ijx} U_{ij}(x) U_{ji}(x) = \frac{1}{\#(G)} \sum_{ijx} U_{ij}(x) \bar{U}_{ji}(x)$$

When U is real, then there always exists a choice such that $U(x) = \bar{U}(x)$, and then the above sum reduces to $1/d_i \sum_{ij} \delta_{ij} \delta_{ji} = 1$. When U is complex, then $\bar{U}$ is by definition an inequivalent

⁴More specifically, take a Householder transformation which acts trivially on the first row and nontrivial on the second to last rows; then it maps the first column to zero except the second element. But as the whole matrix is still unitary, the other elements of the second row have all to be zero.

⁵By a simple variational argument, it can be proven that any (also non-unitary) matrix of even dimensions satisfying $X = -X^T$ can be written in the form $X = W(\epsilon \otimes D)W^T$ with W unitary and D nonnegative diagonal.

representation hence the sum over x always gives zero. Finally, when U is quaternionic, we have $U = X^\dagger \bar{U} X$ with $\bar{X} X = -I$; hence the above sum reduces to

$$\frac{1}{\#G} \sum_x \text{Tr} \left(X^\dagger \bar{U}(x) X \bar{U}(x) \right) = \frac{1}{d_\alpha} \sum_{ij} \bar{X}_{ij} X_{ji} = \frac{1}{d_\alpha} \text{Tr} (\bar{X} X) = -1$$

□

Remark. The quantity (3.88) is sometimes called the **second Schur indicator**. Analogously, the k -th Schur indicator can be defined via

$$\frac{1}{\#(G)} \sum_x \chi(x^k), \quad k = 0, 1, 2, \dots \quad (3.89)$$

The zeroth Schur indicator is for example simply equal to 1.

When is the first Schur indicator non-zero?

Example 3.19. Let's check this for the quaternion group; this group has order 8, and has irreps of dimension $(1, 1, 1, 1, 2)$. Recall its character table 3.80; it can readily be checked for the unique two-dimensional irrep that $\sum_x \chi^2(x^2) = 2 \times 2 + 6 \times (-2) = -8$, so this irrep is quaternionic. Hence the name 'quaternionic' representation.

3.7 Projective Representations

One of the most important characteristics of quantum mechanics is the fact that the central object of study, the **wave function** $|\psi\rangle$, is no physical observable. The wave function only encodes a classical (!) probability distribution $P(x) = \bar{\psi}(x)\psi(x)$ ($\psi(x) := \langle x|\psi\rangle$) from which you can sample, and appears in matrix elements of observables as $\langle \psi|\hat{O}|\psi\rangle$. Since the probability distribution and matrix elements contain always the wave function and its conjugate this means that the wave function is only determined up to a global phase, in other words $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ are physically indistinguishable and encode the same physically accessible information. From a symmetry point of view this observation has a drastic consequence. Imagine your wave function (which is a vector in a let's say finite-dimensional Hilbert space) transforms under a symmetry group G as:

$$|\psi\rangle \mapsto U(g) |\psi\rangle. \quad (3.90)$$

Then according to our reasoning above, it shouldn't matter if the wave function transforms according to $U(g)$ or $e^{i\varphi(g)}U(g)$! Going one step further, under the action of two symmetry transformations, we should allow the $U(g)$'s to multiply according to

$$U(g)U(h) = e^{i\omega(g,h)}U(gh). \quad (3.91)$$

Clearly, these $U(g)$ do not form a (linear) representation of G anymore, but we shall call it a **projective representation** if the phase ω is non-trivial in a sense defined below. In order for the projective representation to still have a matrix representation, a necessary condition is that the $U(g)$ multiply associatively. This immediately leads to the following 2-cocycle condition on the phases ω :

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \omega(h, k), \quad \text{mod } 2\pi. \quad (3.92)$$

Under changing $U(g) \mapsto e^{i\varphi(g)}U(g)$, the cocycle changes according to

$$\omega(g, h) \mapsto \omega(g, h) - \varphi(gh) + \varphi(g) + \varphi(h), \quad (3.93)$$

as you should check.

Choosing the phases $e^{i\varphi(g)}U(g)$ is often called choosing a **gauge** of the projective representation, and via 3.93 of the 2-cocycle. In group cohomology parlance (Appendix 2.B) $\varphi(gh) - \varphi(g) -$

$\varphi(h)$ is called a **coboundary**. In other words, fixing the gauge of the cocycle amounts to adding a coboundary of your choosing.

In this case the solutions of the 2-cocycle equations modulo coboundaries are classified by the second cohomology group $H^2(G, U(1))$, first introduced in the context of the classification of central extensions in 2.4.2, but now appearing in a totally different application. By now you should be convinced of the importance of group cohomology.

Definition 3.14. A **projective representation** of a group G is a collection of matrices X_g multiplying according to

$$X_g X_h = e^{i\omega(g,h)} X_{gh}, \quad (3.94)$$

where ω is a representative of a non-trivial cohomology class $[\omega] \in H^2(G, U(1))$.

A different motivation for the study of projective representations is the following. If a representation of a group is of the form $U(g) \otimes \bar{U}(g)$; then this does not imply that the $U(g)$'s themselves form a representation. Indeed, it is sufficient that the $U(g)$'s form a projective representation, as in that case we have that

$$(U(g) \otimes \bar{U}(g)) (U(h) \otimes \bar{U}(h)) = e^{i\omega(g,h)} U(gh) \otimes e^{-i\omega(g,h)} \bar{U}(gh) = U(gh) \otimes \bar{U}(gh). \quad (3.95)$$

At this point you might think that the existence and appearance of projective representations is merely a mathematical curiosity, but this is far from the truth. Projective representations appear more often than you could think and their existence has drastic physical consequences ranging from the existence of half integer electron spin⁶ to the existence of new so called symmetry protected phases of matter.

Example 3.20. The smallest group with a non-trivial 2-cocycle over $U(1)$ is $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, as it can be computed that $H^2(\mathbb{Z}_2 \oplus \mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$. A representative of the non-trivial class $[\omega]$ in $H^2(\mathbb{Z}_2 \oplus \mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$ is given by

$$e^{i\omega(g,h)} = \begin{matrix} & \begin{matrix} (0,0) & (1,0) & (0,1) & (1,1) \end{matrix} \\ \begin{matrix} (0,0) \\ (1,0) \\ (0,1) \\ (1,1) \end{matrix} & \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \end{matrix}, \quad (3.96)$$

where g labels the rows. A projective representation ρ corresponding to this cocycle reads

$$\rho(0,0) = \mathbb{1}, \quad \rho(0,1) = i\sigma_Y, \quad \rho(1,0) = -\sigma_X, \quad \rho(1,1) = \sigma_Z. \quad (3.97)$$

Eg.:

$$\rho(0,1)\rho(1,0) = i\sigma_Y(-\sigma_X) = -i(-i\sigma_Z) = -\rho(1,1), \quad (3.98)$$

$$\rho(1,1)\rho(0,1) = \sigma_Z(i\sigma_Y) = i(-i\sigma_X) = -\rho(1,0), \quad (3.99)$$

in accordance with $e^{i\omega((0,1),(1,0))} = e^{i\omega((1,1),(0,1))} = -1$.

3.7.1 Constructing projective representations

Given a non-trivial cocycle, a unitary projective representation corresponding to it can readily be constructed by starting from the regular representation $R(g) = \sum_h |gh\rangle\langle h|$, and replacing this by

$$R_\omega(g) = \sum_h e^{i\omega(g,h)} |gh\rangle\langle h|. \quad (3.100)$$

⁶Later we will see this is a consequence of the existence of projective representations of the rotation group $SO(3)$.

We will call this the **projective regular representation** corresponding to ω .

By using the 2-cocycle condition, it can be checked that this indeed gives rise to the relation

$$R_\omega(g)R_\omega(h) = \sum_{x,y} e^{i(\omega(g,x)+\omega(h,y))} |gx\rangle\langle x|hy\rangle\langle y| \quad (3.101)$$

$$= \sum_y e^{i(\omega(g,hy)+\omega(h,y))} |ghy\rangle\langle y| \quad (3.102)$$

$$= \sum_y e^{i(\omega(gh,y)+\omega(g,h))} |ghy\rangle\langle y| \quad (3.103)$$

$$=: e^{i\omega(g,h)} R_\omega(gh) \quad (3.104)$$

It can be shown that $R_\omega(1)$ can be represented by the identity matrix. This is not completely obvious and proving this requires some properties of cocycles:

Proposition 3.12. *A normalized 2-cocycle is a 2-cocycle satisfying $\omega(g, h) = 0$ whenever g or h (or both) are the identity. Every 2-cocycle is (equivalent to) a normalized one.*

Proof: Notice that $\omega(g, 1) = \omega(h, 1)$ for all g, h . This follows from the cocycle equation $\omega(g, g'1) + \omega(g', 1) = \omega(g, g') + \omega(gg', 1)$, hence $\omega(g', 1) = \omega(gg', 1)$. Since for every g, h , there exists a g' such that $gg' = h$, we have indeed the aforementioned property. In particular $\omega(g, 1) = \omega(1, 1)$. Similarly $\omega(1, g) = \omega(1, 1)$.

Now consider the gauge transformation: $\omega(1, 1) \mapsto \omega'(1, 1) = \omega(1, 1) - \varphi(1 \cdot 1) + \varphi(1) + \varphi(1) = \omega(1, 1) - \varphi(1)$. Choosing $\varphi(1) = -\omega(1, 1)$ makes $\omega'(1, 1)$ equal to zero, and thus $\omega'(g, 1) = 0$ for all g . $\square$

Remark. *The previous result is more generally true for n -cocycles, even with a group action.*

Proposition 3.13. *We have that $\omega(g, g^{-1}) = \omega(g^{-1}, g)$ for all $g \in G$, and moreover we can choose a gauge such that $\omega(g, g^{-1}) = \omega(g^{-1}, g) = 0$.*

Proof: Assume (without loss of generality) that the cocycle is normalized: $\omega(g, 1) = \omega(1, g) = 0 \forall g$. We then find from the 2-cocycle equation that $\omega(g, g^{-1}) + \omega(1, g) = \omega(g, 1) + \omega(g^{-1}, g) \implies \omega(g, g^{-1}) = \omega(g^{-1}, g)$.

Now consider adding the coboundary $\omega(g, g^{-1}) \mapsto \omega(g, g^{-1}) - \varphi(1) + \varphi(g) + \varphi(g^{-1})$. Choosing $\varphi(g) = -\frac{1}{2}\omega(g, g^{-1}) = \varphi(g^{-1})$ does the job. $\square$

Corollary 3.13.1. *Note that $R_\omega(1)$ can always be chosen equal to the identity matrix; this follows from Prop. 3.12 and Prop. 3.13.*

Interestingly, a projective representation $\{X_g | g \in G\}$ cannot have 1-dimensional subspaces if ω is a nontrivial cocycle; otherwise, call $|v\rangle$ this invariant subspace, then we have $X_g|v\rangle = e^{i\alpha(g)}|v\rangle$, and hence

$$X_g X_h |v\rangle = e^{i(\alpha(g)+\alpha(h))} |v\rangle \quad (3.105)$$

$$= e^{i\omega(g,h)} X_{gh} |v\rangle \quad (3.106)$$

$$= e^{i\omega(g,h)} e^{i\alpha(gh)} |v\rangle \quad (3.107)$$

and hence $\omega(g, h) = \alpha(g) + \alpha(h) - \alpha(gh)$ which is a trivial coboundary and hence a contradiction.

Actually, all invariant subspaces of the regular projective representation come in multiples equal to their dimensions, just as in the usual case of irreps being repeated in the regular representation, and as 1-d subspaces are excluded, the only groups having nontrivial cocycles must be of the order $\sum_i d_i^2$ with $d_i \geq 2$. This is a corollary of the analogue of the great orthogonality theorem for projective representations:

- Invariant subspaces in projective representations have to span their whole subspace, as they are unitary (cfr. Schur's lemma 1); similarly, Schur's lemma 2 is also true

- The orthogonality theorem can be exactly taken over when we take two irreps corresponding to the same cohomology class $[\omega]$:

$$\begin{aligned}
 X^{(\alpha)}(h) \underbrace{\frac{1}{\#(G)} \sum_{g \in G} X^{(\alpha)}(g) B X^{(\beta)\dagger}(g)}_X &= \frac{1}{\#(G)} \sum_{g \in G} e^{i(\omega(h,g) - \omega(h,g))} X^{(\alpha)}(hg) B X^{(\beta)\dagger}(hg) X^{(\beta)}(h) \\
 &= X X^{(\beta)}(h)
 \end{aligned} \tag{3.108}$$

and hence Schur dictates that $B = c \cdot \mathbb{1}$ with $c = \text{Tr}(B)$, from which the usual orthogonality relations follow.

- The concept of classes also generalizes: if $\text{Tr}(X(x)) = \text{Tr}(X(y)X(x)X(y^{-1}))$, and hence the orthogonality theorem for characters is still in place
- The only non-zero character in the projective regular representation constructed above again corresponds to the identity element, and the multiplicity of each irrep will exactly be its dimension!

3.A The group algebra

It turns out to be useful to consider the algebra $A(G)$ of the matrix representations of a group, i.e. making linear combinations of the matrices obtained from the regular representation.

A natural object is the sum of all those matrices in a given class: $R(C) = \sum_{g \in C} V(g)$. Those matrices form a closed algebra: $R(C_i) \cdot R(C_j) = \sum_k f_{ij}^k R(C_k)$; we are indeed summing over all elements in the different classes. Furthermore, the matrices $R(C_i)$ all commute with each other and actually with all group elements $V(g)$: they represent the center of the group algebra! This means that they can be simultaneously diagonalized⁷. Given the U which diagonalizes all $R(C_i)$ (which is not unique due to the multiplicities of the eigenvalues), it will also block-diagonalize the $R(g)$: the invariant subspaces of the matrices $R(C_i)$ exactly correspond to the blocks obtained by going to the irrep basis (modulo the fact that the unitary cannot resolve the direct sum structure in the multiplicities of the given irreps: indeed, the invariant subspaces have dimension d_i^2 , as each irrep is represented a number of times equal to its dimension).

A projection in the group algebra is an element $R : R^2 = R$. A central projection is a projection in the center of the algebra $A(G)$, and is minimal if it cannot be written as a sum of other projections. The number of minimal central projections is exactly equal to the number of classes, and those are all linear combinations of the $R(C)$ (which all commute!) defined above.

For example, let's take $G = S_3$. There are 3 classes (with 1, 2 and 3 elements), and the corresponding $R(C)$ can easily be constructed. We can now construct the central projections as linear combination of those, and a little algebra gives the minimal central projections

$$P_0 = \frac{1}{6} (R(1) + R(2) + R(3)) \quad \text{1-dimensional} \quad (3.109)$$

$$P_1 = \frac{1}{6} (R(1) + R(2) - R(3)) \quad \text{1-dimensional} \quad (3.110)$$

$$P_2 = \frac{2}{3} R(1) - \frac{1}{3} R(2) \quad \text{4-dimensional} \quad (3.111)$$

Those obviously define the space on which the irreps will live, and the weights coincide with the characters of the different irreps multiplied with their dimension.

All of those statements can easily be proven by writing the minimal projections as

$$P_\alpha = \frac{d_\alpha}{\#G} \sum_g (\chi^\alpha(g))^* V(g) \quad (3.112)$$

for α an irrep with corresponding characters $\chi^\alpha(g)$ and dimension d_α and $V(g)$ the regular representation. Indeed, we know that the regular representation is, up to a basis change, equal to

$$V(g) \simeq \oplus_\beta \oplus_{d_\beta} D^\beta(g) \quad (3.113)$$

and the orthogonality theorem gives us

$$\frac{1}{\#G} \sum_i \sum_g \bar{D}_{ii}^\alpha(g) D_{kl}^\beta(g) = \frac{1}{d_\alpha} \sum_i \delta_{\alpha\beta} \delta_{ki} \delta_{li} = \frac{1}{d_\alpha} \delta_{\alpha\beta} \delta_{kl} \quad (3.114)$$

which proves that P_α as defined above is the $d_\alpha^2 \times d_\alpha^2$ identity matrix in the basis of irrep α .

Given any representation of a group $\{A(g)\}$, the group algebra generated from it has a dimension upper bounded by the number of group elements. Let's assume that the space is spanned by the matrices of a few group elements $\{\mu : \mu \in F \subseteq G\}$. Then all group elements can be written as

⁷The easiest way to do this in practice is to take a random linear combination of the $R(C_i)$ and diagonalize this using the eigenvalue decomposition; the different invariant subspaces then correspond to the eigenvectors with equal eigenvalue.

$A(h) = \sum_{\mu \in F} \psi_{\mu}(h) A(\mu)$, and the transformation rule is

$$A(kh) = \sum_{\mu \in F} \psi_{\mu}(kh) A(\mu) \quad (3.115)$$

$$= A(k) \cdot A(h) \quad (3.116)$$

$$= \sum_{\mu, \nu, \nu' \in F} \psi_{\nu}(k) \psi_{\nu'}(h) \psi_{\mu}(\nu \nu') A(\mu) \quad (3.117)$$

and hence we get the equation

$$\psi_{\mu}(kh) = \sum_{\nu'} \underbrace{\left(\sum_{\nu} \psi_{\mu}(\nu \nu') \psi_{\nu}(k) \right)}_{D_{\mu \nu'}(k)} \psi_{\nu'}(h) \quad (3.118)$$

$D_{\mu \nu}(x)$ clearly forms a representation of the group G with matrix dimension equal to the number of generators.

This is very much related to how transformations work in quantum mechanics. Let us e.g. take a set of wavefunctions $\psi_{\nu}(x, y, z) = \langle x, y, z | \psi_{\nu} \rangle$, and consider how it transforms under $SO(3)$. Then

$$\langle \vec{x} | \hat{U}(g) | \psi_{\nu} \rangle = \psi_{\nu}(O^{-1}(g) \vec{x}) \quad (3.119)$$

$$= \sum_{\mu} D_{\nu \mu}(g) \langle \vec{x} | \psi_{\mu} \rangle \quad (3.120)$$

where we enlarged the set such that we can express the first line as a linear combination of any wavefunction in the set. Note that the inverse is obtained from the fact that $\hat{U}$ works in on the bra as opposed to the ket.

Take for example the set of single particle wavefunctions $\psi_1(x, y, z) = x \exp(-r)$, $\psi_2(x, y, z) = y \exp(-r)$ both defined in an infinite dimensional Hilbert space, and consider rotations and reflections along the z -axis. In this case, both wavefunctions will transform into a linear combination of each other. Indeed, we have $\langle \vec{x} | U(g) | \psi \rangle = \psi(O^{-1}(g) \vec{x})$ with

$$O(g) = \begin{bmatrix} \pm 1 & 0 & 0 \\ 0 & \pm 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.121)$$

and it can readily be checked that the equivalent representation for the given set of wavefunctions is

$$D_{\mu \nu}(g) = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{bmatrix} \quad (3.122)$$

In physics, it is of primordial importance to decompose those $D_{\mu \nu}$ into its irreducible components, as this allows smaller sets of wavefunctions which form a closed vector space under the group multiplication. The different labels that arise like that will exactly correspond to the quantum numbers.

3.B Representation theory of the symmetric group: Young tableaux

Every group with n elements is a subgroup of the permutation group S_n with cardinality $n!$; this is known as Cayley's theorem, and is pretty obvious. Finding irreps of the permutation group is of particular interest in physics, as it plays a central role in e.g. understanding the properties of a global symmetry $U^{\otimes n}$, in the problem of optimal estimation/cloning of a density matrix $\rho^{\otimes N}$ etc.

Every permutation of n objects is a direct product of disjoint cycles (including 1-cycles)

$$\pi = (i_{11}, i_{12}, \dots, i_{1j_1}) (i_{21}, i_{22}, \dots, i_{2j_2}) \dots (i_{m1}, i_{m2}, \dots, i_{mj_m})$$

with $\sum_{k=1}^m j_k = n$. The number of such cycles m is preserved under conjugation, and it is clear that any two permutations with the same sizes of cycles are equivalent: two permutations π and π' are conjugate iff they have the same number k_j of j -cycles. The conjugacy classes are labeled by the integers $k_1, k_2, \dots, k_n$ such that $\sum_k j k_j = n$. The number $p(n)$ of partitions $\{k_j\}$ satisfying this condition asymptotically grows like $\exp\left(\pi\sqrt{\frac{2n}{3}}\right)$, and the number of elements in a certain class is given by $n! / \left(\prod_j (k_j)! j^{k_j}\right)$.

We can represent those partitions using Young frames F , defined as rows with $m_1 \geq m_2 \geq \dots$ ($\sum_i m_i = n$) boxes stacked on top of each other:

Clearly, the number of different Young frames is the number of classes of S_n . We define a Young tableau a Young frame with the numbers $1..n$ written in them, and the Young tableau T is called standard iff the numbers increase in each row and column:

1	3	4	7	8
2	5	9		
6				

The number of standard tableaux for a given frame F is denoted $f(F)$; this function will turn out to be equal to the dimension of the irrep associated to the Young frame, and hence satisfies

$$\sum_F f(F)^2 = n!$$

$f(F)$ can be calculated exactly, and is given by

$$f(F) = n! / \prod (\text{Hook Lengths})$$

where the hook length of a box (a, b) is the total number of boxes such that $(a' \leq a, b' = b)$ or $(a' = a, b' \leq b)$. For example, the hook lengths of the following Young frame are:

6	4	3	1
4	2	1	
1			

and hence its corresponding $f(F) = 8! / (6.4.3.4.2) = 70$. Those formula's can be proved using cumbersome but straightforward combinatorial reasonings.

Now comes an amazing fact for S_n : there is an explicit formula for constructing all irreps, and what is more, all those irreps only contain integer elements! The way it works is as follows: take a particular standard Young tableau T (e.g by filling in the numbers from $1..n$, starting from left to right and then top to bottom). Now we will define two sets of group elements which keep some structure in this particular Young frame

$$R(T) = \{\sigma \in S_n | \sigma \text{ preserves the set of numbers in each row}\} \quad (3.123)$$

$$C(T) = \{\sigma \in S_n | \sigma \text{ preserves the set of numbers in each column}\} \quad (3.124)$$

These group elements can easily be labeled by a product of simple cycles. We now construct the following matrices in the group algebra of S_n w.r.t. the regular representation (and where we attach a parity $\tilde{p}$ to every group element p):

$$P(T) = \sum_{p \in R(T)} R(p) \quad (3.125)$$

$$Q(T) = \sum_{q \in C(T)} (-1)^{\tilde{q}} R(q) \quad (3.126)$$

$$E(T) = P(T) \cdot Q(T) \quad (3.127)$$

Amazingly, $E(T)$ is idempotent (up to a constant factor): $E(T)^2 = c(T) \cdot E(T)$. Furthermore, if T and T' correspond to different Young tables, then $E(T) \cdot E(T') = 0$. And if T and T' correspond to 2 different standard Young tableaux (but to the same Young table, just a different standard ordering), then also $E(T) \cdot E(T') = 0$. This means that we will be able to associate a different Young frame to every different irrep, and a different standard Young tableau to every copy of the irrep in the regular representation (note that every irrep arises a number of times equal to its dimension).

The irreps are now constructed as follows:

1. $E(T)$ is an element of the group algebra, and for all $R(g)$, we can construct the matrices $Y(g) = R(g) \cdot E(T)$. Clearly, $R(h) \cdot Y(g) = Y(h \cdot g)$, and hence the algebra of matrices is closed under left multiplication with $R(h)$.
2. The dimension of this ensuing "vector" space (consisting of linear combinations of matrices) will be equal to the dimension of the irrep. We can choose any linear independent basis $\hat{X}_i$ for this vector space.
3. We now want to see how any vector $x \rightarrow \sum_i x_i \hat{X}_i$ in this space transforms under left multiplication with the regular representation $R(g)$; we get

$$R(g) \cdot \sum_i x_i \hat{X}_i = \sum_i \left(\sum_j A_{i,j}^g x_j \right) \hat{X}_i$$

4. A_{ij}^g is an irrep of the symmetric group, and if we choose the linear independent basis $\hat{X}_i$ equal to some elements $Y(g)$, then the coefficients in this irrep are all integer (note that this will typically not be a unitary irrep!).

Proving this is pretty cumbersome, and can be found in any textbook on representation theory.

Let us give the example of S_3 , which has four possible standard Young tableaux:

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 \\ \hline \end{array} \quad \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array}$$

Let us label our group elements as $1 = (123) \rightarrow (123)$; $2 = (123) \rightarrow (213)$; $3 = (1, 2, 3) \rightarrow (3, 2, 1)$; $4 = (1, 2, 3) \rightarrow (1, 3, 2)$; $5 = (1, 2, 3) \rightarrow (2, 3, 1)$; $6 = (1, 2, 3) \rightarrow (3, 1, 2)$; the Cayley table is

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 5 & 6 & 3 & 4 \\ 3 & 6 & 1 & 5 & 4 & 2 \\ 4 & 5 & 6 & 1 & 2 & 3 \\ 5 & 4 & 2 & 3 & 6 & 1 \\ 6 & 3 & 4 & 2 & 1 & 5 \end{bmatrix}$$

Let us first consider the first standard Young tableau, and the regular representation $R(g)$. Then, $P(T) = \sum_g R(g)$, $Q(T) = R(1)$, $E(T) = P(T)$, and it can easily be checked that $E(T)$ is a rank 1 6×6 matrix with all entries equal to 1. Let us now consider the matrices $Y(g) = R(g) \cdot E(T) = E(T)$; as they are all equal to each other, this corresponds to the trivial irrep mapping all group elements to 1.

Let us next consider the last standard Young tableau. Then, $P(T) = R(1)$, $Q(T) = R(1) - R(2) - R(3) - R(4) + R(5) + R(6)$, $E(T) = Q(T)$, and it can easily be checked that $E(T)$ is a rank 1 6×6 matrix with all entries equal to ± 1 . We now get $R(g)E(T) = (-1)^{\tilde{g}}E(T)$ with $\tilde{g}$ the parity of g , and hence we again get a 1-D irrep corresponding to the parity of the permutation.

Let us now consider the tableau $\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$; we have $R(T) = \{1, 2\}$ and $C(T) = \{1, 3\}$, and hence $P(T) = R(1) + R(2)$; $Q(T) = R(1) - R(3)$; $E(T) = R(1) + R(2) - R(3) - R(5)$. $E(T)$ turns out to have rank 2. If we now consider the algebra $R(g)E(T)$, then we see that it is completely spanned by $R(1)E(T)$ and $R(3)E(T)$, which means that any $R(g)E(T)$ can be expressed as a linear combination of those two. Taking $\hat{X}_1 = R(1)E(T)$, $\hat{X}_2 = R(3)E(T)$, we now calculate A_{ij}^g as explained above and get

$$\begin{aligned} A^1 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & A^2 &= \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} & A^3 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ A^4 &= \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} & A^5 &= \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} & A^6 &= \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \end{aligned}$$

Finally, let us look at the tableau $\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$. The discussion parallels completely the previous one, except that $E(T)$ is orthogonal to the one of $\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$.

It is hence clear that all irreps can be constructed starting from standard Young tableaux, and that the working horse to prove this is the concept of group algebra's.

It happens that the 2-dimensional representation obtained is the so-called standard representation of S_3 . In general, the standard representation is obtained by starting from $n \times n$ matrices $R(g) = I(:, \text{permutation}(g))$ with the permutation of the form $[2, 3, 1, \dots]$. By projecting those matrices on the subspace $I - \text{ones}(n, n)$, we obtain a $n - 1 \times n - 1$ irrep!

Frobenius has worked out an explicit formula for all characters of the obtained irreps; his proof is very complicated but can be deduced in a much easier way by looking at continuous groups defined in the next section.

3.C Induced Representations

In the first part of these lectures, we went to great length discussing subgroups and also how subgroups H could be extended to a larger group G . From the point of representation theory, a related question is to “lift” a representation of a subgroup H to a representation of the full group G . This is used extensively in the context of band structure calculations in solid state physics.

If H is a subgroup of G , and given a matrix representation (π, V) of H , we can construct a representation of G by first taking one representative element $\tilde{g}_\alpha$ out of every left coset of G/H . Then any element in G can uniquely be represented as $g = \tilde{g}_\alpha \cdot h_i$. This follows from the fact that the cosets of a group form a partition of that group. Or in other words, every group element $g \in G$ is contained in exactly one coset $\tilde{g}_\alpha H$. Given a group element $g \in G$ we can then write with some abuse of notation

$$g \cdot \tilde{g}_\alpha \cdot = \tilde{g}_{\phi_1(g, \alpha)} \cdot h_{\phi_2(g, \alpha)} \quad (3.128)$$

for certain functions $\phi_1 : G \times G/H \rightarrow G/H$ and $\phi_2 : G \times G/H \rightarrow H$.

Given a representation (π, V) for H , the induced representation denoted by $(\text{Ind}_H^G(\pi), W)$ or simply $\text{Ind}_H^G(\pi)$ then has as underlying representation space the direct sum space

$$W = \bigoplus_{\alpha} g_{\alpha} V \quad (3.129)$$

where the direct sum is over all the (representatives of the) cosets G/H , and every direct summand $g_{\alpha} V$ is a copy of V for which the elements are (formally) written as $g_{\alpha} v$, $v \in V$. The complex dimension of W is thus $|G/H| \cdot \dim_{\mathbb{C}}(V) = |G|/|H| \cdot \dim_{\mathbb{C}}(V)$. The action of a given group element $g \in G$ on a vector $\sum_{\alpha} \tilde{g}_{\alpha} v_{\alpha}$ (every vector in W is of this form) is then given by:

$$g : W \mapsto W : \sum_{\alpha} \tilde{g}_{\alpha} v_{\alpha} \mapsto \sum_{\alpha} \tilde{g}_{\phi_1(g, \alpha)} \pi(h_{\phi_2(g, \alpha)}) v_{\alpha} \quad (3.130)$$

Example 3.21. *The induced representation of the (trivial representation of the) trivial subgroup $\{e\}$ of G is just the left regular representation of G .*

Example 3.22. *More generally, the induced representation of the trivial representation of any subgroup $H \leq G$ is the permutation representation on the cosets. Indeed, according to (3.130) we get for $(\mathbb{C}, \underline{0}_H)$, the trivial representation of H , that $g \in G$ acts on $W = \mathbb{C}[G/H] = \bigoplus_{\alpha \in G/H} \mathbb{C}$ by permuting the direct summands in $\bigoplus_{\alpha \in G/H} \mathbb{C}$ which correspond to the cosets G/H .*

3.D Application in physics: selection rules

The first and most successful application of group theory to quantum mechanics was discovered by Wigner and involves the prediction of selection rules for atomic transitions between eigenstates of the Hamiltonian labelled by the irrep according to which they transform. We are concerned with calculating matrix elements as occurring in Fermi's golden rule

$$\langle l_3, m_3 | \hat{O}_{m_2}^{l_2} | l_1, m_1 \rangle \quad (3.131)$$

where l_i labels the irrep according to which the state transforms and m the precise label within that subspace. The states thus transform as $|l, m\rangle \mapsto U(g)|l, m\rangle = \sum_{m'} D^l(g)_{m'm} |l, m'\rangle$ and we assumed here that $\hat{O}$ transforms according to the irrep D^{l_2} , i.e. $U(g) \hat{O}_{m_2}^{l_2} U^{\dagger}(g) = \sum_{m'_2} D_{m'_2 m_2}^{l_2}(g) \hat{O}_{m'_2}^{l_2}$, then the expectation value of interest can only be nonzero iff $D^{l_1} \otimes D^{l_2}$ contains the irrep l_3 , or equivalently if $D^{l_1} \otimes D^{l_2} \otimes (D^{l_3})^*$ contains the trivial irrep. Indeed, by adding resolutions of the identity, we get

$$\langle l_3, m_3 | \hat{O}_{m_2}^{l_2} | l_1, m_1 \rangle = \langle l_3, m_3 | U^{\dagger}(g) U(g) \hat{O}_{m_2}^{l_2} U^{\dagger}(g) U(g) | l_1, m_1 \rangle \quad (3.132)$$

$$= \sum_{m'_1 m'_2 m'_3} D_{m'_1 m_1}^{l_1}(g) D_{m'_2 m_2}^{l_2}(g) \bar{D}_{m'_3 m_3}^{l_3}(g) \langle l_3, m'_3 | \hat{O}_{m'_2}^{l_2} | l_1, m'_1 \rangle \quad (3.133)$$

We can now take the sum of the right hand side over all group elements and use the Great Orthogonality Theorem

$$\frac{1}{\#G} \sum_g D_{m'_1 m_1}^{l_1}(g) D_{m'_2 m_2}^{l_2}(g) \bar{D}_{m'_3 m_3}^{l_3}(g) \quad (3.134)$$

$$= \frac{1}{\#G} \sum_g \sum_{l_4 \alpha m'_4 m'_4} \langle l_4, m'_4, \alpha | l_1 m'_1; l_2 m'_2 \rangle \langle l_1 m_1; l_2 m_2 | l_4, m_4, \alpha \rangle D_{m'_4 m_4}^{l_4}(g) \bar{D}_{m'_3 m_3}^{l_3}(g) \quad (3.135)$$

$$= \sum_{l_4 \alpha m'_4 m'_4} \langle l_4, m'_4, \alpha | l_1 m'_1; l_2 m'_2 \rangle \langle l_1 m_1; l_2 m_2 | l_4, m_4, \alpha \rangle d_{l_3} \delta_{l_4, l_3} \delta_{m_4, m_3} \delta_{m'_4, m'_3} \quad (3.136)$$

where we used the Clebch Gordan notation $|l_1 m_1; l_2 m_2\rangle = \sum_{l_4 m_4 \alpha} |l_4 m_4 \alpha\rangle \langle l_4 m_4 \alpha | l_1 m_1; l_2 m_2\rangle$ and α labelling the multiplicity index of l_4 in $l_1 \otimes l_2$. We hence get

$$\langle l_3, m_3 | \hat{O}_{m_2}^{l_2} | l_1, m_1 \rangle = \sum_{\alpha} \langle l_1 m_1; l_2 m_2 | l_3, m_3, \alpha \rangle \underbrace{\sum_{m'_1 m'_2 m'_3} \langle l_3, m'_3, \alpha | l_1 m'_1; l_2 m'_2 \rangle \langle l_3, m'_3 | \hat{O}_{m'_2}^{l_2} | l_1, m'_1 \rangle}_{\langle l_3, \alpha | |O^{l_2}| | l_1 \rangle} \quad (3.137)$$

$$= \sum_{\alpha} \langle l_1 m_1; l_2 m_2 | l_3, m_3, \alpha \rangle \langle l_3, \alpha | |O^{l_2}| | l_1 \rangle \quad (3.138)$$

where we defined the reduced element $\langle l_3, \alpha | |O^{l_2}| | l_1 \rangle$. Although we have proven this in generality, this expression is most known in the context of representations of $SO(3)$ and is then called the Wigner-Eckart theorem: in that case, there are no degeneracies and hence no α . From a practical point of view, this means that we only have to calculate the matrix element $\langle l_3, m_3 | \hat{O}_{m_2}^{l_2} | l_1, m_1 \rangle$ for one particular instance (e.g. $\langle l_3, 0 | \hat{O}_0^{l_2} | l_1, 0 \rangle$) and then we know the values of all other matrix elements.

Those selection rules are of enormous importance in spectroscopy and perturbation theory. In the former case, it controls which transitions are possible (e.g. electric dipole transition rules). In the latter case, it controls which degeneracies will be lifted by a perturbation (e.g. spin orbit coupling or Zeeman splitting).

In the case of expectation values of the form $|\psi\rangle = \sum_m c_m |l, m\rangle$

$$\langle \psi | \hat{O} | \psi \rangle \quad (3.139)$$

there is a further symmetry which is important, namely the way in which $\hat{O}$ transforms under time reversal (complex conjugation) $O \rightarrow \bar{O}$. if $\hat{O}$ is invariant under conjugation, we have that

$$\langle lm | \hat{O}^j | lm' \rangle = \langle lm' | \hat{O}^j | lm \rangle \quad (3.140)$$

and hence only the symmetric part of $D^l \otimes (D^l)^*$ can contribute; hence we get an extra selection rule dictating that the irrep D^j must be contained in the symmetric irreps of $D^l \otimes (D^l)^*$. We can of course repeat the same when $\hat{O}$ is antisymmetric with respect to time reversal, for which the irrep should be D^j should be contained in the antisymmetric irreps of $D^l \otimes (D^l)^*$.

3.E Application in physics: Kramers' theorem

The concept of complex conjugation plays a fundamental role in describing topological insulators and systems of fermions in general. Indeed, if we consider a Hamiltonian which is invariant under time-reversal (which includes conjugation and reversal of magnetic fields), and the wavefunctions under consideration transform nontrivially under conjugation, then this will automatically lead to a degeneracy of the spectrum. This was first observed by Kramer in the case of systems with an odd number of electrons and time-reversal invariant nonrelativistic Schrödinger Hamiltonian (i.e. without magnetic fields).

To present Kramers argument, we first have to understand how time reversal is implemented on a system of fermions. As the momentum operator $p = i\partial/\partial x$ is odd under time reversal, so must be the spin 1/2 operator associated to the spins of the electrons: $T\sigma_\alpha = -(\sigma_\alpha)^* T$. It can be shown that the time reversal operator must be of the form $T = UK$ with K complex conjugation in some basis, which we take the Pauli Z basis, and U a unitary. We have

$$T\sigma_\alpha = UK\sigma_\alpha = U(\sigma_\alpha)^* U^\dagger T \quad (3.141)$$

The only operator U which has this property is σ_y , hence the time reversal operator for n electrons is

$$\hat{T} = \left(\otimes_i \sigma_y^{(i)} \right) \cdot \hat{K} \quad (3.142)$$

An odd number n of electrons transform nontrivially under conjugation:

$$T^2|\psi\rangle = \left(\otimes_i \sigma_y^{(i)}\right) \cdot \hat{K} \cdot \left(\otimes_i \sigma_y^{(i)}\right) \cdot \hat{K}|\psi\rangle = (-1)^n |\psi\rangle \quad (3.143)$$

as $\sigma_y^* = -\sigma_y$ and $K^2 = \mathbb{1}$. Hence $T|\psi\rangle$ must be linearly independent of $|\psi\rangle$: if it were an eigenstate with eigenvalue λ , then $T^2|\psi\rangle = |\lambda|^2|\psi\rangle$ and hence a contradiction as $|\lambda|^2$ cannot be -1 . This is called Kramers theorem: in the absence of a magnetic field, every energy level of a system with an odd number of electrons has an even degeneracy. The fact that this has to be even is because $T^2 = -\mathbb{1}$, so for every eigenstate $|\psi\rangle$ there is a partner $T|\psi\rangle$.

More generally, assume that our system has a larger symmetry group (such as a point group symmetry) represented by the unitaries $\{U_g\}$ acting on the Hilbert space, and let's assume that those unitaries are invariant under time reversal (which is true for all rotations and point group symmetries). Let us focus on a subset $\{|\psi_i\rangle\}$ which transform under an irrep of the group $\hat{U}_g|\psi_i\rangle = \sum_j |\psi_j\rangle D_{ji}(g)$. We would like to see for which irreps time-reversal symmetry gives rise to extra degeneracies, in other words, we would like to find under which conditions the set $\{T|\psi_i\rangle\}$ is linearly independent of $\{|\psi_i\rangle\}$. This will depend on whether the irrep is real, complex or quaternionic, and also on the parity of the number of electrons.

To proceed, let us make some preliminary remarks. U_g commutes with T , and we have the equation

$$\hat{U}_g T|\psi_i\rangle = T \hat{U}_g |\psi_i\rangle = T(|\psi_j\rangle D_{ji}(g)) = \bar{D}_{ji}(g) (T|\psi_j\rangle) \quad (3.144)$$

Note that in the last equality we made use of $T(\sum_i x_i |\psi_i\rangle) = \sum_i \bar{x}_i T|\psi_i\rangle$. The anti-unitary operation maps any vector $|\psi\rangle$ in the Hilbert space to another one, so the notation $T|\psi\rangle = |T.\psi\rangle$ is well defined. Anti-unitarity implies

$$\langle \psi_a | \psi_b \rangle = \langle T.\psi_b | T.\psi_a \rangle \quad (3.145)$$

without a sign as this is about vectors in the Hilbert space. Let's next consider the inner product $\langle \psi_i | T|\psi_j \rangle$ where T is acting on the right and make use of the previous equality:

$$\langle \psi_i | T|\psi_j \rangle = \langle \psi_i | T\psi_j \rangle = \langle T^2\psi_j | T.\psi_i \rangle = \pm \langle \psi_j | T|\psi_i \rangle \quad (3.146)$$

with $+1$ for the n even case (hence symmetric) and -1 for the odd case (hence antisymmetric). Finally, let us consider a given irrep α and the corresponding set of vectors $\{|\psi_a\rangle\}$ which transform under α :

$$\langle \psi_a | T|\psi_b \rangle = \frac{1}{\#G} \sum_g \langle \psi_a | U_g^\dagger T U_g |\psi_b \rangle \quad (3.147)$$

$$= \sum_{cd} \frac{1}{\#G} \underbrace{\sum_g \bar{D}_{ca}(g) \bar{D}_{db}(g)}_{Q_{abcd}} \langle \psi_c | T|\psi_d \rangle \quad (3.148)$$

Let us now consider the three cases.

- The complex case is easiest: because of the great orthogonality theorem, $Q_{abcd} = 0$ and hence the set of vectors $\{T|\psi_a\rangle\}$ are all linearly independent of the set $\{|\psi_i\rangle\}$. This means that we will always get a degeneracy, independent of the even or odd case.
- Let's next consider the real case; then $Q_{abcd} = \delta_{ab}\delta_{cd}/d_\alpha$ and we get

$$\langle \psi_a | T|\psi_b \rangle = \frac{\delta_{ab}}{d_\alpha} \sum_c \langle \psi_c | T|\psi_c \rangle$$

In the case of an even number of spins, we cannot deduce anything from this, so no extra degeneracy will emerge. However, Kramers' argument proved that $\langle \psi_c | T|\psi_c \rangle = 0$ whenever we have an odd number of spins, and hence a degeneracy will emerge in the spectrum due to time reversal.

- The quaternionic case is opposite: there a degeneracy will emerge in the even case, and no degeneracy in the odd case. We make use of the fact that the matrix X for which $\bar{D} = XDX^{-1}$ is antisymmetric:

$$\sum_g \bar{D}_{ca}(g) \bar{D}_{db}(g) = \sum_{g b' d'} \bar{D}_{ca}(g) X_{dd'} D_{d' b'}(g) X_{b' b} \quad (3.149)$$

$$= \#G/d_\alpha \sum_{b' d'} \delta_{cd'} \delta_{ab'} X_{dd'} X_{b' b} \quad (3.150)$$

$$= \#G/d_\alpha X_{dc} X_{ab} \quad (3.151)$$

We can now fill this in:

$$\langle \psi_a | T | \psi_b \rangle = X_{ab} \cdot \frac{1}{d_\alpha} \sum_{cd} X_{dc} \langle \psi_c | T | \psi_d \rangle \quad (3.152)$$

The clue is the fact that X_{dc} is antisymmetric, and $\langle \psi_c | T | \psi_d \rangle$ symmetric or antisymmetric depending on even or odd number of electrons. In the even case, we get the trace of a symmetric with an antisymmetric matrix, which is clearly zero: $\langle \psi_a | T | \psi_b \rangle = 0$ and we hence get a two-fold degeneracy. In the odd case, we get the trace of 2 antisymmetric matrices and as such there is no reason for expecting extra degeneracies.

Probably the most amazing application of Kramer's theorem is in the context of superconductors. In the normal BCS theory, you need perfect translational invariance etc such as to make sure that momentum is a good quantum number. Then you can form Cooper pairs from electrons with quantum numbers $\vec{k}, \uparrow$ and $\vec{k}, \downarrow$. However, a real superconductor is dirty and breaks translational invariance. Anderson realized that Kramers theorem provides the exact reason why pairing is still exactly possible in that case: just pair the electron with its time reversed counter part which has exactly the same energy, independent of the random fields around as long as they respect time reversal. From the moment a time reversal term is added, superconductivity breaks down, which happens e.g. when adding magnetic impurities. Also, the argument only works for s-wave superconductors; others are very susceptible to disorder.

Part II

Lie groups and Lie algebras

Chapter 4

The rotation groups $SU(2)$ and $SO(3)$

In previous courses you already encountered many physical systems with a rotational symmetry. Think for example of a particle in a spherical symmetric gravitational potential or of the hydrogen atom. The goal of this chapter is to give a very hands on approach to the groups $SO(3)$ and $SU(2)$ which describe rotations in three-dimensional space. Contrary to the groups encountered in previous sections (with the exception of $U(1)$) these groups have an continuous infinite number of group elements, since there are an infinite (continuous) number of rotations one can carry out. Both are examples of **Lie groups**, which will be studied in more generality in the next chapter.

We will define and introduce these groups in great detail and compare the subtle differences between the two. We then do the full representation theory of the Lie algebra $\mathfrak{su}(2)$, or rather $\mathfrak{sl}(2)$, and connect it to the representation theory of $SO(3)$ and $SU(2)$. The Lie algebra $\mathfrak{sl}(2)$ has a very concrete physical interpretation as it is used to describe the angular momentum operators in quantum mechanics. This will be discussed in some detail in the exercises and put to use in many of future courses. Finally, spherical harmonics will be discussed. These also appear all over the place in electromagnetism, quantum mechanics and atomic - and molecular physics.

This chapter also serves as an introduction to the next chapter in which we will study Lie algebras and Lie groups from a more formal point of view. Many of the important concepts we will introduce there will already appear naturally in this chapter.

Let us summarize some of the most important conclusions of this chapter.

Synopsis $SO(3)$ is the symmetry group of rotations in three-dimensional space. The **special unitary group** $SU(2)$ plays a very important role in quantum physics, where it is understood to give rise to the concept of spin. ‘Small’ symmetry transformations in $SO(3)$ and $SU(2)$ are described by a **Lie algebra** which turns out to be isomorphic for both groups: $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$. The **irreducible representations** of $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ are labeled by an integer or half integer $l \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ and the irrep labeled by l is $2l + 1$ dimensional. All irreps of $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$ can be ‘lifted’ via the **exponential map** to be irreps of $SU(2)$. Only those corresponding to integer l will correspond to irreps of $SO(3)$, those corresponding to half integer l will turn out to be **projective representations** of $SO(3)$.

Remark. *Throughout this chapter and the next, we will often apply the theorems previously proven in the context of finite groups and their representation theory to infinite Lie groups. Most of the times the proof from the finite case immediately generalizes to the infinite case.*

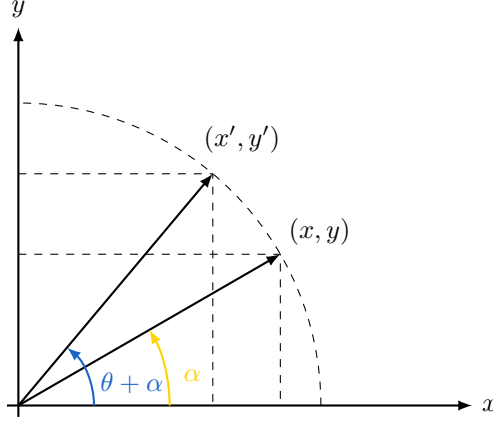


Figure 4.1: A rotation in the plane.

4.1 Warming up with $SO(2)$ and $U(1)$

In this section we explore the description of rotations in the 2D plane using rotation matrices. These rotation matrices constitute a group, $SO(2)$ which is shown to be isomorphic to the group $U(1)$ encountered previously in Def. 2.7. We also derive their irreducible representations and give some physical interpretation. The groups $SO(2)$ and $U(1)$ are the first and simplest examples of matrix **Lie groups**.

4.1.1 Rotations in the plane and group theory

Consider a point in the 2D plane with polar coordinates

$$\begin{cases} x &= r \cos \alpha, \\ y &= r \sin \alpha. \end{cases} \quad (4.1)$$

The angle α is measured counterclockwise from the horizontal x-axis (Fig. 4.1).

A counterclockwise rotation with an angle θ (Fig. 4.1) brings this point to a new point with coordinates (x', y') . We can write

$$\begin{cases} x' &= r \cos(\alpha + \theta) = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta, \\ y' &= r \sin(\alpha + \theta) = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta. \end{cases} \quad (4.2)$$

Or in terms of the old coordinates:

$$\begin{cases} x' &= \cos \theta x - \sin \theta y, \\ y' &= \sin \theta x + \cos \theta y. \end{cases} \quad (4.3)$$

In matrix notation:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (4.4)$$

For obvious reasons, the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (4.5)$$

is called a **rotation matrix**.

Remark. Notice that this we took an **active** point of view which means that it is understood that the vector (x, y) is really geometrically moved to a new position with coordinates (x', y') in the

fixed coordinate frame. This action is implemented by acting with $R(\theta)$ on the coordinate vector of the point, namely

$$\begin{cases} x \\ y \end{cases} \rightarrow \begin{cases} x' = \cos \theta x - \sin \theta y, \\ y' = \sin \theta x + \cos \theta y. \end{cases} \quad (4.6)$$

while keeping the basis vectors

$$\{e_x, e_y\} \equiv \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad (4.7)$$

fixed. A **passive** transformation on the other hand, keeps the vector put and rotates the basis according to

$$\{e_x, e_y\} \rightarrow \{R(\theta)^{-1}e_x, R(\theta)^{-1}e_y\}. \quad (4.8)$$

However, as you can readily check the coordinates of the vector in the rotated frame (passive point of view) are the same as the coordinates of the rotated vector (active point of view) in the fixed coordinate frame. Indeed:

$$v = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = R(\theta)^{-1} \begin{pmatrix} x' \\ y' \end{pmatrix} = x' R(\theta)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y' R(\theta)^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (4.9)$$

The collection of all matrices

$$\{R(\theta) | \theta \in [0, 2\pi)\} \quad (4.10)$$

form a group with group multiplication the matrix multiplication. Let us check this explicitly:

- The identity obviously corresponds to $R(\theta = 0) = \mathbb{1}_2$. Geometrically this means that a ‘rotation’ with an angle of 0 is equivalent to not rotating at all. Makes sense.
- For every rotation $R(\theta)$ there is an inverse rotation $R(2\pi - \theta) = R(-\theta)$:

$$R(\theta)R(2\pi - \theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \quad (4.11)$$

$$= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = R(2\pi - \theta)R(\theta) = \mathbb{1}_2. \quad (4.12)$$

Notice that this is equivalent to saying that the rotation matrix $R(\theta)$ is **orthogonal**: $R(\theta)^{-1} = R(\theta)^T = R(2\pi - \theta)$.

- The matrix multiplication is by construction associative.

Notice also that $\det R(\theta) = \cos^2 \theta - (-\sin \theta) \sin \theta = 1$. This leads to the following definition:

Definition 4.1. The **2D rotation group** also called the **special orthogonal group** $SO(2)$ is the matrix **Lie group**

$$SO(2) := \left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} | \theta \in [0, 2\pi) \right\}. \quad (4.13)$$

Remark. Notice that we have defined $SO(2)$ in terms of a 2D representation. We will see below that this representation is reducible. Many Lie groups, in fact almost all of them and certainly those most relevant in physics, are exactly defined in this way: through a representation. This representation is then called the **defining representation**. When we will study the more general theory of Lie groups in the next chapter, we will restrict to this class of Lie groups that can be defined via a matrix representation: the so called **linear Lie groups**.

An completely equivalent definition is:

Definition 4.2. The **2D rotation group** also called the **special orthogonal group** $SO(2)$ is the matrix **Lie group**

$$SO(2) := \{O \in \mathbb{R}^{2 \times 2} | OO^T = O^T O = \mathbb{1}, \det O = 1\}. \quad (4.14)$$

Here, ‘orthogonal’ refers to the fact that $O^T = O^{-1}$ means that O is an orthogonal matrix, ‘special’ refers to the fact that the determinant of all matrices in $SO(2)$ have determinant one.

Notice that $\det O = 1$ is not necessary for the orthogonal matrices to form a group. In fact, $SO(2)$ is a proper subgroup of the group $O(2)$:

Definition 4.3. *The orthogonal group $O(2)$ is the matrix Lie group*

$$O(2) := \{O \in \mathbb{R}^{2 \times 2} | OO^T = O^T O = \mathbb{1}\}. \quad (4.15)$$

By virtue of the properties $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$, all matrices in $O(2)$ have determinant ± 1 . Those with determinant $+1$ obviously form the subgroup $SO(2) < O(2)$. While the matrices with determinant -1 are of the form

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (4.16)$$

The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ corresponds to a reflection around the y-axis:

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix} \quad (4.17)$$

Another way of giving meaning to the group $O(2)$ is by considering the Hilbert space $\mathbb{R}^2$ endowed with the canonical inner product:

$$\langle \vec{v}_1 | \vec{v}_2 \rangle_{\mathbb{R}^2} := \sum_{i=1}^2 v_1^i v_2^i. \quad (4.18)$$

The linear operators, matrices in $O(2)$ are then exactly those that preserve this inner product:

$$\langle O\vec{v}_1 | O\vec{v}_2 \rangle_{\mathbb{R}^2} = \langle \vec{v}_1 | \vec{v}_2 \rangle_{\mathbb{R}^2}, \quad \forall v_1, v_2 \in \mathbb{R}^2 \iff O \in O(2). \quad (4.19)$$

The reasons that the definitions 4.1 and 4.2 are equivalent is because all 2×2 orthogonal matrices with unit determinant can be written in the form of a rotation matrix $R(\theta)$ (exercise).

Intuitively it is clear that $SO(2)$ is **abelian**: first rotating with an angle θ_1 followed by a rotation with an angle θ_2 is the same as first doing a rotation with an angle θ_2 and then with an angle θ_1 . You can simply work this out:

$$\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix} = \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix} \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \quad (4.20)$$

$$= \begin{pmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{pmatrix}. \quad (4.21)$$

Another way to conclude that $SO(2)$ is abelian is as follows. Every point in the plane can also be written as a complex number:

$$(x, y) \leftrightarrow z = x + iy = re^{i\alpha}. \quad (4.22)$$

Here $r = \sqrt{x^2 + y^2}$ and $\alpha = \text{atan}\left(\frac{y}{x}\right)$. The rotation with an angle θ from above then simply amounts to multiplying z with the complex phase $e^{i\theta}$:

$$z' = x' + iy' = re^{i(\alpha+\theta)}. \quad (4.23)$$

Obviously, $e^{i\theta_1}e^{i\theta_2} = e^{i\theta_2}e^{i\theta_1}$. Recall that the complex phases form an abelian group under the group multiplication the multiplication of complex numbers, $U(1)$ (Def. 2.7), this suggests the following theorem:

Theorem 4.1. *The rotation group $SO(2)$ is isomorphic to $U(1)$: $SO(2) \simeq U(1)$.*

PROOF: : The isomorphism is given by

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in SO(2) \rightarrow e^{i\theta} \in U(1), \quad \forall \theta \in [0, 2\pi). \quad (4.24)$$

□

It is interesting to see that the group $O(2)$ is not Abelian; this will of course have a big influence on its representations.

4.1.2 Irreducible representations $U(1)$ and $SO(2)$

Let us compute the irreps of $U(1) \simeq SO(2)$.

From Schur's lemma it follows that the irreducible representations of an abelian group are one-dimensional. Moreover all irreps are unitary. In other words, we are looking for homomorphisms $\varphi : U(1) \rightarrow U(1)$. Since $\varphi(1) \stackrel{!}{=} 1$ this means that such homomorphisms are of the form

$$\varphi_n(e^{i\theta}) = e^{in\theta}, \quad n \in \mathbb{Z}. \quad (4.25)$$

That these representations are irreducible and inequivalent can simply be proven with a generalization of character theory for Lie groups (next chapter).

Every irrep is thus labeled by an integer $n \in \mathbb{Z}$. This is in stark contrast with the finite group case where we only had a finite number of irreps equal to the number of conjugacy classes of the group.

In physics literature it is usual to refer to the integer labeling an irrep of $U(1)$ as the (electric) **charge**. You are probably aware that electric charge conservation is a fundamental feature of nature. Due to Noether's theorem (Sec. 1.1) it can be related to the invariance of physical laws under a mathematical transformation called $U(1)$ **gauge transformations**. These transformations involves multiplying charged fields by $U(1)$ phase factors $e^{i\theta(x)}$, where $\theta(x)$ is a function of the spacetime coordinates x . Exactly this principle lies at the heart of gauge theories which are a main topic of study in theoretical particle physics and courses on quantum field theory.

Example 4.1. *Recall that $SO(2)$ was defined in terms of a representation! Namely, every element of $SO(2)$ corresponds to a rotation matrix*

$$\begin{pmatrix} e^{in\theta} & 0 \\ 0 & e^{-in\theta} \end{pmatrix}. \quad (4.26)$$

We can construct other reducible (!) representations of $SO(2)$ as follows

$$\begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix} \simeq \begin{pmatrix} e^{in\theta} & 0 \\ 0 & e^{-in\theta} \end{pmatrix}. \quad (4.27)$$

The similarity transformation is induced by the matrix

$$\begin{pmatrix} 1/\sqrt{2} & i/\sqrt{2} \\ i/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \quad (4.28)$$

which, as should be, does not depend on θ .

The representation theory of $O(2)$ is very similar. But as the group is non-Abelian, some irreps must have a dimension larger than 1. It is a nice exercise to see that there are 2 1-dimensional irreps (the trivial one $R_g = 1$, and another one $R_g = \det(O)$), and all other irreps are labeled by an element of $\mathbb{Z}_2$ and $n \in \mathbb{N}_0$, and are 2-dimensional:

$$\begin{pmatrix} 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \cdot \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix} \quad (4.29)$$

4.2 Definitions and parametrizations of $SO(3)$ and $SU(2)$

In this section we will define the matrix Lie groups $(S)O(3)$ and $(S)U(2)$ and provide different parametrizations of the group elements. This way lay the foundation for all subsequent sections.

4.2.1 Definitions

First some definitions:

Definition 4.4. *The orthogonal group $O(3)$ is the group of real 3×3 orthogonal matrices:*

$$O(3) := \{M \in \mathbb{R}^{3 \times 3} | MM^T = M^T M = \mathbb{1}\}. \quad (4.30)$$

From the fact that $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$, it follows that all matrices in $O(3)$ have determinant ± 1 . This means in particular that all matrices of $O(3)$ with determinant 1 form a subgroup. This subgroup of $O(3)$ is exactly $SO(3)$:

Definition 4.5. *The 3D rotation group or special orthogonal group $SO(3)$ is the group of real 3×3 orthogonal matrices with unit determinant:*

$$SO(3) := \{M \in \mathbb{R}^{3 \times 3} | MM^T = M^T M = \mathbb{1}, \det M = 1\} < O(3). \quad (4.31)$$

Obviously, the group $SO(3)$ is the 3D analogue of the group $SO(2)$ from the previous section and describes all 3D rotations. In the same spirit $O(3)$ describes all rotations plus reflections.

Definition 4.6. *The unitary group $U(2)$ is the group of complex unitary 2×2 matrices:*

$$U(2) := \{U \in \mathbb{C}^{2 \times 2} | UU^\dagger = U^\dagger U = \mathbb{1}\}. \quad (4.32)$$

Similarly, by virtue of $\det(AB) = \det(A)\det(B)$ and $\det(A^\dagger) = \overline{\det(A)}$ the determinant of every matrix in $U(2)$ is a phase $e^{i\varphi}$. So, we can again restrict $U(2)$ to the subgroup of unitary matrices with unit determinant:

Definition 4.7. *The special unitary group $SU(2)$ is the group of complex unitary 2×2 matrices with unit determinant:*

$$SU(2) := \{U \in \mathbb{C}^{2 \times 2} | UU^\dagger = U^\dagger U = \mathbb{1}, \det U = 1\} < U(2). \quad (4.33)$$

Another way of giving meaning to the groups $O(3)$ and $U(2)$ is by considering the Hilbert spaces $\mathbb{R}^3$ and $\mathbb{C}^2$. They both can be endowed with a canonical inner product:

$$\langle \vec{v}_1 | \vec{v}_2 \rangle_{\mathbb{R}^3} := \sum_{i=1}^3 v_1^i v_2^i, \quad (4.34)$$

$$\langle \vec{w}_1 | \vec{w}_2 \rangle_{\mathbb{C}^2} := \sum_{i=1}^2 \bar{w}_1^i w_2^i. \quad (4.35)$$

The linear operators, matrices in $O(3)$ and $U(2)$ are then exactly those that preserve these inner products:

$$\langle O\vec{v}_1 | O\vec{v}_2 \rangle_{\mathbb{R}^3} = \langle \vec{v}_1 | \vec{v}_2 \rangle_{\mathbb{R}^3}, \quad \forall v_1, v_2 \in \mathbb{R}^3 \iff O \in O(3), \quad (4.36)$$

$$\langle U\vec{w}_1 | U\vec{w}_2 \rangle_{\mathbb{C}^2} = \langle \vec{w}_1 | \vec{w}_2 \rangle_{\mathbb{C}^2}, \quad \forall w_1, w_2 \in \mathbb{C}^2 \iff U \in U(2). \quad (4.37)$$

4.2.2 Parametrizations

In this subsection we will construct different parametrizations of the matrices contained in the groups $SO(3)$ and $SU(2)$. Depending on the context and application, some of these parametrizations will be more convenient than others.

Parametrizations $SO(3)$

The most obvious way to describe rotations in 3-dimensional space is by considering every rotation as subsequent rotations around the (fixed) x-, y- and z-axis. Hence:

$$R(\theta_x, \theta_y, \theta_z) = R_z(\theta_z)R_y(\theta_y)R_x(\theta_x), \quad (4.38)$$

where

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}, \quad (4.39)$$

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}, \quad (4.40)$$

$$R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.41)$$

Notice that the order $x - y - z$ matters! Rotations around different axis don't commute!

Another way to construct 3D rotation matrices is by realizing that every rotation in 3D leaves one axis fixed, this is the **axis of rotation**. Every rotation can thus be described by an axis of rotation defined via a unit vector $\vec{n}$ and an angle $\theta \in [0, \pi]$ around that axis. By convention a positive rotation is given by the right-hand rule.¹ We will denote this rotation by $R_{\vec{n}}(\theta)$. The rotation matrices from above (4.39-4.41) are for example passive rotations corresponding to rotations around the axes $\vec{e}_x, \vec{e}_y, \vec{e}_z$ respectively, with $\vec{e}_i$ the coordinate axis in the i direction.² Notice that $R_{\vec{n}}(\pi)$ and $R_{-\vec{n}}(\pi)$ describe the same rotation, this will be of crucial observation in Sec. 4.4.

Rotations in 3-dimensional space can also be parametrized in terms of **Euler angles**. There are a gazillion number of different conventions to parametrize these angles, so we won't even bother trying to discuss different conventions and just pick one for illustrative purposes. The underlying idea of Euler angles is however relatively simple: we parametrize a rotation by subsequently rotating around the z-axis, the (new) x-axis and finally the (new) z-axis. This procedure is illustrated in Fig. 4.2. Note that what is depicted in this figure is a **passive** coordinate transformation: the rotation is interpreted as a rotation of the coordinate axis, rather than the actual rotation of a point. In this parametrization the rotation matrix describing this rotation is then given by:

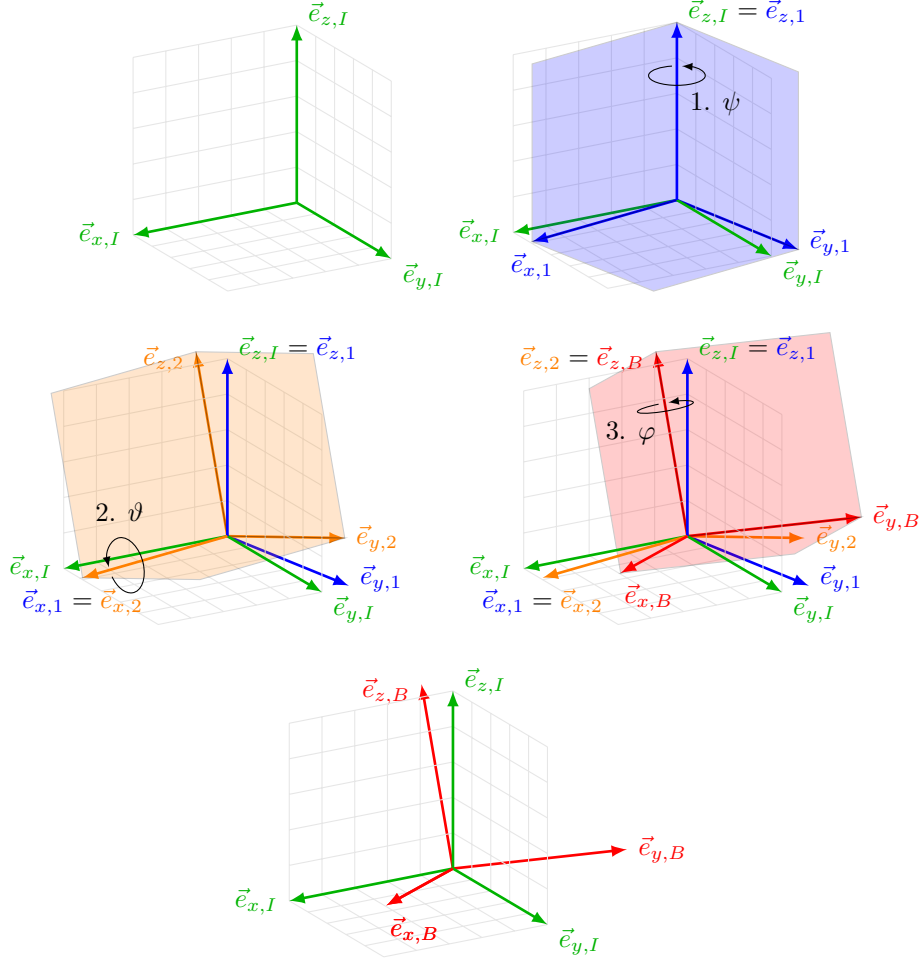
$$R(\psi, \theta, \varphi) = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.42)$$

$$= \begin{pmatrix} \cos \varphi \cos \psi - \cos \theta \sin \varphi \sin \psi & -\cos \varphi \sin \psi - \cos \theta \sin \varphi \cos \psi & \sin \varphi \sin \theta \\ \sin \varphi \cos \psi + \cos \theta \cos \varphi \sin \psi & -\sin \varphi \sin \psi + \cos \theta \cos \varphi \cos \psi & -\cos \varphi \sin \theta \\ \sin \psi \sin \theta & \cos \psi \sin \theta & \cos \theta \end{pmatrix}, \quad (4.43)$$

$$\psi \in [0, 2\pi), \theta \in [0, \pi), \varphi \in [0, 2\pi). \quad (4.44)$$

¹Concretely, if your right thumb points along the axis of rotation, the curl of your fingers is in the direction of a positive rotation.

²Notice the difference in the sign in the off-diagonal terms in $R_y(\theta)$ vs. $R_x(\theta)$ and $R_z(\theta)$. This is explained by the right-hand rule.

Figure 4.2: Euler angles at work, taken from [here](#).

Parametrizations $SU(2)$

The first parametrization is:³

$$U = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (4.45)$$

Writing $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ with α_i and β_i real allows us to also write every matrix in $SU(2)$ in terms of the by now familiar Pauli matrices:

$$\begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} = \alpha_1 \mathbb{1}_2 + i\beta_2 \sigma_X + i\beta_1 \sigma_Y + i\alpha_2 \sigma_Z, \quad \alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}, \quad (4.46)$$

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.47)$$

Another way to write the $SU(2)$ matrices in terms of the Pauli matrices is by using the matrix exponential (see Vector and Function spaces and Def. 4.9 below). If we write $t_i := \frac{\sigma_i}{2}$ and $\sigma_{\vec{n}} := n_x t_x + n_y t_y + n_z t_z$, $\vec{n}$ being a real unit vector. We can show that every matrix in $SU(2)$

³See exercise session.

can be written in the form:⁴

$$R_{\vec{n}}(\theta) = e^{i\theta\sigma_{\vec{n}}} = \cos\left(\frac{\theta}{2}\right)\mathbb{1}_2 + 2i\sin\left(\frac{\theta}{2}\right)\sigma_{\vec{n}}, \quad \theta \in [0, 2\pi]. \quad (4.48)$$

Notice the resemblance with the parametrization of $SO(3)$ in terms of $R_{\vec{n}}(\theta)$ ($\theta \in [0, \pi]$!). This is not a coincidence! Even though we will use the same notation $R_{\vec{n}}(\theta)$ for both $SO(3)$ and $SU(2)$, it will be clear from the context which of the two groups we are dealing with.

4.3 The Lie algebras $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$

In the previous section we defined the rotation group $SO(3)$. As we already saw, every rotation can be parametrized by subsequent rotations around the coordinate axes x, y, z . In this section we will have a closer look at ‘infinitesimal’ rotations around these axes. Why would one care about such small rotations? For one, and this will also be our main motivation, infinitesimal rotations will lead to the notion of the **Lie algebra** of the rotation group, denoted $\mathfrak{so}(3)$. As it will turn out in a next section, all irreps of $SO(3)$ will correspond to irreps of $\mathfrak{so}(3)$ (but not all irreps of $\mathfrak{so}(s)$ are irreps of the group!). Similarly, we will construct the Lie algebra $\mathfrak{su}(2)$ and it will turn out that $\mathfrak{su}(2) = \mathfrak{so}(3)$! In the case of $SU(2)$, all irreps of $\mathfrak{su}(2)$ are irreps of the group! This will have deep implications that will become clear later on.

Let us investigate these rotations around the principal axes a bit more closely. Rotations around these axes correspond thus to the rotation matrices (4.39-4.41).

For small angles, we find to first order in a small angle $\delta\theta$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \delta\theta & \sin \delta\theta \\ 0 & -\sin \delta\theta & \cos \delta\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \delta\theta \\ 0 & -\delta\theta & 1 \end{pmatrix} + \mathcal{O}(\delta\theta^2) = \mathbb{1}_3 + i\delta\theta \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}}_{L_x} + \mathcal{O}(\delta\theta^2). \quad (4.49)$$

Here we used the Taylor expansion of $\cos$ and $\sin$: $\cos \delta\theta = 1 + \mathcal{O}(\delta\theta^2)$, $\sin \delta\theta = \delta\theta + \mathcal{O}(\delta\theta^2)$. Factoring out i is usual in physics literature, as it makes the matrix L_x Hermitian.

Similarly we find:

$$L_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad (4.50)$$

$$L_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.51)$$

This means that any very small rotation in 3D can be written in the form

$$\mathbb{1}_3 + i\delta\theta (n_x L_x + n_y L_y + n_z L_z). \quad (4.52)$$

Clearly, the matrices L_x, L_y, L_z do not commute. This is in correspondence with our intuition because also rotations around different axes do not commute. Rather, they satisfy the commutation relations

$$[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k, \quad (4.53)$$

with ε_{ijk} denotes the Levi-Civita symbol, and with the understanding that $x, y, z \equiv 1, 2, 3$. This brings us to the concept of **Lie algebras**.

⁴See exercise session.

In eq. 4.52 we observed that every small rotation can be parametrized as a **real** linear combination of the matrices L_1, L_2, L_3 that satisfy the above commutation relations. Hence, in mathematical parlance, we say that the L_i span a three-dimensional real vector space that is endowed with a **Lie bracket**, together forming the structure of a **Lie algebra**. Every vector in the Lie algebra thus corresponds to a small rotation. Lie algebras are often denoted by ‘mathfrak’ font: $\mathfrak{g}, \mathfrak{h}, \dots$. Formally, a Lie bracket is a function $[-, -] : \mathfrak{g} \times \mathfrak{g} \mapsto \mathfrak{g}$ ⁵ that is:

1. bilinear $[\lambda a + b, c + \lambda' d] = \lambda[a, c] + \lambda\lambda'[a, d] + [b, c] + \lambda'[b, d]$,
2. antisymmetric $[a, b] = -[b, a]$,
3. and satisfies the Jacobi identity

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0, \quad (4.54)$$

0 being the zero vector.

Unsurprisingly, the matrices L_1, L_2, L_3 form the 3 basis vectors of the Lie algebra called $\mathfrak{so}(3)$. The Lie bracket is then simply the matrix commutator. A simple exercise which you should certainly do, is that the matrix commutator automatically satisfies the Jacobi identity. Obviously it’s bilinear and antisymmetric as well by definition.

The matrices L_1, L_2, L_3 are in this context called the **generators of $\mathfrak{so}(3)$** and are said to ‘generate’ small rotations around the coordinate axes.

Remark. *Even though $\mathfrak{so}(3)$ is a real vector space, its generators are represented by complex matrices. Even though this might come across as somewhat counterintuitive, this is not a contradiction: we only consider real linear combinations of these generators.*

Let us do the same exercise for $SU(2)$. Take the parametrization (4.48):

$$R_{\vec{n}}(\theta) = e^{i\theta\sigma_{\vec{n}}} = \cos\left(\frac{\theta}{2}\right)\mathbb{1}_2 + 2i\sin\left(\frac{\theta}{2}\right)\sigma_{\vec{n}}, \quad \theta \in [0, 2\pi[. \quad (4.55)$$

Again using the Taylor expansion of $\cos$ and $\sin$ allows us to write for small angles:

$$R_{\vec{n}}(\delta\theta) \approx \mathbb{1}_2 + i\delta\theta \left(n_x \frac{\sigma_x}{2} + n_y \frac{\sigma_y}{2} + n_z \frac{\sigma_z}{2} \right). \quad (4.56)$$

So, the infinitesimal rotations of $SU(2)$ are generated by:

$$L_x = \frac{\sigma_x}{2} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (4.57)$$

$$L_y = \frac{\sigma_y}{2} = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (4.58)$$

$$L_z = \frac{\sigma_z}{2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.59)$$

If we now again compute the commutation relations we obtain exactly the same as those of $\mathfrak{so}(3)$ from above:

$$\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i \sum_k \varepsilon_{ijk} \frac{\sigma_k}{2}. \quad (4.60)$$

This means that the Lie algebra of $SU(2)$, written as $\mathfrak{su}(2)$, is ‘the same’, isomorphic to $\mathfrak{so}(3)$! Of course in the case of $\mathfrak{so}(3)$ we had 3×3 matrices spanning the Lie algebra, while here for $\mathfrak{su}(2)$ the generators are 2×2 matrices. In the next section we will argue that these two are two distinct irreducible **representations** of the same Lie algebra $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$.

⁵So technically speaking, since $\mathfrak{so}(3)$ is a real Lie algebra, the commutation relation $[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k$ brings you outside the Lie algebra but this extra factor i is taken care of by an extra compensating factor i in the exponential map (see below). Again, this factor i guarantees we are dealing with Hermitian generators.

4.4 Topological aspects $SU(2)$ and $SO(3)$

Even though the Lie algebras of both $SO(3)$ and $SU(2)$ are isomorphic: $\mathfrak{so}(3) \simeq \mathfrak{su}(2)$. $SO(3)$ and $SU(2)$ are not isomorphic groups! This implies that the structure of these groups (or more generally of any Lie group) cannot be fully understood by looking only at transformations infinitesimally close to the identity. To unveil the difference between $SU(2)$ and $SO(3)$ we should also consider ‘large’ rotations ‘far from’ the identity. These large rotations give us information about the **global** or **topological** properties of these groups, and it will turn out that $SU(2)$ and $SO(3)$, even though locally they are indistinguishable, are topologically very different. However, let us rephrase this observation in a bit more general terms, as this observation is generally true for all Lie groups and lies at the heart of the Lie algebra-Lie group correspondence discussed in the next chapter.

A Lie group is described by two pieces of information: local information about the small symmetry transformations close to the identity, and global or topological information. The first piece of information is fully captured by the Lie algebra, whereas the second piece of information is independent of this local information. Both pieces of information are needed to fully define a Lie group!

Let us first consider $SU(2)$. Taking (4.45) and writing $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ with real α_i and β_i , it is clear that $SU(2)$ is parametrized by four *real* parameters that define a 3-sphere S^3 :

$$SU(2) = \{(\alpha_1, \alpha_2, \beta_1, \beta_2) \in \mathbb{R}^4 | \alpha_1^2 + \alpha_2^2 + \beta_1^2 + \beta_2^2 = 1\}. \quad (4.61)$$

Due to the constraint that all the points $(\alpha_1, \alpha_2, \beta_1, \beta_2)$ lie on a sphere, only three of these four parameters are independent. We say that **topologically** $SU(2)$ is a 3-sphere. As a topological space S^3 is **compact**, meaning it is a closed and bounded subset of $\mathbb{R}^4$, and **connected**, meaning every two points of S^3 can be connected by a smooth line interpolating between those points. Another way of parametrizing $SU(2)$, namely (4.48) reveals, that we can equally well think about $SU(2)$ as a ball $\mathbb{B}(2\pi)$ in three dimensions of radius 2π : every point in $\mathbb{B}(2\pi)$ can be written as $r\vec{n}$ where $\vec{n}$ is a normalized vector around which we rotate and r is an angle between 0 and 2π . However there is something funny going on with this ball. Every point on the boundary of this ball, namely all points which have $r = 2\pi$ define the same element of $SU(2)$, namely minus the identity. So, we can equally well say that $SU(2)$ is, topologically speaking, a ball of radius 2π where all boundary points should be identified, which we try to translate in the notation:

$$SU(2) = \mathbb{B}(2\pi) / \{2\pi\vec{n} | \|\vec{n}\| = 1\} := \{(rn_x, rn_y, rn_z) | \|\vec{n}\| = 1, r \in [0, 2\pi]\} / \{2\pi\vec{n} | \|\vec{n}\| = 1\} \quad (4.62)$$

This makes it even clearer that $SU(2)$ is compact.

Now, let us repeat the exercise for $SO(3)$. In this case every rotation can now be parametrized as $R_{\vec{n}}(\theta)$ where $\vec{n}$ is again a normalized vector defining the axis of rotation, and θ is now an angle between 0 and π . In this case however, rotations with an angle π cannot all be identified, but those corresponding to $\vec{n}$ and $-\vec{n}$ can! So, in conclusion, $SO(3)$ is, again topologically speaking, a ball of radius π with antipodal points identified. Such a space is sometimes called a **real projective space of dimension 3** denoted $\mathbb{RP}^3$:

$$SO(3) = \mathbb{RP}^3. \quad (4.63)$$

This reveals the global or topological difference between $SO(3)$ and $SU(2)$ anticipated above. So, if $SO(3)$ and $SU(2)$ are locally identical, but globally very different, what is then the exact correspondence between $SO(3)$ and $SU(2)$? From a group theoretical perspective, we will argue in the next chapter that $SO(3) \simeq SU(2)/\mathbb{Z}_2$, again as consequence of the Lie algebra-Lie group correspondence. In that context $SU(2)$ will be called the **universal cover** or **double cover** of $SO(3)$: with every rotation in $SO(3)$ there correspond *two* rotations in $SU(2)$ and rotations in $SU(2)$ which differ only by a minus sign should be identified in $SO(3)$. From a more representation

theoretical perspective we will see that all irreps of $SO(3)$ are also irreps of $SU(2)$, but the converse is however not true. The irreps of $SU(2)$ come in two flavors: so called integer spin and half integer spin representations. The integer spin representations are also irreps of $SO(3)$ or more strictly speaking can be lifted to representations of $SO(3)$, but the half integer spin representations of $SU(2)$ are only **projective representations** of $SO(3)$! This is the topic of the next section.

4.5 Irreducible representations $SU(2)$ and $SO(3)$

The goal of this section is to describe all irreducible representations of the Lie groups $SU(2)$ and $SO(3)$ which share the same Lie algebra, namely $\mathfrak{su}(2) \simeq \mathfrak{so}(3)$. As it will turn out, all irreps of $SU(2)$ are in one-to-one correspondence with the irreps of its Lie algebra $\mathfrak{su}(2)$. This can be attributed to the fact that as a Lie group $SU(2)$ is simply connected (next chapter). The explicit map from the irreps of $\mathfrak{su}(2)$ to $SU(2)$ is via the so called exponential map.

First we will consider the irreps of the Lie algebra $\mathfrak{su}(2)$, or rather its complexification $\mathfrak{sl}(2)$, we will give some concrete examples and then come back to the irreps of $SU(2)$ and $SO(3)$.

4.5.1 Irreducible representations $\mathfrak{sl}(2)$

Recall that the Lie algebra $\mathfrak{su}(2)$ is a *real* 3-dimensional Lie algebra with 3 generators $L_1, L_2, L_3 \equiv L_x, L_y, L_z$ that satisfy the commutation relations

$$[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k. \quad (4.64)$$

These generators are Hermitian $L_i = L_i^\dagger$.

We already saw that these commutation relations can be **represented** by the 2-dimensional Pauli matrices $\sigma_1, \sigma_2, \sigma_3 \equiv \sigma_X, \sigma_Y, \sigma_Z$ by identifying L_i with $\sigma_i/2$:

$$\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i \sum_k \varepsilon_{ijk} \frac{\sigma_k}{2}. \quad (4.65)$$

Even though we will only formally introduce representations of Lie algebras in the next chapter, it is clear what a representation of a Lie algebra will turn out to be. It is a collection of linear operators (matrices) that satisfy the commutation relations of the Lie algebra.

The Pauli matrices (up to a global scale factor $1/2$) thus constitute a **2-dimensional irreducible representation** of $\mathfrak{su}(2)$. Of course there is also a 1-dimensional trivial irrep given by $L_i = 0$.⁶ The central question of this section is: given the Lie algebra $\mathfrak{su}(2)$, what are all its irreducible representations? The approach we will follow is a standard approach found in almost every book on quantum mechanics and/or Lie groups and algebras, for example [2].

Finite dimensional irreducible representations

The ladder algebra The standard approach to constructing the irreps of $\mathfrak{su}(2)$ is by introducing **ladder operators**

$$L_\pm := L_1 \pm iL_2, \quad L_0 := 2L_3. \quad (4.66)$$

You should check that these operators satisfy the commutation rules

$$[L_0, L_\pm] = \pm 2L_\pm, \quad (4.67)$$

$$[L_+, L_-] = L_0. \quad (4.68)$$

Notice that the ladder operators $L_\pm$ are not Hermitian, they rather satisfy $L_\pm^\dagger = L_\mp$.

⁶A notable difference between the representations of groups and Lie algebras is that the trivial representation of a group is given by $X_g = 1$ and that of a Lie algebra by $L_g = 0$! Do you understand why?

Let us also introduce the **quadratic Casimir** operator. This operator is defined as the sum of the squares of all $\mathfrak{su}(2)$ generators⁷:

$$\vec{L}^2 := L_x L_x + L_y L_y + L_z L_z = \frac{1}{2}(L_+ L_- + L_- L_+) + \frac{L_0^2}{4}. \quad (4.69)$$

Notice that $L_+ L_- + L_- L_+ \neq 2L_+ L_-$ because L_+ and L_- do not commute! This operator has the special property that it commutes with all generators:

$$[\vec{L}^2, L_i] = 0, \quad \forall i. \quad (4.70)$$

You should check this, and you can use the property $[AB, C] = A[B, C] + [A, C]B$ and $[A, BC] = B[A, C] + [A, B]C$. $\vec{L}^2$ obviously also commutes with the ladder operators $L_{\pm}$ then.

We will also need the identities

$$L_{\pm} L_{\mp} = \vec{L}^2 - L_3^2 \pm L_3. \quad (4.71)$$

These follow from

$$L_{\pm} L_{\mp} =: (L_1 \pm iL_2)(L_1 \mp iL_2) \quad (4.72)$$

$$= L_1^2 \pm i(L_2 L_1 - L_1 L_2) + L_2^2 \quad (4.73)$$

$$= L_1^2 + L_2^2 + L_3^2 - L_3^2 \pm i(-i)L_3 \quad (4.74)$$

$$= \vec{L}^2 - L_3^2 \pm L_3. \quad (4.75)$$

Where we used $[L_2, L_1] = -iL_3$.

Already, there is something finicky going on. Notice That $L_{\pm}$ are *complex* rather than real combinations of the $\mathfrak{su}(2)$ generators. This means that $L_{\pm}$ don't belong to the Lie algebra $\mathfrak{su}(2)$ anymore but rather to a *complex* Lie algebra, namely $\mathfrak{sl}(2)$. In mathematical parlance, $\mathfrak{sl}(2)$ is said to be the **complexification** of $\mathfrak{su}(2)$, whereas $\mathfrak{su}(2)$ is a **real form** of $\mathfrak{sl}(2)$. One writes:

$$\mathfrak{sl}(2) \simeq \mathfrak{su}(2) \oplus i\mathfrak{su}(2) \simeq \mathfrak{su}(2) \otimes_{\mathbb{R}} \mathbb{C}. \quad (4.76)$$

So, as stressed above, what we *actually* will be computing are the irreps of $\mathfrak{sl}(2)$ instead of those $\mathfrak{su}(2)$. However, as it turns out, there is a one-to-one correspondence between the irreps of these two Lie algebras! This is a very non-trivial fact, and has to do with the fact that $\mathfrak{su}(2)$ is a very special real form of $\mathfrak{sl}(2)$ called the **compact real form**. All irreps of $\mathfrak{su}(2)$ can be obtained from those of $\mathfrak{sl}(2)$ by restricting $\mathfrak{sl}(2)$ to $\mathfrak{su}(2)$.

Definition 4.8. $\mathfrak{sl}(2)$ is the **complex 3-dimensional Lie algebra** with generators $L_{\pm}$ and L_0 with commutation relations

$$[L_0, L_{\pm}] = \pm 2L_{\pm}, \quad (4.77)$$

$$[L_+, L_-] = L_0. \quad (4.78)$$

Sometimes we write $\mathfrak{sl}(2) \equiv \mathfrak{sl}(2, \mathbb{C})$, also the notation $\mathfrak{su}(2) \equiv \mathfrak{sl}(2, \mathbb{R})$ is being used.

⁷Notice that the quadratic Casimir is not in the Lie algebra, as generally speaking the product of generators is not defined in a Lie algebra. The quadratic Casimir lives in something called the **universal enveloping algebra** of in this case $\mathfrak{su}(2)$. Informally speaking, the universal enveloping algebra of a Lie algebra $\mathfrak{g}$, denoted $U(\mathfrak{g})$, is an associative algebra with unit in which you can in a precise sense embed $\mathfrak{g}$ in such a way that the Lie bracket $\mathfrak{g}$ corresponds to the commutator in $U(\mathfrak{g})$.

Highest weight representations All finite-dimensional irreps of $\mathfrak{sl}(2)$ can be proven to be **highest weight representations**. These representations are constructed as follows.⁸

Since L_3 is Hermitian, this operator can be diagonalized and has real eigenvalues. The same goes for $\vec{L}^2$. Since $\vec{L}^2$ commutes with L_3 , there exists a simultaneous eigenbasis of L_3 and $\vec{L}^2$. Since the eigenvalues of $\vec{L}^2$ are non-negative we can write them as $l(l+1)$, the eigenvalues of L_3 will be denoted by m . We will denote this common eigenbasis then as $\{|l, m\rangle\}$, where

$$\vec{L}^2 |l, m\rangle = l(l+1) |l, m\rangle, \quad (4.79)$$

$$L_3 |l, m\rangle = m |l, m\rangle. \quad (4.80)$$

Moreover we assume that these basis states are orthonormal:

$$\langle l, m | l', m' \rangle = \delta_{l,l'} \delta_{m,m'}. \quad (4.81)$$

The eigenvalues l are called **weights** in the mathematical literature. From the commutation relations we then have that

$$L_3 L_{\pm} |l, m\rangle = (L_{\pm} L_3 \pm L_{\pm}) |l, m\rangle = (m \pm 1) L_{\pm} |l, m\rangle, \quad (4.82)$$

motivating the name ‘ladder operators’: given some eigenvector $|l, m\rangle$, acting with $L_{\pm}$ on this eigenvector brings you to an eigenvector with eigenvalue $m \pm 1$. The underlying idea of highest weight representations can now be explained in the following steps:

1. Start from an eigenvector with a ‘maximal’ eigenvalue M , called the **highest weight**, which is annihilated by L_+ ,

$$L_+ |l, M\rangle = 0, \quad (4.83)$$

2. Sequentially act with L_- on this highest weight vector. The commutation relations then guarantee that this string of vectors, $\{|l, M\rangle, |l, M-1\rangle, |l, M-2\rangle, \dots\}$ is closed under the action of $\mathfrak{sl}(2)$ and thus forms a **representation** of $\mathfrak{sl}(2)$.
3. Importantly, the value of l is constant in a given representation! This was to be expected because l (rather $l(l+1)$) is the eigenvalue of $\vec{L}^2$ and since $\vec{L}^2$ commutes with the ladder operators, the ladder operators cannot change this eigenvalue.
4. Another necessary condition for this representation to be finite-dimensional is the existence of some M' corresponding to an eigenvector $|l, M'\rangle$ this is annihilated by L_- :

$$L_- |l, M'\rangle = 0. \quad (4.84)$$

This property implies that the string $S_l := \{|l, M\rangle, |l, M-1\rangle, |l, M-2\rangle, \dots, |l, M'\rangle\}$ only has a finite number of vectors and thus constitutes a **finite-dimensional representation**. The **dimension** of the irrep $V_l := \text{span}_{\mathbb{C}}(S_l)$ is then of course $\dim_{\mathbb{C}} V_l = M - M' + 1$.

We can now relate M to l as follows:

$$L_- L_+ |l, M\rangle = (\vec{L}^2 - L_3^2 - L_3) |l, M\rangle \quad (4.85)$$

$$= (l(l+1) - M^2 - M) |l, M\rangle \quad (4.86)$$

$$= 0. \quad (4.87)$$

Here we used (4.71). Hence,

$$l(l+1) = M(M+1). \quad (4.88)$$

⁸As it turns out, all finite-dimensional irreps of ‘nice’ (=semi-simple and finite-dimensional, see next chapter) Lie algebra’s are of this type.

From (4.84) on the other hand it follows that

$$L_+ L_- |l, M'\rangle = (\vec{L}^2 - L_3^2 + L_3) |l, M'\rangle \quad (4.89)$$

$$= (l(l+1) - M'^2 + M') |l, M'\rangle \quad (4.90)$$

$$= 0. \quad (4.91)$$

We find:

$$l(l+1) = M'(M' - 1). \quad (4.92)$$

(4.88) and (4.92) require that:

$$M(M+1) = M'(M' - 1) \iff M' = -M \quad \text{or} \quad M' = M + 1. \quad (4.93)$$

The second solution should be discarded as by assumption M is the highest weight of the representation. Furthermore, since $M \geq 0$ and l is non-negative we thus find from (4.88)

$$M = l \quad (4.94)$$

$$M' = -l. \quad (4.95)$$

The dimension of the irrep is thus

$$\boxed{\dim_{\mathbb{C}} V_l = M - M' + 1 = 2l + 1}. \quad (4.96)$$

Conclusion This leads to the following conclusion. All finite-dimensional irreps of $\mathfrak{sl}(2)$ are labeled by **one integer/quantum number** l , and the dimension of the corresponding irrep is $2l + 1$, and spanned by the eigenvectors of L_3 , $\{|l, m\rangle, m = l, l-1, \dots, -l\}$. In physical literature, l is often called the **spin** or **angular momentum** quantum number, and m is called a **magnetic quantum number**. On the irrep labeled by l the quadratic Casimir takes the value $l(l+1)$. A $2l+1$ dimensional irrep is also sometimes called a spin l **multiplet**. The low dimensional ones are of fundamental importance in physics and have often due to technical or historical reasons special names that we summarize in the table below. Since there is exactly one irrep of each dimension, irreps are also sometime labeled as **d**, where d is the dimension of the irrep.

Dimension	1	1(1+1)	d	Common names
1	0	0	1	Trivial, singlet, scalar
2	$\frac{1}{2}$	$\frac{3}{4}$	2	Fundamental, defining, doublet, spinor
3	1	2	3	Adjoint, triplet, vector
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 4.1: Low dimensional irreducible representations of $\mathfrak{sl}(2)$.

Infinite dimensional representation $\mathfrak{sl}(2)$

Let us finally comment on the existence of infinite-dimensional representations of $\mathfrak{sl}(2)$.

We argued in the previous section that the existence of a integral or half-integral highest weight $M = l \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ is a necessary condition to have a finite-dimensional irrep of $\mathfrak{sl}(2)$. Relaxing this condition and starting from a highest weight which is an arbitrary (complex) number leads to the notion of **Verma modules**. Infinite dimensional representations Verma modules (which are not unitary!) can be constructed by starting from a highest weight vector with a highest weight $M \in \mathbb{C} \setminus \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, as in this case the string $\{|l, M\rangle, |l, M-1\rangle, \dots\}$ never terminates. Exotic as these may seem, these representations find applications in physics, most notably in the study of conformal field theories. But let us stop here.

4.5.2 Computing matrix representations of $\mathfrak{sl}(2)$

This section is devoted to finding the matrix representations of the operators $\vec{L}^2$, L_3 and $L_{\pm}$. From the latter, also the matrix operators of L_1 and L_2 can readily be obtained.

Recall that the basis states $|l, m\rangle$ are orthonormal:

$$\langle l', m' | l, m \rangle = \delta_{l,l'} \delta_{m,m'}. \quad (4.97)$$

We then immediately find the matrix representation of $\vec{L}^2$ as

$$\langle l', m' | \vec{L}^2 | l, m \rangle = l(l+1) \delta_{l,l'} \delta_{m,m'}. \quad (4.98)$$

And L_3 :

$$\langle l', m' | L_3 | l, m \rangle = m \delta_{l,l'} \delta_{m,m'}. \quad (4.99)$$

We know from the previous section that $L_{\pm} |l, m\rangle$ are proportional to $|l, m \pm 1\rangle$. Let us write:

$$L_{\pm} |l, m\rangle = N_{\pm} |l, m \pm 1\rangle. \quad (4.100)$$

Using (4.71, 4.97) we then find

$$|N_{\mp}|^2 = \langle l, m | L_{\mp}^{\dagger} L_{\mp} | l, m \rangle \quad (4.101)$$

$$= \langle l, m | L_{\pm} L_{\mp} | l, m \rangle \quad (4.102)$$

$$= (l(l+1) - m(m \mp 1)). \quad (4.103)$$

hence:

$$N_{\pm} = \sqrt{l(l+1) - m(m \pm 1)}. \quad (4.104)$$

A couple of remarks. For $m = l$ we simply compute that $N_+ = 0$ as it should, and likewise for $m = -l$, $N_- = 0$. Notice that the argument of the square root is non-negative as $-l \leq m \leq l$ and also notice that we fixed the phase freedom of $N_{\pm}$ by making the choice that $N_{\pm}$ are real and non-negative.

With this equation at our disposal, we can compute

$$\langle l', m' | L_+ | l, m \rangle = N_+ \delta_{l,l'} \delta_{m+1,m'} \quad (4.105)$$

$$= \sqrt{l(l+1) - m(m+1)} \delta_{l,l'} \delta_{m+1,m'}, \quad (4.106)$$

similarly:

$$\langle l', m' | L_- | l, m \rangle = N_- \delta_{l,l'} \delta_{m-1,m'} \quad (4.107)$$

$$= \sqrt{l(l+1) - m(m-1)} \delta_{l,l'} \delta_{m-1,m'}, \quad (4.108)$$

4.5.3 Examples of irreducible representations of $\mathfrak{sl}(2)$

The $l=1/2$ representation

The spinor representation of $\mathfrak{sl}(2)$ is two-dimensional and has a basis $\{|\frac{1}{2}, +\frac{1}{2}\rangle, |\frac{1}{2}, -\frac{1}{2}\rangle\}$. This basis is often also written as the ‘spin up and down’ basis states (see Ex. 3.9) $\{|\uparrow\rangle, |\downarrow\rangle\}$ which diagonalize L_3 :

$$L_3 |\uparrow\rangle = \frac{L_0}{2} |\uparrow\rangle = \frac{1}{2} |\uparrow\rangle \quad (4.109)$$

$$L_3 |\downarrow\rangle = \frac{L_0}{2} |\downarrow\rangle = -\frac{1}{2} |\downarrow\rangle. \quad (4.110)$$

The basis states can also be represented by column matrices

$$|\uparrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (4.111)$$

Vectors of the form $a|\uparrow\rangle + b|\downarrow\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$ are sometimes called **two component spinors**.

In this basis L_3 is thus given by

$$L_3 = \frac{\sigma_Z}{2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.112)$$

The ladder operators are then given by (4.106,4.108):

$$L_+ = \frac{1}{2}(\sigma_X + i\sigma_Y) =: \sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad (4.113)$$

$$L_- = \frac{1}{2}(\sigma_X - i\sigma_Y) =: \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (4.114)$$

exactly what we expect since:

$$L_+|\uparrow\rangle \equiv \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \equiv 0, \quad L_-|\uparrow\rangle \equiv \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \equiv |\downarrow\rangle, \quad (4.115)$$

and similarly for $|\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. And finally,

$$\vec{L}^2 = \frac{3}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (4.116)$$

The $l=1$ representation

A basis for the 3-dimensional spin 1 representation is $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$. The basis states are represented by following column matrices:

$$|1, 1\rangle \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1, 0\rangle \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1, -1\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (4.117)$$

The ladder operators are computed to be

$$L_+ = \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (4.118)$$

$$L_- = \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (4.119)$$

The $\mathfrak{su}(2)$ generators in this basis then read:

$$L_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (4.120)$$

$$L_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad (4.121)$$

$$L_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (4.122)$$

And finally:

$$\vec{L}^2 = 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.123)$$

Verify that all these matrices indeed satisfy the $\mathfrak{su}(2)$ and $\mathfrak{sl}(2)$ commutation relations!

Now, these matrices clearly don't correspond to the matrices we derived in Sec. 4.3, in order to obtain those we need to carry out the following basis transformation:

$$L_i^c = U^\dagger L_i U, \quad U = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \end{pmatrix}. \quad (4.124)$$

One then indeed finds:

$$L_x^c = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad (4.125)$$

$$L_y^c = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad (4.126)$$

$$L_z^c = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.127)$$

From $L_i^c L_j^c = U^\dagger L_i U U^\dagger L_j U = U^\dagger L_i L_j U$ one concludes that these generators also satisfy the same $\mathfrak{su}(2)$ commutation relations. Here the subscript c stands for 'Cartesian', the basis states $|1, m\rangle$ are sometimes called the **spherical basis**.

4.5.4 Irreducible representations $SU(2)$, the exponential map and the BCH formula

Now that we have all irreps of $\mathfrak{sl}(2)$ labeled by $l = 0, \frac{1}{2}, 1, \dots$, we can construct all irreps of $\mathfrak{su}(2)$ by restriction. Concretely this means that given an irreducible (matrix) representation of the generators $L_\pm, L_0$ of $\mathfrak{sl}(2)$, an irreducible (matrix) representation of $\mathfrak{su}(2)$ is given by regarding $L_1 \equiv \frac{1}{2}(L_+ + L_-)$, $L_2 \equiv \frac{1}{2i}(L_+ - L_-)$, $L_3 \equiv \frac{L_0}{2}$ as generators of $\mathfrak{su}(2)$.

As it turns out, every irrep of $\mathfrak{su}(2)$ also corresponds to an irrep of the group $SU(2)$. The irreps of $SU(2)$ can then be constructed via the **exponential map** on $\mathfrak{su}(2)$: given an irrep of the Lie algebra, exponentiating it gives an irrep of the group. Let us first define the exponential map and consider some of its properties.

Definition 4.9. Given a square matrix M , the **matrix exponential** is defined via its series expansion:

$$\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}. \quad (4.128)$$

This series can be shown to converge.

From the definition it is clear that if A and B commute, $AB = BA$, that

$$\exp(A) \exp(B) = \exp(A + B), \quad (4.129)$$

just like for real and complex numbers. However, since matrices in general don't commute we will have corrections to the above formula. Surprisingly, it can be shown that the corrections can all be expressed in terms of commutators. This famous result is the **Baker–Campbell–Hausdorff (BCH) formula**:

Theorem 4.2. *Given matrices A, B :*

$$\exp(A)\exp(B) = \exp(C), \quad (4.130)$$

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] - \frac{1}{12}[B, [A, B]] + \dots \quad (4.131)$$

where $\dots$ are all terms that involve commutators of A, B .

PROOF: Let us try to obtain the first three terms of C . From the definition:

$$\exp(A)\exp(B) = \left(1 + A + \frac{A^2}{2} + \mathcal{O}(A^3)\right) \left(1 + B + \frac{B^2}{2} + \mathcal{O}(B^3)\right) \quad (4.132)$$

$$= 1 + A + B + AB + \frac{1}{2}(A^2 + B^2) + \mathcal{O}(A^3, B^3) \quad (4.133)$$

$$= 1 + A + B + \frac{1}{2}(A + B)^2 + \frac{[A, B]}{2} + \mathcal{O}(A^3, B^3), \quad (4.134)$$

$$(4.135)$$

Hence, up to second order:

$$C = A + B + \frac{1}{2}[A, B]. \quad (4.136)$$

The higher order terms can be derived using recursive methods. $\square$

4.5.5 Examples of irreducible representations of $SU(2)$

Let us apply the exponential map to the irreducible representations of $\mathfrak{su}(2)$.

$l=0$

The smallest irrep of $\mathfrak{su}(2)$ is given by the one-dimensional irrep $l = 0$:

$$L_x = L_y = L_z = 0. \quad (4.137)$$

Obviously the (matrix) exponential of 0 is $\exp(0) = 1$. So this irrep of $\mathfrak{su}(2)$ gives rise to the trivial irrep of $SU(2)$.

$l=1/2$

A much more interesting example is the spinor representation. Recall from above that in this representation

$$L_x = \frac{\sigma_X}{2} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L_y = \frac{\sigma_Y}{2} = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad L_z = \frac{\sigma_Z}{2} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.138)$$

The corresponding irrep of $SU(2)$ can then be obtained by exponentiation of $R_{\vec{n}}(\theta) = \exp(i\theta(n_x L_x + n_y L_y + n_z L_z))$ where $\vec{n}$ is a (real) unit vector. This gives⁹:

$$R_{\vec{n}}(\theta) = e^{i\theta\sigma_{\vec{n}}} = \cos\left(\frac{\theta}{2}\right) \mathbb{1}_2 + 2i \sin\left(\frac{\theta}{2}\right) \sigma_{\vec{n}} \quad (4.139)$$

$$= \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) + in_z \sin\left(\frac{\theta}{2}\right) & n_y \sin\left(\frac{\theta}{2}\right) + in_x \sin\left(\frac{\theta}{2}\right) \\ -n_y \sin\left(\frac{\theta}{2}\right) + in_x \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) - in_z \sin\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad \theta \in [0, 2\pi]. \quad (4.140)$$

This is exactly the defining representation that appeared first in (4.48)! Since this is the smallest faithful representation, it is also called the **fundamental representation** of $SU(2)$.

⁹See exercise session.

4.5.6 Examples of irreducible and projective representations of $SO(3)$

As stressed already a couple of times the Lie algebras of $SU(2)$ and $SO(3)$ are isomorphic. So we can wonder, can all irreps of $SO(3)$ be obtained from the irreps of $\mathfrak{so}(3)$ via the exponential map? The answer is yes. However, not every irrep of $\mathfrak{so}(3)$ will when we take its exponential give an irrep of $SO(3)$! Only those with integer $l = 0, 1, 2, \dots$ will correspond to irreps of $SO(3)$!

$l=1$

Let us take the $l=1$ representation of $\mathfrak{so}(3)$ of above:

$$L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad (4.141)$$

$$L_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad (4.142)$$

$$L_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.143)$$

Let us only focus on rotations about the z-axis. The matrix exponential of L_z can be computed by diagonalizing L_z with the basis transformation encountered in (4.124):

$$L_z = U \text{diag}(1, 0, -1) U^\dagger, \quad (4.144)$$

so that

$$\exp(i\theta L_z) = \exp(U \text{diag}(i\theta, 0, -i\theta) U^\dagger) \quad (4.145)$$

$$= U \exp(\text{diag}(i\theta, 0, -i\theta)) U^\dagger \quad (4.146)$$

$$= U \text{diag}(\exp(i\theta), 0, \exp(-i\theta)) U^\dagger \quad (4.147)$$

$$= \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.148)$$

Exactly what we already had in (4.41)! The same goes for rotations around the x and y axes. We thus obtain the 3-dimensional irrep of $SO(3)$ from the 3-dimensional irrep of its Lie algebra.

It goes without saying that this representation of $SO(3)$ is also a representation of $SU(2)$.

$l=1/2$

Let us do the same for $l = 1/2$. In that case the generators are given by (4.57-4.59). Let us again take the rotation around the z-axis since this generator is already diagonal and immediately gives away its matrix exponential:

$$\exp(i\theta L_z) = \exp \begin{pmatrix} i\frac{\theta}{2} & 0 \\ 0 & -i\frac{\theta}{2} \end{pmatrix} \quad (4.149)$$

$$= \begin{pmatrix} \exp(i\frac{\theta}{2}) & 0 \\ 0 & \exp(-i\frac{\theta}{2}) \end{pmatrix}. \quad (4.150)$$

This is funny. In this representation a rotation about $\theta = 2\pi$ does not correspond to the identity, but rather to minus the identity! The way to understand this is as follows. Starting from a spinor

in the ‘up’ state $|\uparrow\rangle$, a small rotation around the z-axis brings this spinor to the state

$$\propto \left(1 + i \frac{\delta\theta}{2} \sigma_Z\right) |\uparrow\rangle = \left(1 + i \frac{\delta\theta}{2}\right) |\uparrow\rangle \quad (4.151)$$

repeating this rotation an infinite number of times until we reach a rotation of $\int \delta\theta = 2\pi$, we eventually end up with the state $-|\uparrow\rangle$:

$$\lim_{N \rightarrow +\infty} \left(1 + i \frac{2\pi}{2N}\right)^N |\uparrow\rangle = e^{i\pi} |\uparrow\rangle = -|\uparrow\rangle. \quad (4.152)$$

Here we applied (4.151) an infinite number of times for $\delta\theta = 2\pi/N$. And so to reobtain the state $|\uparrow\rangle$ from $-|\uparrow\rangle$ we have to repeat the entire procedure a second time, so that we have to conclude that a spinor only gets back to itself after a 4π rotation!

It can be shown that this observation generalizes in the following way. All irreps of $\mathrm{SO}(3)$ correspond to the **integer l** ($=0,1,2,\dots$) representations of $\mathfrak{so}(3)$, while the **half integer l** ($1/2,3/2,\dots$) representations of $\mathfrak{so}(3)$ only correspond to **projective representations** of $\mathrm{SO}(3)$. All of these can be obtained via the exponential map. In the case of $\mathrm{SU}(2)$, the irreps are all obtained from the irreps of $\mathfrak{su}(2)$.

4.6 Tensor product representations $\mathfrak{sl}(2)$ and Clebsch-Gordan coefficients

In this section we will study how tensor products of irreducible representations of $\mathfrak{sl}(2)$ decompose in irreducible representations.

Suppose we have two sets of $\mathfrak{su}(2)$ generators and their quadratic Casimirs which are distinguished by a superscript 1/2:

$$[L_i^{(1)}, L_j^{(1)}] = i \sum_k \varepsilon_{ijk} L_k^{(1)}, \quad (4.153)$$

$$[L_i^{(1)}, \overrightarrow{L^{(1)}}^2] = 0, \quad (4.154)$$

$$[L_i^{(2)}, L_j^{(2)}] = i \sum_k \varepsilon_{ijk} L_k^{(2)}, \quad (4.155)$$

$$[L_i^{(2)}, \overrightarrow{L^{(2)}}^2] = 0. \quad (4.156)$$

Then we know from the previous section that a $2l_1 + 1$ -dimensional irrep of the first set of generators is given by the basis states $V_{l_1} := \{|l_1, m_1\rangle \mid m_1 = l_1, l_1 - 1, \dots, -l_1\}$:

$$L_3^{(1)} |l_1, m_1\rangle = m_1 |l_1, m_1\rangle, \quad (4.157)$$

$$\overrightarrow{L^{(1)}}^2 |l_1, m_1\rangle = l_1(l_1 + 1) |l_1, m_1\rangle. \quad (4.158)$$

$$(4.159)$$

And similarly a $2l_2 + 1$ -dimensional irrep of the second set of generators is given by $V_{l_2} := \{|l_2, m_2\rangle \mid m_2 = l_2, l_2 - 1, \dots, -l_2\}$:

$$L_3^{(2)} |l_2, m_2\rangle = m_2 |l_2, m_2\rangle, \quad (4.160)$$

$$\overrightarrow{L^{(2)}}^2 |l_2, m_2\rangle = l_2(l_2 + 1) |l_2, m_2\rangle. \quad (4.161)$$

We can now wonder if we can combine these two irreps as so to form a tensor product representation of $\mathfrak{sl}(2)$. The general theory of representations of Lie algebras and tensor product representations is one of the topics of the next chapter, but we will illustrate this here already for the particular case at hand of $\mathfrak{sl}(2)$. First we define operators which have a natural action on the tensor product $V_{l_1} \otimes V_{l_2}$:

$$L_{1,i} := L_i^{(1)} \otimes \mathbb{1}, \quad (4.162)$$

$$L_{2,i} := \mathbb{1} \otimes L_i^{(2)}, \quad (4.163)$$

and

$$\vec{L}_1^2 := \overrightarrow{L^{(1)}}^2 \otimes \mathbb{1}, \quad (4.164)$$

$$\vec{L}_2^2 := \mathbb{1} \otimes \overrightarrow{L^{(2)}}^2. \quad (4.165)$$

It can readily be checked that operators acting on different tensor factors all commute. Concretely:

$$[L_{1,i}, L_{2,i}] = 0, \quad (4.166)$$

$$[L_{1,i}, \vec{L}_2^2] = 0, \quad (4.167)$$

$$[\vec{L}_1^2, L_{2,i}] = 0, \quad (4.168)$$

$$[\vec{L}_1^2, \vec{L}_2^2] = 0. \quad (4.169)$$

This means that there is a basis in which the operators $\{L_{1,3}, L_{2,3}, \vec{L}_1^2, \vec{L}_2^2\}$ can be simultaneously diagonalized. This simultaneous eigenbasis is of course the tensor product basis

$$\{|l_1, m_1\rangle |l_2, m_2\rangle |m_1 = l_1, l_1 - 1, \dots, -l_1, m_2 = l_2, l_2 - 1, \dots, -l_2\rangle\}. \quad (4.170)$$

This basis spans a Hilbert space of dimension $(2l_1 + 1)(2l_2 + 1)$. On the basis states the operators defined above act according to:

$$L_{1,3} |l_1, m_1\rangle |l_2, m_2\rangle = m_1 |l_1, m_1\rangle |l_2, m_2\rangle, \quad (4.171)$$

$$L_{2,3} |l_1, m_1\rangle |l_2, m_2\rangle = m_2 |l_1, m_1\rangle |l_2, m_2\rangle, \quad (4.172)$$

$$\vec{L}_1^2 |l_1, m_1\rangle |l_2, m_2\rangle = l_1(l_1 + 1) |l_1, m_1\rangle |l_2, m_2\rangle, \quad (4.173)$$

$$\vec{L}_2^2 |l_1, m_1\rangle |l_2, m_2\rangle = l_2(l_2 + 1) |l_1, m_1\rangle |l_2, m_2\rangle. \quad (4.174)$$

This basis will henceforth be called the **uncoupled basis** and forms the basis of the **tensor product representation** that we can write as

$$l_1 \otimes l_2. \quad (4.175)$$

Let us now define the following operators

$$L_i := L_{1,i} + L_{2,i}, \quad (4.176)$$

$$\vec{L}^2 := \vec{L}_1^2 + \vec{L}_2^2. \quad (4.177)$$

We leave it as an exercise to check that these operators again satisfy the $\mathfrak{su}(2)$ commutation relations:

$$[L_i, L_j] = i \sum_k \varepsilon_{ijk} L_k, \quad (4.178)$$

$$[L_i, \vec{L}^2] = 0. \quad (4.179)$$

Applying the result of the previous section, this means that the irreps of these operators have a basis

$$\{|l, m\rangle \mid m = l, l-1, \dots, -l\}. \quad (4.180)$$

On this basis the action of the operators (4.176) and (4.177) is given by

$$L_3 |l, m\rangle = m |l, m\rangle \quad (4.181)$$

$$\vec{L}^2 |l, m\rangle = l(l+1) |l, m\rangle. \quad (4.182)$$

This basis is called the **coupled basis**.

Let us first state the result without proof.

Theorem 4.3. *Given the spin l_1 and l_2 representation of $\mathfrak{sl}(2)$, their tensor product decomposes as a direct sum as*

$$l_1 \otimes l_2 \simeq \bigoplus_{l=|l_1-l_2|}^{l_1+l_2} l. \quad (4.183)$$

As usual, $\simeq$ means ‘up to a basis transformation’.

PROOF: See eg. [2]. □

Notice that the dimensions of the representations that appear at both sides of the isomorphism (4.183) are the same: the dimension of $l_1 \otimes l_2$ is $(2l_1+1)(2l_2+1)$ and so is the dimension of $\bigoplus_{l=|l_1-l_2|}^{l_1+l_2} l$.¹⁰

The central question we now have to deal with is how **the coupled basis states (spanning a basis for the irreps appearing in the right hand side of (4.183))** can be expressed **in terms of the uncoupled basis states (spanning a basis for the left hand side of (4.183))**? This question can be rephrased as: what are the **Clebsch-Gordan coefficients** between the coupled and the uncoupled basis?

4.6.1 Examples

Simple and important examples illustrating Theorem 4.3 are

Example 4.2. *The tensor product of the trivial irrep with any other irrep is isomorphic to that irrep: $\mathbf{0} \otimes \mathbf{d} \simeq \mathbf{d}$.*

Example 4.3. *The tensor product of two spinor representations is isomorphic to a spin singlet and a spin triplet: $\mathbf{2} \otimes \mathbf{2} \simeq \mathbf{1} \oplus \mathbf{3}$. The isomorphism is computed below in Sec. 4.6.3*

Corollary 4.3.1. *A simple observation is then that every irrep of $\mathfrak{sl}(2)$ can be obtained by taking tensor products of the $\mathbf{2}$ irrep with itself. Eg. both the $\mathbf{1}$ and $\mathbf{3}$ are obtained from $\mathbf{2} \otimes \mathbf{2} \simeq \mathbf{1} \oplus \mathbf{3}$. The $\mathbf{4}$ arises in $\mathbf{2} \otimes \mathbf{2} \otimes \mathbf{2} \simeq \mathbf{2} \otimes (\mathbf{1} \oplus \mathbf{3}) \simeq (\mathbf{2} \otimes \mathbf{1}) \oplus (\mathbf{2} \otimes \mathbf{3}) \simeq \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{4}$, and so on. This explains why the $\mathbf{2}$ representation is sometimes called the **fundamental** representation (see table 4.1): every representation can be obtained from this one!*

4.6.2 Computing Clebsch-Gordan coefficients

The task we are confronted with is thus the computation of the Clebsch-Gordan coefficients C in

$$|l, m\rangle = \sum_{m_1+m_2=m} C_{m_1 m_2 m}^{l_1 l_2 l} |l_1, m_1\rangle |l_2, m_2\rangle \quad (4.184)$$

Using the notation

$$|l_1, m_1\rangle \otimes |l_2, m_2\rangle \equiv |l_1, m_1\rangle |l_2, m_2\rangle \equiv |l_1, m_1; l_2, m_2\rangle, \quad (4.185)$$

¹⁰See exercise session

$$C_{m_1 m_2 m}^{l_1 l_2 l} = \langle l_1, m_1; l_2, m_2 | l, m \rangle. \quad (4.186)$$

Notice that in the above formula we sum only over the allowed values of m

$$m \stackrel{!}{=} m_1 + m_2. \quad (4.187)$$

This follows from the fact that $L_3 := L_{1,3} + L_{2,3}$. We say that the magnetic quantum numbers are ‘additive’. From the Theorem 4.3 it also follows that for fixed l_1, l_2 the only allowed values of l are

$$l \in \{l_1 + l_2, l_1 + l_2 - 1, \dots, |l_1 - l_2|\}. \quad (4.188)$$

Given these constraint it is straightforward although tedious to compute the CGs. We will do this explicitly for the case $l_1 = l_2 = \frac{1}{2}$ below and explain how to deal with CGs for arbitrary spins l_1, l_2 .

Moreover it should be noticed that the Clebsch-Gordan coefficients form a **unitary** basis transformation. The CGs have a phase freedom that in quantum mechanics is conventionally chosen such that the matrix elements are real, this convention is called the **Condon-Shortley convention**. In this convention the **orthonormality** and **completeness** read:

$$\begin{aligned} \sum_{m_1, m_2} \langle l_1, m_1; l_2, m_2 | l, m \rangle \langle l_1, m_1; l_2, m_2 | l', m' \rangle &= \delta_{l, l'} \delta_{m, m'}, & (\text{orthonormality}) \\ \sum_{l, m} \langle l_1, m_1; l_2, m_2 | l, m \rangle \langle l_1, m'_1; l_2, m'_2 | l, m \rangle &= \delta_{m_1, m'_1} \delta_{m_2, m'_2}. & (\text{completeness}) \end{aligned}$$

4.6.3 Coupling of two spinors

As a concrete example we will consider the coupling of two spin 1/2 particles in this section.

From the above considerations we are thus looking for a basis transformation between the uncoupled direct product basis

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad (4.189)$$

which we will again denote by

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle, \quad (4.190)$$

and the coupled basis given by the basis states

$$|1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle. \quad (4.191)$$

The constraint (4.187) already results in

$$|1, 1\rangle = |\uparrow\uparrow\rangle, \quad (4.192)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle. \quad (4.193)$$

Notice that indeed the Clebsch-Gordan coefficient in this case is real (+1).

To obtain $|1, 0\rangle$ we use (4.108) applied to $l = 1, m = 1$:

$$L_- |1, 1\rangle = \sqrt{1(1+1) - 1(1-1)} |1, 0\rangle \quad (4.194)$$

$$= \sqrt{2} |1, 0\rangle. \quad (4.195)$$

On the other hand, since $L_- = L_{1-} + L_{2-}$ we can also write

$$L_- |1, 1\rangle = (L_{1-} + L_{2-}) |\uparrow\uparrow\rangle \quad (4.196)$$

$$= (L_{1-} |\uparrow\rangle) |\uparrow\rangle + |\uparrow\rangle (L_{2-} |\uparrow\rangle) \quad (4.197)$$

$$= |\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle. \quad (4.198)$$

Relating this with the result from (4.195) yields:

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle). \quad (4.199)$$

Again due to $m = m_1 + m_2$ (4.187) the state $|0, 0\rangle$ can only be of the form

$$|0, 0\rangle = \alpha |\uparrow\downarrow\rangle + \beta |\downarrow\uparrow\rangle. \quad (4.200)$$

α and β can then be found by requiring orthonormality of the basis states. $\langle 1, 0|0, 0\rangle = 0$ results in

$$\langle 1, 0|0, 0\rangle = \frac{1}{\sqrt{2}} (\alpha \langle \uparrow\downarrow|\uparrow\downarrow\rangle + \alpha \langle \uparrow\downarrow|\downarrow\uparrow\rangle + \beta \langle \uparrow\downarrow|\uparrow\downarrow\rangle + \beta \langle \uparrow\downarrow|\downarrow\uparrow\rangle) \quad (4.201)$$

$$= \frac{1}{\sqrt{2}} (\alpha \cdot 1 + \alpha \cdot 0 + \beta \cdot 0 + \beta \cdot 1) \quad (4.202)$$

$$= \frac{1}{\sqrt{2}} (\alpha + \beta) \quad (4.203)$$

$$\stackrel{!}{=} 0. \quad (4.204)$$

Hence, $\alpha = -\beta$. Normalization of the state $|0, 0\rangle$ then requires

$$|\alpha|^2 + |\beta|^2 \stackrel{!}{=} 1, \quad (4.205)$$

or thus, fixing the global phase of $|0, 0\rangle$ as well:

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (4.206)$$

Since this is probably one of the most important results of this chapter we put it in a box:

$$|1, 1\rangle = |\uparrow\uparrow\rangle, \quad (4.207)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad (4.208)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle, \quad (4.209)$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (4.210)$$

We can then also express the uncoupled basis states in terms of the coupled basis states:

$$|\uparrow\uparrow\rangle = |1, 1\rangle, \quad (4.211)$$

$$|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}} (|1, 0\rangle + |0, 0\rangle), \quad (4.212)$$

$$|\downarrow\uparrow\rangle = \frac{1}{\sqrt{2}} (|1, 0\rangle - |0, 0\rangle), \quad (4.213)$$

$$|\downarrow\downarrow\rangle = |1, -1\rangle. \quad (4.214)$$

Digression: scalar and vector mesons

A **meson** is a particle consisting of a quark and an anti-quark. Since quarks transform in the spin 1/2 representation of $\mathfrak{su}(2)$, the combined system quark + anti-quark transforms in the $\frac{1}{2} \otimes \frac{1}{2}$ representation. From the above considerations, this means that we get a refined classification of mesons by considering only the states of the quark and anti-quark which span irreps of $\mathfrak{su}(2)$. Mesons in the total spin $l = 0$ irrep are called **scalar mesons** whereas mesons in the spin triplet $l = 1$ are **vector mesons**. Other symmetries leading to other quantum numbers allow for an even more refined classification of mesons, resulting in an entire [meson zoo](#). Notice the short lifetimes of these particles.

4.6.4 Coupling of arbitrary spins

In the same spirit as for the case $l_1 = l_2 = \frac{1}{2}$ one can obtain the Clebsch-Gordan coefficients for arbitrary l_1 and l_2 . These coefficients are tabulated in many different places, for example [here](#). This table is reproduced in Appendix 4.A

How do you use this table?

Every table corresponds to a combination l_1, l_2 . For example the table labeled $1/2 \times 1/2$ corresponds to coupling two spin $1/2$'s. This is the case discussed at length above. The rows are then labeled by $m_1 \in \{l_1, l_1 - 1, \dots, -l_1\}, m_2 \in \{l_2, l_2 - 1, \dots, -l_2\}$ and then correspond to the uncoupled basis states

$$m_1, m_2 \leftrightarrow |l_1, m_1\rangle \otimes |l_2, m_2\rangle \equiv |l_1, m_1\rangle |l_2, m_2\rangle. \quad (4.215)$$

The columns are labeled by J and M and correspond to l and $m = m_1 + m_2$ in our notation:

$$J, M \leftrightarrow |J, M\rangle \equiv |l, m\rangle. \quad (4.216)$$

They represent the coupled basis states. The Clebsch-Gordan coefficients are then written on the gray background. The coefficients in a certain column allow you to read of the coefficients to express the coupled basis states in terms of the uncoupled basis and vice versa for the rows. Notice that all the coefficients require an extra square root. So, a coefficient $-8/15$ means that the Clebsch-Gordan coefficient is $-\sqrt{8/15}$! As a first example you can read of the coefficients (4.207-4.210) and (4.211-4.214) in the table $1/2 \times 1/2$. Let us also consider some other examples.

Example: coupling $l_1 = 1, l_2 = 1/2$

According to Theorem 4.3 a spin 1 and a spin $1/2$ couple according to

$$\mathbf{3} \otimes \mathbf{2} \simeq \mathbf{2} \oplus \mathbf{4}. \quad (4.217)$$

The six uncoupled basis states of $\mathbf{3} \otimes \mathbf{2}$ read:

$$|1, 1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle, |1, 1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad (4.218)$$

$$|1, 0\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle, |1, 0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad (4.219)$$

$$|1, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle, |1, -1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle. \quad (4.220)$$

The basis states of the doublet $\mathbf{2}$ with total spin $l = 1/2$ will be denoted

$$|\uparrow\rangle, |\downarrow\rangle. \quad (4.221)$$

Those of $\mathbf{4}$ by:

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle, \left| \frac{3}{2}, \frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{3}{2} \right\rangle. \quad (4.222)$$

What is for example the state $|\uparrow\rangle$ written as a linear combination of uncoupled basis states? Looking at the table for $1 \times 1/2$ and the column labeled $J = 1/2 = M$, $|\uparrow\rangle$, we immediately read of

$$|\uparrow\rangle = \sqrt{\frac{2}{3}} |1, 1\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} |1, 0\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle. \quad (4.223)$$

By looking at the rows instead of the columns, we can also express the uncoupled basis states in terms of the coupled basis. Take for example the state

$$|1, 0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \quad (4.224)$$

by looking at the row labeled $m_1 = 0, m_2 = -1/2$ we read of:

$$|1, 0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} |\downarrow\rangle. \quad (4.225)$$

Example: coupling $l_1 = l_2 = 1$

Let us couple two spin 1's. According to Theorem 4.3 two spin 1's can couple to either a singlet ($l=0$), a triplet ($l=1$) or a spin $l=2$ irrep:

$$\mathbf{3} \otimes \mathbf{3} \simeq \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}. \quad (4.226)$$

The nine basis states of $\mathbf{3} \otimes \mathbf{3}$ are given by

$$|1, 1\rangle |1, 1\rangle, |1, 1\rangle |1, 0\rangle, |1, 1\rangle |1, -1\rangle, \quad (4.227)$$

$$|1, 0\rangle |1, 1\rangle, |1, 0\rangle |1, 0\rangle, |1, 0\rangle |1, -1\rangle, \quad (4.228)$$

$$|1, -1\rangle |1, 1\rangle, |1, -1\rangle |1, 0\rangle, |1, -1\rangle |1, -1\rangle. \quad (4.229)$$

The one basis state of the singlet is given by

$$|0, 0\rangle \quad (4.230)$$

The three basis states of the $l = 1$ irrep are given by

$$|1, 1\rangle, |1, 0\rangle, |1, -1\rangle. \quad (4.231)$$

The five basis states of $\mathbf{5}$ are given by

$$|2, 2\rangle, |2, 1\rangle, |2, 0\rangle, |2, -1\rangle, |2, -2\rangle. \quad (4.232)$$

Looking at the table for 1×1 , we find for example that the state $|l = 0, m = 0\rangle$ is given by:

$$|0, 0\rangle = \frac{1}{\sqrt{3}} |1, 1\rangle |1, -1\rangle - \frac{1}{\sqrt{3}} |1, 0\rangle |1, 0\rangle + \frac{1}{\sqrt{3}} |1, -1\rangle |1, 1\rangle. \quad (4.233)$$

These Clebsch-Gordans can be read of from the column labeled by $J = 0, M = 0$.

We can also look at the rows. For example from the row labeled by $m_1 = 1, m_2 = 0$ we find

$$|1, 1\rangle |1, 0\rangle = \frac{1}{\sqrt{2}} |2, 1\rangle + \frac{1}{\sqrt{2}} |1, 1\rangle \quad (4.234)$$

4.7 Spherical harmonics

4.7.1 Motivation

Spherical harmonics appear everywhere in both classical and quantum physics, and appear as solutions of wave-equations when there is a spherical (rotation) symmetry through the Laplacian. In this section, we will demonstrate that there is a basis in which this operator (i.e. matrix) can be written as a block-diagonal matrix where all blocks are labelled by the irreps of $SO(3)$, or in other words, the integer representations of $SU(2)$. The spherical harmonics $Y_l^m(\theta, \phi)$ are exactly the unitary transformations ("generalized Clebsch-Gordan coefficients") which bring the Laplacian into this block-diagonal form, and hence also represent eigenfunctions of the Laplacian.

The Laplace operator, acting on functions of the form $f(x, y, z)$, is given by

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Written in spherical coordinates (r, θ, ϕ) , $r \geq 0, \theta \in [0, \pi], \phi \in [0, 2\pi]$, we have

$$\nabla^2 = \frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2} - \frac{\hat{J}^2}{r^2}$$

with

$$\begin{aligned}\hat{J}^2 &= \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 \\ &= -\frac{1}{\tan(\theta)} \frac{\partial}{\partial \theta} - \frac{\partial^2}{\partial \theta^2} - \frac{1}{\sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \\ \hat{J}_x &= -i \left(-\sin(\phi) \frac{\partial}{\partial \theta} - \frac{\cos(\phi)}{\tan(\theta)} \frac{\partial}{\partial \phi} \right) \\ \hat{J}_y &= -i \left(\cos(\phi) \frac{\partial}{\partial \theta} - \frac{\sin(\phi)}{\tan(\theta)} \frac{\partial}{\partial \phi} \right) \\ \hat{J}_z &= -i \frac{\partial}{\partial \phi}\end{aligned}$$

All eigenstates of the Laplacian are of the form $f_{ln}(r)Y_l^m(\theta, \phi)$ with Y_l^m the spherical harmonics, the first ones given by the following table:

l	m	Y_l^m
0	0	$Y_0^0(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{1}{\pi}}$
1	-1	$Y_1^{-1}(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta = \frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{(x-iy)}{r}$
	0	$Y_1^0(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta = \frac{1}{2} \sqrt{\frac{3}{\pi}} \frac{z}{r}$
	1	$Y_1^1(\theta, \varphi) = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin \theta = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{(x+iy)}{r}$
2	-2	$Y_2^{-2}(\theta, \varphi) = \frac{1}{4} \sqrt{\frac{15}{2\pi}} e^{-2i\varphi} \sin^2 \theta = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \frac{(x-iy)^2}{r^2}$
	-1	$Y_2^{-1}(\theta, \varphi) = \frac{1}{2} \sqrt{\frac{15}{2\pi}} e^{-i\varphi} \sin \theta \cos \theta = \frac{1}{2} \sqrt{\frac{15}{2\pi}} \frac{(x-iy)z}{r^2}$
	0	$Y_2^0(\theta, \varphi) = \frac{1}{4} \sqrt{\frac{5}{\pi}} (3 \cos^2 \theta - 1) = \frac{1}{4} \sqrt{\frac{5}{\pi}} \frac{(3z^2 - r^2)}{r^2}$
	1	$Y_2^1(\theta, \varphi) = -\frac{1}{2} \sqrt{\frac{15}{2\pi}} e^{i\varphi} \sin \theta \cos \theta = -\frac{1}{2} \sqrt{\frac{15}{2\pi}} \frac{(x+iy)z}{r^2}$
	2	$Y_2^2(\theta, \varphi) = \frac{1}{4} \sqrt{\frac{15}{2\pi}} e^{2i\varphi} \sin^2 \theta = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \frac{(x+iy)^2}{r^2}$

We note that the labels l exactly correspond to the integer labels of the irreps of $SO(3)$, and the number of spherical harmonics with a given l is exactly $(2l+1)$, the dimension of the corresponding irrep. This is of course no coincidence. Consider an arbitrary rotation R around the center, which maps the coordinates (θ, ϕ) to (θ', ϕ') ; it can be shown that the spherical harmonics transform as

$$Y_l^m(\theta', \varphi') = \sum_{m'=-l}^l D_{m'm}^l(R) Y_l^{m'}(\theta, \varphi) \quad (4.235)$$

where $D_{m'm}^l(R)$ is the **irreducible** representation l of the rotation group:

$$\sum_{m'=-l}^l D_{mm'}^l(R) D_{m'm''}^l(R') = D_{mm''}^l(R \cdot R') \quad (4.236)$$

$$D^l(R) D^{l'}(R') = D^l(R \cdot R') \quad (4.237)$$

4.7.2 Generators

How can we understand this from the point of view of Lie groups / algebra's? The spherical harmonics define a unitary transformation which block-diagonalizes the Laplacian:

$$U = \sum_{lm} \int \sin(\theta) d\theta d\phi Y_l^m(\theta, \phi) |l, m\rangle \langle \theta, \phi|$$

with U a unitary operator mapping the Hilbert space of functions over variables $f(\theta, \phi)$ to vectors in the countable vector space spanned by $|l, m\rangle$. This is obvious from the following observation: it can easily be checked that the operators J_x, J_y, J_z form a representation of the Lie algebra of $SO(3)$:

$$\begin{aligned} [J_\alpha, J_\beta] &= i\epsilon_{\alpha\beta\gamma} J_\gamma \\ [J_\alpha, J^2] &= 0 \end{aligned}$$

Just as in the case of finite groups, any representation of a group is equivalent to a direct sum of irreps of that group. For the case at hand, this group is $SO(3)$, we have identified all irreps of $SO(3)$ in the previous sections. Translated to the case of generators of a Lie group, this means that the generators J_α must be equivalent to sums of generators of those irreps; indeed:

$$U J_\alpha U^\dagger = \bigoplus_l \sum_{m, m'} (L_\alpha^l)_{mm'} |l, m\rangle \langle l, m'|$$

with $\alpha \in (x, y, z)$ and L_α^l the generator of the irrep l of the rotation group. Equivalently, we obtain for the Laplacian:

$$U (\nabla^2) U^\dagger = \frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2} - \frac{1}{r^2} \sum_{l, m} l(l+1) |l, m\rangle \langle l, m|$$

There is hence a fascinating connection between special functions and representation theory of groups: they encode the basis change to map a representation of a Lie group defined on a continuous Hilbert space to the irrep basis.

42. Clebsch-Gordan coefficients 1

Note: A square-root sign is to be understood on every coefficient, e.g., for $-8/15$ read $-\sqrt{8/15}$. Notation: J M $\bar{M}$ $\dots$

$1/2 \times 1/2$ $\begin{bmatrix} 1 & 1 \\ +1/2 & -1/2 \\ -1/2 & +1/2 \\ -1/2 & -1/2 \end{bmatrix}$ $Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta$ $2 \times 1/2$ $\begin{bmatrix} 5/2 & 5/2 \\ +2 & +2 \\ -1/2 & -1/2 \end{bmatrix}$ $Y_1^1 = \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$ $Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi}$ $Y_2^0 = \sqrt{\frac{5}{4\pi}} \frac{3}{2} \cos^2 \theta - \frac{1}{2}$ $Y_2^1 = \sqrt{\frac{5}{8\pi}} \sin \theta \cos \theta e^{i\phi}$ $Y_2^{-1} = \sqrt{\frac{5}{8\pi}} \sin \theta \cos \theta e^{-i\phi}$ $Y_2^2 = \frac{1}{4} \sqrt{\frac{5}{2\pi}} \sin^2 \theta e^{2i\phi}$ $3/2 \times 1/2$ $\begin{bmatrix} 5/2 & 5/2 \\ +3/2 & +1/2 \\ -1/2 & -1/2 \end{bmatrix}$ $3/2 \times 1$ $\begin{bmatrix} 5/2 & 5/2 \\ +3/2 & +1/2 \\ -1/2 & -1/2 \end{bmatrix}$ 1×1 $\begin{bmatrix} 2 & 2 \\ +1 & +1 \\ -1 & -1 \end{bmatrix}$ $Y_\ell^{-m} = (-1)^m Y_\ell^{m*}$ $d_{\ell,0}^m = \sqrt{\frac{4\pi}{2\ell+1}} Y_\ell^m e^{-im\phi}$ $(j_1 j_2 m_1 m_2 | j_1 j_2 J M) = (-1)^{-j_1-j_2} (j_2 j_1 m_2 m_1 | j_2 j_1 J M)$ $d_{1,0}^0 = \cos \theta$ $d_{1/2,1/2}^0 = \cos \frac{\theta}{2}$ $d_{1,1}^1 = \frac{1 + \cos \theta}{2}$ $d_{1/2,-1/2}^0 = -\sin \frac{\theta}{2}$ $d_{1,0}^1 = \frac{\sin \theta}{\sqrt{2}}$ $d_{1,-1}^1 = \frac{1 - \cos \theta}{2}$ $d_{3/2,3/2}^0 = \frac{1 + \cos \theta}{2} \cos \frac{\theta}{2}$ $d_{3/2,1/2}^0 = -\sqrt{\frac{1}{2}} \frac{1 + \cos \theta}{2} \sin \frac{\theta}{2}$ $d_{3/2,-1/2}^0 = \sqrt{\frac{1}{2}} \frac{1 - \cos \theta}{2} \sin \frac{\theta}{2}$ $d_{3/2,-3/2}^0 = \frac{1 - \cos \theta}{2} \cos \frac{\theta}{2}$ $d_{3/2,1/2}^1 = -\frac{1 + \cos \theta}{2} \sin \theta$ $d_{3/2,0}^1 = \frac{\sqrt{6}}{4} \sin^2 \theta$ $d_{3/2,-1/2}^1 = -\frac{1 - \cos \theta}{2} \sin \theta$ $d_{3/2,-3/2}^1 = -\frac{1 - \cos \theta}{2} \cos \theta$ $d_{3/2,1/2}^2 = \frac{3 \cos \theta - 1}{2} \cos \frac{\theta}{2}$ $d_{3/2,0}^2 = -\frac{1 - \cos \theta}{2} \sin \theta$ $d_{3/2,-1/2}^2 = \frac{3 \cos \theta + 1}{2} \sin \frac{\theta}{2}$ $d_{3/2,-3/2}^2 = \frac{3 \cos \theta - 1}{2} \sin \frac{\theta}{2}$ $d_{5/2,5/2}^0 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^0 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^0 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^0 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^0 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^0 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,3/2}^1 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^1 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^1 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^1 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^1 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^1 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^2 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^2 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^2 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^2 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^2 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^2 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^2 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^3 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^3 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^3 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^3 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^3 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^3 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^3 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^4 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^4 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^4 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^4 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^4 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^4 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^4 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^5 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^5 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^5 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^5 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^5 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^5 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^5 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^6 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^6 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^6 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^6 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^6 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^6 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^6 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^7 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^7 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^7 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^7 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^7 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^7 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^7 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^8 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^8 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^8 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^8 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^8 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^8 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^8 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^9 = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^9 = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^9 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^9 = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^9 = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^9 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^9 = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^{10} = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^{10} = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^{10} = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^{10} = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^{10} = \frac{1 - \cos \theta}{2} \cos \theta$ $d_{5/2,-3/2}^{10} = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,-5/2}^{10} = \frac{1 - \cos \theta}{2} \sin \theta$ $d_{5/2,5/2}^{11} = \frac{1}{2} \frac{1 + \cos \theta}{2}$ $d_{5/2,3/2}^{11} = \frac{1 + \cos \theta}{2} \sin \theta$ $d_{5/2,1/2}^{11} = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,0}^{11} = \frac{1 + \cos \theta}{2} \cos \theta$ $d_{5/2,-1/2}^{11} = \frac$

Figure 42.1: The sign convention is that of Wigner (*Group Theory*, Academic Press, New York, 1959), also used by Condon and Shortley (*The Theory of Atomic Spectra*, Cambridge Univ. Press, New York, 1953), Rose (*Elementary Theory of Angular Momentum*, Wiley, New York, 1957), and Cohen (*Tables of the Clebsch-Gordan Coefficients*, North American Rockwell Science Center, Thousand Oaks, Calif., 1974).

Figure 4.3: Taken from [Particle Data Group](#).

Chapter 5

Crystallography: Frieze Groups, Wallpaper Groups, Space Groups

Before embarking in crystallography, let us consider the simpler problem of finding all finite subgroups of $SO(3)$.

5.1 Finite subgroups of $SO(3)$

Let's try to find all finite subgroups of $SO(3)$. We will do this by finding all possible representations of orthogonal 3×3 matrices $\in SO(3)$, as those provide a faithful representation of $SO(3)$ (meaning that there is a 1 to 1 correspondence between group elements and matrices), and then by finding discrete families of those forming a subgroup.

- Given any matrix $A \in SO(3)$, its eigenvalues are $e^{\pm i\theta}$ and 1, hence leaving an axis of rotation $|\alpha\rangle$ invariant (corresponding to eigenvalue 1).
- Given a finite subgroup $G \subset SO(3)$ with n elements, take an element $X \neq e$ out of it with corresponding axis of rotation $|\alpha\rangle$, and define $G^\alpha \subseteq G$ as the subgroup having this axis of rotation. As this group is finite and Abelian, the corresponding eigenvalues of all those elements must be of the form $\theta = \pm 2\pi m/n_\alpha$ for some integer n_α , and hence this subgroup is isomorphic to Z_{n_α} and we call the axis n_α -fold. W.l.o.g. we can assume that X has eigenvalues $1, e^{\pm i2\pi/n_\alpha}$.
- Take this n_α -fold axis of rotation, and any other element $Y \in G$; then $Y|\alpha\rangle$ is also an n_α fold axis of rotation: indeed, $YXY^{-1}|\alpha\rangle = Y|\alpha\rangle$. Let us therefore consider the complete orbit associated to $|\alpha\rangle$ by applying all group elements Y to it and call this orbit O_1 with associated integer $n_1 = n_\alpha$. The number of distinct elements in this orbit is $\#G/n_1$; indeed, if $Y_1|\alpha\rangle = Y_2|\alpha\rangle$, $\exists m : Y_2^{-1}Y_1 = X^m$, and hence Y_1 and Y_2 give rise to the same point in the orbit iff they are related to each other by a power of X , and there are precisely n_1 distinct powers. We can now repeat this for any other n_j fold axis of rotation of all elements of the group G lying on the unit sphere (including the vectors $-|\alpha\rangle$ if those were not yet generated!), and we end up with k orbits O_j with each $\#G/n_j$ elements. The union of all orbits is the set S of all axis of rotation on the unit sphere.
- The following now holds:

$$\sum_{i=1}^k \left(1 - \frac{1}{n_j}\right) = 2(1 - 1/n) \quad (5.1)$$

Proof: Let us consider the set $P = \{(|\alpha\rangle, A), |\alpha\rangle \in S, A \in G, A \neq I, A|\alpha\rangle = |\alpha\rangle\}$. There are two ways of counting the number of elements in this set:

k	n_1	n_2	n_3	$\#G$	Name
2	n	n	—	n	$C_n = Z_n$
3	2	2	n	$2n$	D_n
3	2	3	3	12	T
3	2	3	4	24	C
3	2	3	5	60	I

(5.2)
Table 5.1: Finite subgroups of $SO(3)$

1. For every element in the group $A \in G$, there are two associated axis of rotation ($\pm|\alpha\rangle$), and we have to exclude I , hence $\#P = 2(\#G - 1)$.
 2. To every element of an orbit O_i , we can associate $n_i - 1$ group elements in P ; hence $\#P = \sum_{i=1}^k \#O_i(n_i - 1) = \#G \sum_i (1 - 1/n_i)$.
- As $n, n_i \geq 2$, we can easily figure out that the only possible cases are $k = 2$ or $k = 3$, as otherwise the left hand side would be larger than the right hand side, readily leading to the following table: C_n corresponds to two orbits $\pm|\alpha\rangle$ each with 1 point. This group is just Z_n . D_n has 1 n -fold orbit consisting of 2 points, and two 2-fold orbits with n points. Hence there is one n -fold axis, and the n 2-fold rotations interchange the two ends of this axis. This group is called the dihedral group D_n and is isomorphic to $Z_n \rtimes Z_2$ (indeed, $\forall n$, Z_2 is a subgroup of $A(C_n)$). Note also that D_1 is isomorphic to C_2 with respect to conjugation with elements in $SO(3)$, although they cannot be converted into each other with respect to conjugations in $O(2)$.
 T denotes the symmetries of a tetrahedron. It turns out that $T \cong A_4 \cong (Z_2 \times Z_2) \rtimes Z_3$
 C denotes the symmetries of the cube and its dual the octahedron. It holds that $C \cong S_4 \cong (Z_2 \times Z_2) \rtimes S_3$.
 I denotes the symmetries of the icosahedron and its dual the dodecahedron. We have $T \cong A_5$.
This classifies all platonic solids.

A more refined classification can be obtained if we also include reflections, it is when considering $O(3)$ instead of $SO(3)$. In this case, the larger groups emerging are the direct products of C_n, D_n, A_4, S_4, A_5 with the reflection Z_2 , plus the so-called “mixed groups” $C_{2n}C_n, D_nC_n, S_4A_4, D_nD_n$ of order $2n, 2n, 24, 4n$. Those groups form the starting point for the classification of the point groups in crystallography.

5.2 Crystallography

Crystallography is an immense subject; its main achievement is the classification of all finite subgroups of the translational group combined with rotations and reflections. This can be done along the lines of the previous example, but clearly rotations and translations can interact in very non-trivial ways. This leads to the classification of all space groups in 3D, giving rise to 230 different groups. The amazing thing is that all of those different symmetries can be found in crystals. Those 230 groups are built up by combining the 32 point groups (point symmetries such as rotations and reflections through a point) with the 14 Bravais lattices; jointly they give rise to glide reflections, symmetries involving both translations and reflections at the same time.

For the case of 2 dimensions, the problem is to identify the so-called wallpaper groups, and the same classification gives rise to 17 possibilities. People seem to be as inventive as nature, as the Egyptians had already made decorations involving all of those 17 different groups.

A key theorem in both of those classifications involves the following **crystallographic restriction**¹: Let a crystal be invariant under rotations $2\pi/n$ around an axis, then $n = 2, 3, 4, 6$.

¹Apparently, the Egyptian artists spent thousands of years to make wallpaper symmetries with a five-fold axis

The proof is surprisingly simple and is given in bullet point 3 of the subsequent discussion.

There is a beautiful systematic way of understanding that there are 17 wallpaper groups and 230 space groups by making use of relatively simple ideas of linear algebra. Let us present here the main ideas in doing this classification; the example of wallpaper groups is then explicitly worked out.

1. A lattice is a periodic structure of vertices, and the vertices are located at the coordinates $n_1\vec{e}_1 + n_2\vec{e}_2 + n_3\vec{e}_3$ with $\vec{e}_i$ the lattice vectors and $n_i \in \mathbb{Z}$. Note that all points must be included. Let us denote the matrix of lattice vectors E . Note that the hexagonal lattice is for example not a lattice, but becomes a lattice after blocking two sites; the goal of crystallography is to be able to deal with such cases in a systematic way.
2. We are interested in finding lattice structures with rotational and/or reflection symmetries, represented by 3×3 matrices O_g in the group $O(3)$. We are looking for all subgroups of $O(3)$ for which there exists an integer representation N_g , it is for which there is a basis E such that

$$O_g \cdot E = E \cdot N_g \quad (5.3)$$

and N_g only contains integer coefficients. This is indeed a necessary and sufficient condition for this rotation/reflection to map lattice vectors to integer linear combinations of those lattice vectors. Two solutions N_g which are equivalent to each other by conjugation with an element in $GL(3, \mathbb{Z})$ (the 3×3 matrices with integer coefficients for which the determinant is ± 1) correspond to the same symmetry, so we are only interested in the inequivalent matrix representations.

3. We have already classified all subgroups of $SO(3)$ and the subgroups of $O(3)$. In the case of space groups we have the extra restriction that the elements O_g must be such that they are equivalent to an integer matrix (where the conjugation is the matrix of basis vectors: $E^{-1}O_gE = N_g$). The eigenvalues of an integer matrix corresponding to an element with order n must be of the form $(\pm 1, \pm e^{i2\pi/n}, \pm e^{-i2\pi/n})$; it can directly be seen that the trace of this matrix, which has to be an integer because it is equivalent to a matrix with only integers, can only be integer iff $n = 1, 2, 3, 4, 6$, hence we have a simple proof for the crystallographic restriction. The remaining groups are the so-called point groups, and there are 32 of them.
4. The problem is hence reduced to finding inequivalent (w.r.t. $SL(3, \mathbb{Z})$, it is just integer combinations of the unit vectors) integer representations of all rotation/reflection groups in the list of 32 point groups. It turns out that there are only a few such representations. In 3 dimensions, there are exactly 73 inequivalent integer representation N_g (note that there are only 32 inequivalent representations with respect to $SL(3, \mathbb{R})$). These define the **73 symmorphic space groups**, space groups for which there is one point which remains invariant under all possible rotations. To obtain those, we have to do some painstaking linear algebra for every separate pointgroup, and this is done in the appendix for the 2D case.
5. Given such a solution N_g , we can easily construct the basis vectors E associated to them by solving the linear equation $O_g \cdot E = E \cdot N_g$. Many of those 73 solutions correspond to the same lattice, and indeed there are only 14 distinct solutions E , which define the **Bravais lattices**. The distinction between the 73 cases corresponds to the fact that the Bravais lattices can be decorated with objects with more or less symmetry. From a mathematics point of view, there are 73 representations which are inequivalent from the point of view of transformation in $SL(3, \mathbb{Z})$, but only 14 if we allow conjugation with $SL(3, \mathbb{R})$. This marks the difference between the "arithmetic" and "geometric" classes.
6. The 73 symmorphic space groups only form a small part of the full classification, as it is possible to have lattices with symmetries which do not leave any point invariant. In 2D for

of rotation, but never managed to do this!

example, the brickwall lattice contains special symmetries that can only be revealed by doing half-shifts and reflections. In 3D, such extra symmetries are called glide plane symmetries and screw axis symmetries. To obtain all of those, we have to introduce group elements of the form $(\vec{\omega}(g), g)$ with $g \in O(3)$ labeling the rotation/reflection and $\omega(g) = \vec{\omega}(g) \in R^3$ a translation vector in space (e.g. a half shift in some direction). The meaning of this element is that we first do a point group operation, and then follow this up with a translation with the vector $\vec{\omega}(g)$. For a given set of lattice vectors transforming according to one of the 73 possible integer representations N_g , the following group structure emerges:

$$(\omega(g_2), g_2) \cdot (\omega(g_1), g_1) = (\omega(g_2) + N_{g_2} \cdot \omega(g_1), g_2 \cdot g_1)$$

where the matrix N_g is acting on the vector ω . We indeed have to rotate the shifted vector ω_1 , and then add a new shift ω_2 to it. For these group elements to form a closed group, we must have that the right hand side of this equation is equal to

$$(\omega(g_2) + N_{g_2} \cdot \omega(g_1), g_2 \cdot g_1) \equiv (\omega(g_2 \cdot g_1), g_2 \cdot g_1)$$

up to integer shifts with the lattice vectors. Consistency hence dictates the vector equation

$$\omega(g_2) + N_{g_2} \cdot \omega(g_1) - \omega(g_2 \cdot g_1) = 0 \quad \text{mod } Z^3$$

This is a linear set of $3n^2$ equations with n the order of the group for $3 \cdot n$ unknowns, which is of course overdetermined. Using the Smith normal form (see appendix) we can easily solve for this problem and find all inequivalent solutions ω_g . The null vectors of the above equation over the real numbers as opposed to R/Z correspond to global shifts specified by a vector ω , and those so-called co-boundaries and of the form $\omega_{cb}(g) = N_g \cdot \omega - \omega$. Obviously, two solutions related to each other by a global shift are equivalent, and we have to mod out those co-boundaries. There can also be nontrivial solutions of the cocycle condition which are only solutions over R/Z but not over R . As will become clear in the discussion on cocycles, topologically distinct solutions of those equations exist due to the fact that they are characterized by an integer index. The nontrivial solutions of this equation are called 1-cocycles of the group G with respect to the group action N_g on $(R/Z)^{\otimes 3}$: $H_{N_g}^1(G, (R/Z)^{\otimes 3})$ with R/Z isomorphic to $U(1)$, and those solutions can readily be found by doing linear algebra over the integers using the Smith normal form. In conclusion, the topologically distinct solutions of the 1-cocycle condition yield different groups with group action $(\omega_g, g) \cdot (\omega_h, h) = (\omega_{g \cdot h}, g \cdot h)$.

7. There is only 1 subtlety left: solutions corresponding to different cocycles might correspond to the same space group, as an automorphism of the point group might transform different cocycles into each other. The allowed automorphisms in the context of the space groups can always be effected through a conjugation with an arbitrary dilation $\in GL(3, R) \times Z^{\otimes 3}$ (it is with respect to dilations outside of the space group). A dilation is composed as a product of a translation and an element in $GL(3, R)$ (general linear Lie group over the real numbers). Translations are completely characterized by the co-boundaries of the 1-cocycle, and have already been taken into account. It only remains to look at point transformations X_q in $GL(3, R)$ which generate inner automorphisms on the point group elements g : $\beta_q(g) = h : X_q N_g X_q^{-1} = N_h$ with N_g the integer representation of the point group symmetry, but with X_q not bound to be integers but just elements of $GL(3, R)$. For a given group specified by the group elements $(\omega(g), g)$, a conjugation of all elements with q amounts to the operation

$$(0, q) \cdot (\omega(g), g) \cdot (0, q^{-1}) = (X_q \cdot \omega(g), \beta_q(g))$$

(we are just adopting the group multiplication rule over all dilations $\in GL(3, R) \times Z^{\otimes 3}$). As X_q has a nontrivial action, it might indeed transform the cocycle $\omega_1(g)$ into a different one $\omega_2(g)$. Two different cocycles $\omega_1(g), \omega_2(g)$ are then physically equivalent iff there exists

a dilation of the type discussed which transforms them into each other up to a trivial co-boundary. Formally, they are equivalent when $\exists q, \tilde{q}$ such that $X_q N_g X_q^{-1} = N_{\beta_q(g)}$ and $X_{\tilde{q}} N_h X_{\tilde{q}}^{-1} = N_{\tilde{\beta}_{\tilde{q}}(h)}$ for which $\beta_q(g)$ and $\tilde{\beta}_{\tilde{q}}(h)$ are automorphisms and for which $\forall g, h$:

$$\begin{aligned} X_q \omega_1(g) &\simeq \omega_2(\beta_q(g)) \\ X_{\tilde{q}} \omega_2(h) &\simeq \omega_1(\tilde{\beta}_{\tilde{q}}(h)) \end{aligned}$$

with $\simeq$ meaning equal up to trivial coboundaries.

8. It is actually pretty straightforward (but a lot of work) to implement this while program. Doing this for the 73 symmorphic groups leads to the 230 space groups.

The mathematics underlying this classification is the one of cohomology theory and exact sequences: we are trying to find all possible ways to merge the Abelian group A of lattice translations with the rotation/reflection group H :

$$1 \rightarrow A \rightarrow G \rightarrow H \rightarrow 1$$

as will be discussed in the next section. As an illustration, the specific example of the space groups for 2-dimensional lattices (i.e. the wallpaper groups), is specifically worked out here.

5.2.1 Wallpaper groups

The crystallographic restriction obviously also applies in the 2-dimensional case, and it is easy to derive that the only possible finite subgroups of $O(2)$ are $1, Z_2, Z_3, Z_4, Z_6, D_1, D_2, D_3, D_4, D_6$. Note that D_1 and Z_2 have to be treated differently, as one involves a reflection while the other one does not; the representations for those groups are hence certainly not equivalent. The first step in the program is to identify all inequivalent integer representations of those 10 groups, where inequivalent means with respect to $SL(2, Z)$, the integer 2×2 matrices with determinant equal to $+1$ (if we allowed for determinant -1 , we would get a more coarse grained classification). Let us consider the 10 different cases, where in each case we are looking for interger representations N_g for which $O_g.E = E.N_g$. We will write the generators of the representation N_g as G .

1. Z_2 : As the orthogonal group element corresponding to the generator is just $O_g = -I$, we are looking for integer matrices with trace equal to -2 and determinant $+1$ that square to the identity:

$$G = \begin{bmatrix} a & b \\ c & -2-a \end{bmatrix}$$

with $G^2 = I$. After some trivial algebra, we get the condition $b = c = 0$, and hence the only solution is $G = -I$.

2. D_1 : Here $O_g = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, an orthogonal matrix with trace equal to zero and determinant -1 . Hence G must be of the form

$$G = \begin{bmatrix} x & (1-x^2)/y \\ y & -x \end{bmatrix}$$

Let us consider a similarity transformation of the form XX^{-1} with

$$X = \begin{bmatrix} 1 & 0 \\ -x+1 & 1 \end{bmatrix}, \quad X^{-1} = \begin{bmatrix} 1 & 0 \\ x-1 & 1 \end{bmatrix},$$

then we get immediately that

$$XX^{-1} = \begin{bmatrix} 1 & x \\ 0 & -1 \end{bmatrix}$$

Using a similarity transform $\begin{bmatrix} 1 & -[x/2] \\ 0 & 1 \end{bmatrix}$ we get two possibilities:

$$G = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{or} \quad G = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

Doing a similarity transform with $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ for the second case, we obtain two inequivalent possibilities for D_1 :

$$G = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{or} \quad G = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The first one corresponds to a reflection along a main crystal axis, the second one along an axis which is rotated over $\pi/4$. Note that those matrices are equivalent with respect to $GL(2, R)$ but not w.r.t. $GL(2, Z)$. This marks the distinction between a geometric and an arithmetic crystal class. It is the more refined arithmetic crystal class that is relevant here.

3. Z_4 : Here $O_g = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, so we want to find a normal form for matrices with trace equal to zero and determinant 1. We hence have

$$G = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

with $a^2 + b.c = -1$. We will reduce this matrix recursively. Assume first that $|a| < |b|$ and $|a| < |c|$. As $a^2 + b.c = -1$, the only possibility is that $a = 0$ and $b, c = \pm 1$. We will now do operations that are guaranteed to yield $|a| < |b|$ and $|a| < |c|$, such that there is a unique case.

Assume first that $|a| \geq |b|$. Then there exists a $n \in Z$ and corresponding $X = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$ which brings me to the form

$$\begin{bmatrix} a - n.b & b \\ \tilde{c} & n.b - a \end{bmatrix}$$

for some $\tilde{c}$; let's write $\tilde{a} = a - n.b$. Clearly, I can always find a n such that $|\tilde{a}| < |b|$. As $\tilde{a}^2 + b.\tilde{c} = -1$, this guarantees that $|\tilde{c}| < |c|$. We now repeat the story with $X = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$. This makes the new a smaller than c , while also reducing b . This recursive algorithm has to stop, which is only possible when $a = 0$. The Z_4 case hence yields a unique solution:

$$G = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

4. Z_3 : The proof is similar to the case of Z_4 ; here we are looking for matrices for which the trace is -1 and the determinant $+1$, hence of the form

$$\begin{bmatrix} -a & b \\ c & a - 1 \end{bmatrix}$$

with $a - a^2 - b.c = 1$. We need a small lemma, which can readily be proved: given fixed b and two solutions with $0 \leq a_1 < a_2$, then the corresponding c_1, c_2 satisfy $c_1 < c_2$. W.l.o.g. we can indeed choose $a > 0$. Repeating the same recursive algorithm as before, we will converge to the case $a = 0$, which leads to the representation

$$G = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$$

5. Z_6 : here we want the trace to be equal to +1 and the determinant also +1. The case is exactly as in the Z_3 situation, and we readily find

$$G = \left[\begin{array}{cc} 0 & 1 \\ -1 & 1 \end{array} \right]$$

6. D_2 : here we need two generators, as it is equivalent to $Z_2 \times D_1$. A subgroup is certainly Z_2 , so we can easily use the unique representation of Z_2 for those elements. As $-I$ commutes with everything, there will be 2 solutions for the reflection part as derived for D_1 :

$$G = \left\{ \left[\begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array} \right], \left[\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array} \right] \right\}$$

$$G = \left\{ \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], \left[\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array} \right] \right\}$$

7. D_3 : a subgroup of D_3 is Z_3 , so w.l.o.g. we can take the integer representation of Z_3 for those elements. We now need to find a matrix T which stands for the reflection with the property that $T.R = R^2.T$. This can readily be found using the Smith normal form of the matrix $Q = R^T \otimes I - I \otimes R^2$ and looking at its null space. The null space is spanned by 2 matrices, namely $T_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$ and $T_2 = \left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array} \right]$. We have to look for any integer linear combination of those matrices, and check whether any of them forms a representation of D_1 . The condition that the determinant is -1 yields the equation $\det(aT_1 + bT_2) = -1$ or $a^2 + b^2 + a.b = -1$. This equation has as only solutions $(\pm 1, 0), (0, \pm 1), \pm(1, -1)$, corresponding to $T_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$, $T_2 = \left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array} \right]$ and $T_3 = \left[\begin{array}{cc} 1 & -1 \\ 0 & -1 \end{array} \right]$ up to a sign. The first and last one are related to each other by $T_3 = R.T_1.R^{-1}$ hence yields an equivalent representation. We hence obtain two $SL(2, Z)$ inequivalent solutions of with generators:

$$G = \left\{ \left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array} \right], \left[\begin{array}{cc} 0 & -1 \\ 1 & -1 \end{array} \right] \right\}$$

$$G = \left\{ \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], \left[\begin{array}{cc} 0 & -1 \\ 1 & -1 \end{array} \right] \right\}$$

8. D_4 : similar to the D_3 case, we have the Z_4 subgroup which we can represent with $R = \left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array} \right]$. Forming the matrix $Q = R^T \otimes I - I \otimes R^3$, we find that the null space is spanned by the matrices $T_1 = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$ and $T_2 = \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right]$. In this case, no new representations are obtained by taking integer linear combinations of the two, so these are the only two cases. However, the two representations obtained are equal to each other up to an automorphism (no similarity transformation needed, just a relabeling of the matrices), hence only 1 case survives:

$$G = \left\{ \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], \left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array} \right] \right\}$$

9. D_6 : similar as before, we start from Z_6 and look for solutions $T.R = R^5.T$. The space is spanned by the two vectors as obtained in the D_3 case, so the same analysis goes through. But just as in the case D_4 , the two solutions obtained are equivalent up to automorphisms:

$$G = \left\{ \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], \left[\begin{array}{cc} 0 & 1 \\ -1 & 1 \end{array} \right] \right\}$$

In total, we have thus found 12 subgroups of $O(3)$ with integer representations which are inequivalent with respect to $SL(2, Z)$ transformations. Additionally, there is 1 with no symmetries at all. Those correspond to the point group symmetries, and we now want to add glide planes to the picture through the use of 1-cocycles as explained in the main text. We find only nontrivial cocycles for the cases D_1 (first case), D_2 (first case) and D_4 . Note that the cocycles $\omega(g)$ are vectors in R^2 , and are solutions of $N_g\omega(h) - \omega(g.h) + \omega(g) = 0$ modulo co-boundaries of the form $N_g\omega_0 - \omega_0$, where N_g forms one of the 13 possible solutions just obtained. Those solutions are readily found using the Smith normal form.

- Case D_1 : the only nontrivial element of the group is the reflection operator, and the nontrivial Z_2 cocycle

$$(\omega(g), g) = \left(\begin{bmatrix} 1/2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right)$$

is a nontrivial solution of the equation $N_g\omega(h) - \omega(g.h) + \omega(g) = 0$.

- Case D_2 : there are only nontrivial cocycles for the first solution of D_2 with the reflection operator equal to $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. By making use of the Smith normal form, we see that there are 2 nontrivial cocycles ($H^1(G, A) = Z_2 \times Z_2$). Here we write the 2 solutions for the four group elements corresponding to the identity element, reflection, rotation and rotation+reflection:

$$\omega_1 = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} \right\}$$

$$\omega_2 = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

As cocycles can be summed, we get 3 nontrivial solutions. Note that the co-boundaries for the cocycles are given by $\omega_{cb}(g) = (N_g - I)\omega_0$ or

$$\omega_{cb} = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} x \\ 0 \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} 0 \\ y \end{bmatrix} \right\}$$

with $x, y \in R$.

As discussed in the main text, it might be that some of those 3 nontrivial solutions can be mapped into each other by a conjugation of a general dilation. There is indeed such a dilation, not part of the group elements, which induces an automorphism on the group elements, and its integer representation is $X_g = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; this dilation swaps the matrices corresponding to reflection and rotation+reflection under conjugation. We therefore have to check whether $X_g\omega(g^{-1}hg) = \omega'(h)$ up to a co-boundary for all pairs of h, h' :

$$X_g \cdot \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} \simeq \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (5.4)$$

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$$X_g \cdot \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} \simeq \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} \quad (5.6)$$

Note that the last term in every line is a co-boundary with $\tilde{x} = 0, \tilde{y} = 1/2$. This dilation therefore identifies 2 of the three nontrivial solutions. It turns out that the sum of the two cocycles cannot be converted into this one, so we end up with 3 space group classes for the

Über die Analogie der Kristallsymmetrie in der Ebene.

281

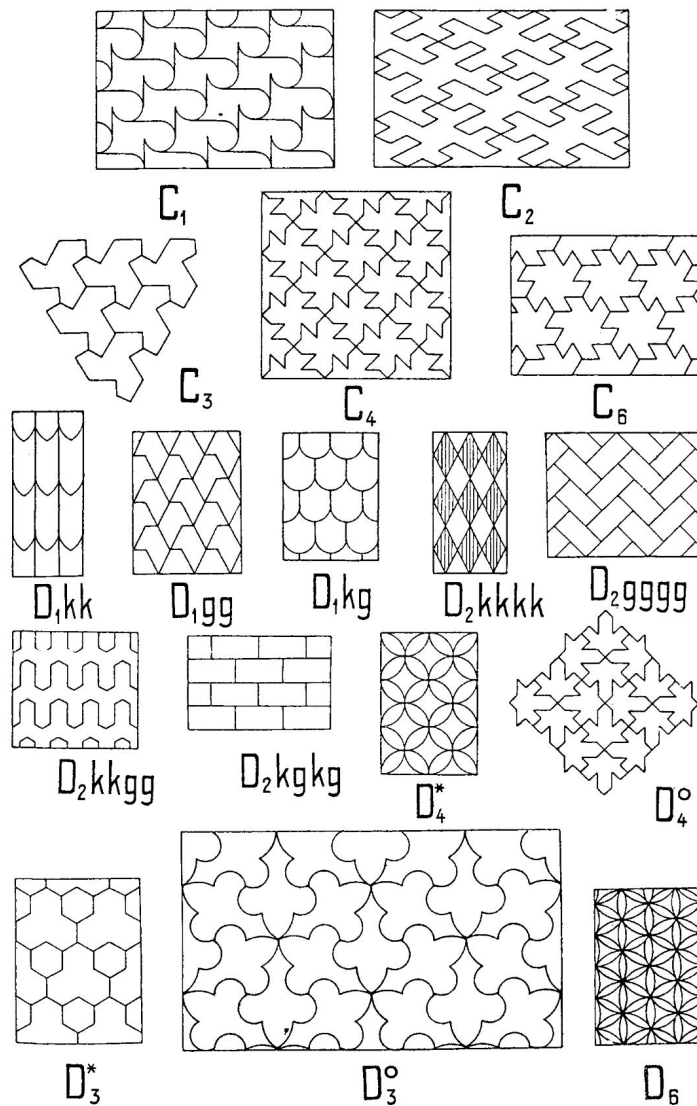


Figure 5.1: All 17 wallpaper groups. From the original article of Polya.

first class of solutions of $D2$ (namely the trivial one and 2 nontrivial glide planes). The second class of solutions N_g for $D2$ does not exhibit cocycles, so in total there are 4 distinct wallpaper groups with a $D2$ symmetry.

- Case D_4 : there is one Z_2 cocycle, for which the generator of reflections is

$$\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right)$$

and for rotations is

$$\left(\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right)$$

This means that we have created $1 + 2 + 1$ extra wallpaper groups associated with glide planes, and so we discovered all 17 wallpaper groups! Hereby a picture of the 17 cases; I leave it as an exercise to match the corresponding representations!

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